

WSQ COURSE SLIDES

Data Mining and Machine Learning Fundamentals for Beginners

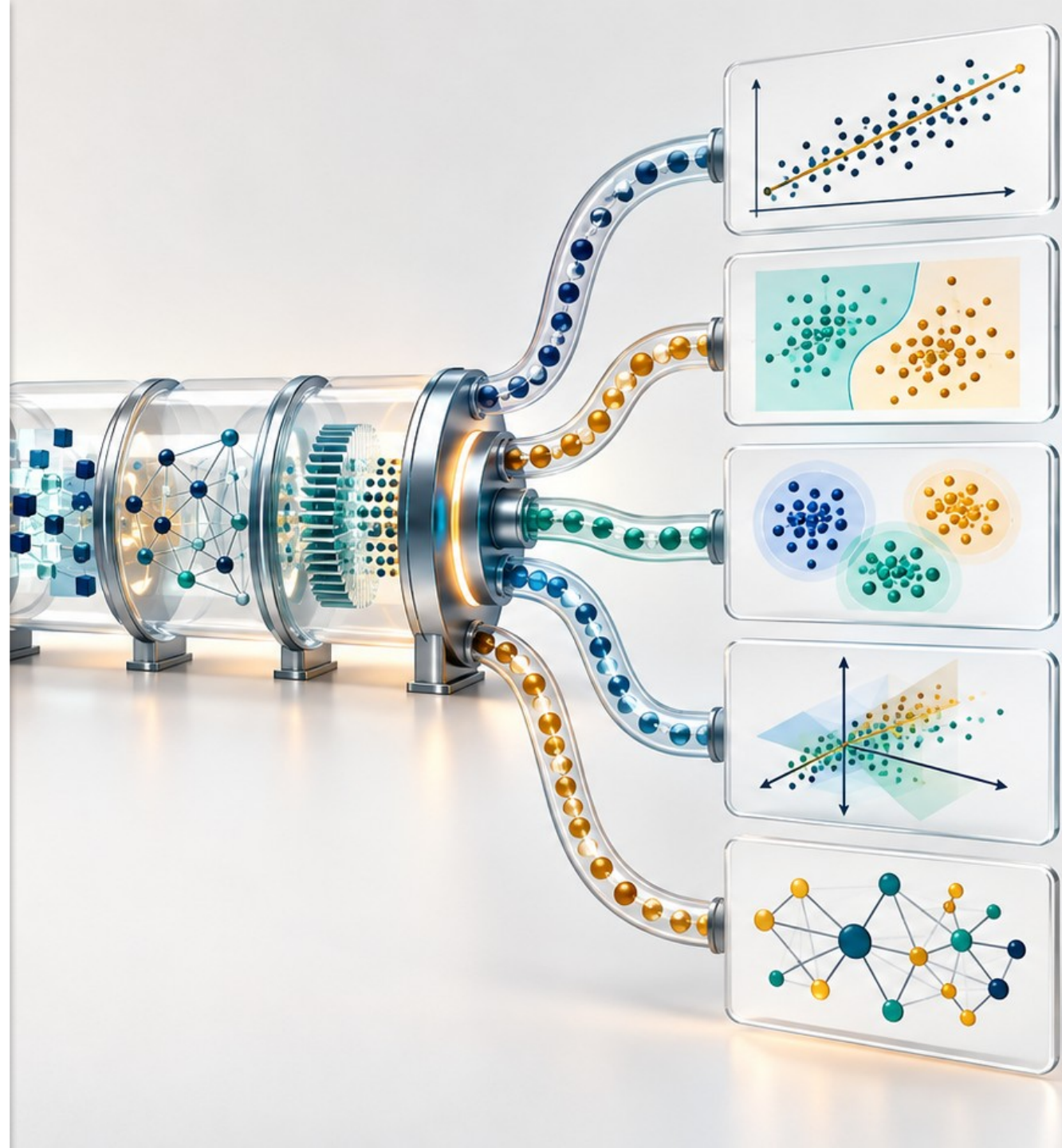
Course Code TGS-2020503264

Data Analytics · MED-ACE-3018-1

PYTHON + STREAMLIT · 15 LABS

Version 17.0 · 10 September 2026

Tertiary Infotech Academy Pte Ltd · UEN 201200696W



Digital attendance · evidence trail

1

AM check-in

2

PM check-in

3

**Assessment
check-in**

4

**Attendance
≥75%**

Your trainer



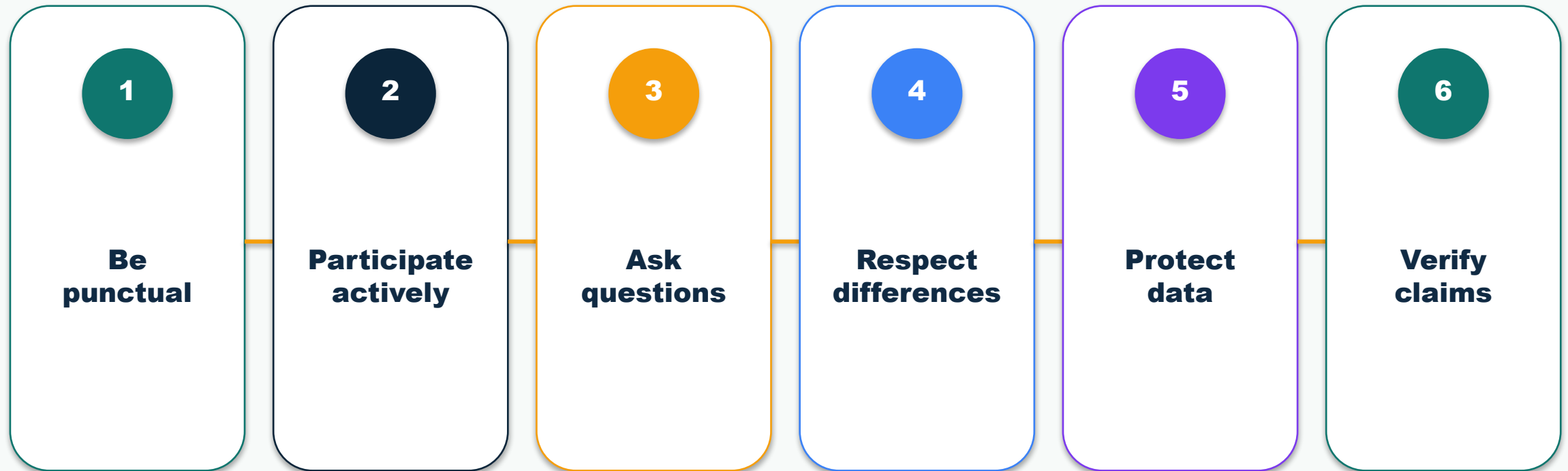
Dr Alfred Ang

Principal trainer · data science, machine learning, optimisation, and applied AI education

Facilitation focus

Mechanisms · reproducible evidence · business interpretation · responsible decisions

Ground rules for an evidence-led lab



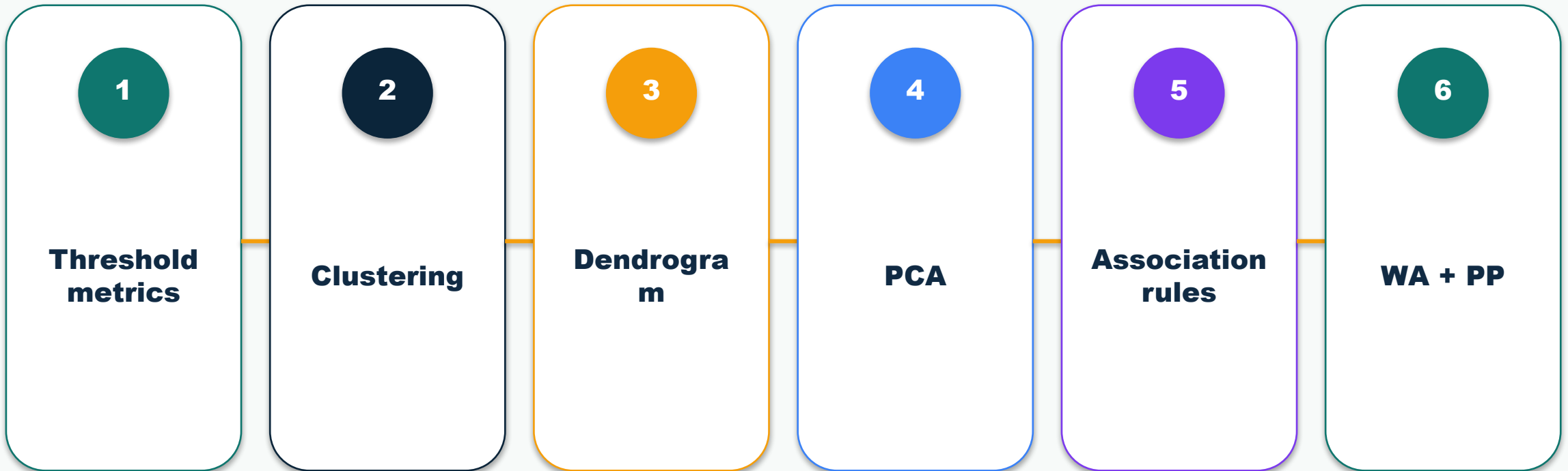
Trace the named state at each boundary; a missing state breaks reproducibility.

Day 1 · prepare, predict, and classify



Trace the named state at each boundary; a missing state breaks reproducibility.

Day 2 · discover, compress, and associate



Trace the named state at each boundary; a missing state breaks reproducibility.

Learning outcomes mapped to evidence

LO1: Apply data mining and machine learning principles to assess business insights.

LO2: Integrate and aggregate information from multiple datasets to build data models.

LO3: Apply predictive data modelling techniques to identify underlying trends in data.

LO4: Apply machine learning classification techniques to gain new insights from data.

LO5: Apply clustering techniques to discover patterns, visualise evidence, and make decisions.

LO6: Develop prototype algorithms with dimension-reduction and feature-ranking techniques.

LO7: Construct association rules to identify patterns across transactional datasets.

Assessment evidence chain



Trace the named state at each boundary; a missing state breaks reproducibility.

TSC abilities A1–A4

#

A1 Integrate information from multiple datasets

#

A2 Apply predictive modelling

#

A3 Identify patterns across datasets

#

A4 Develop prototype algorithms

TSC abilities A5–A8

#

A5 Decide which patterns warrant further analysis

#

A6 Assemble aggregations into data models

#

A7 Use machine learning for new insight

#

A8 Assess organisational impact

TSC ability A9 · interactive evidence

#

A9 Support interactive data visualisation

{ }

proof

readable chart + labels

{ }

interaction

selected fields + parameters

{ }

decision

interpretation + limitation

TSC knowledge K1–K4

#

K1 Data collection, mining, cleaning and analysis

#

K2 Statistical principles

#

K3 Communicating data effectively

#

K4 Organisational domains and processes

TSC knowledge K5–K8

#

K5 Exploratory visual analysis and predictive modelling

#

K6 Machine-learning algorithms

#

K7 Building a data model

#

K8 Established and novel analytical techniques

TSC knowledge K9–K10

#

K9 Applying software tools to analyse data

#

K10 Statistical and experimental techniques

{

evidence

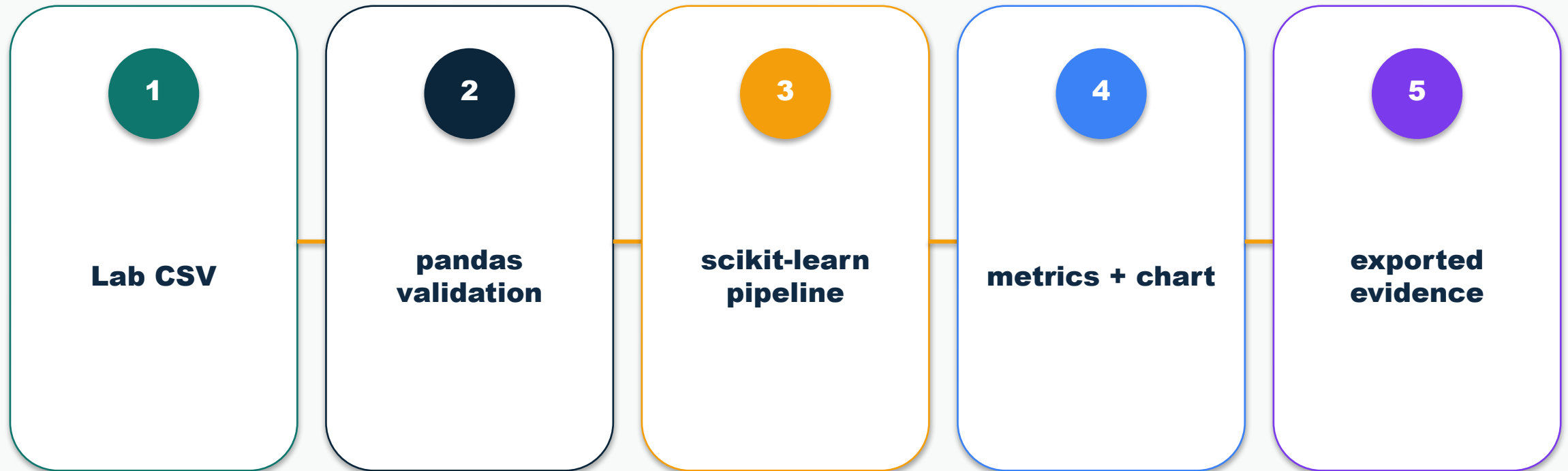
runnable central app

{

mapping

WA Q1–Q10

Central app architecture



Trace the named state at each boundary; a missing state breaks reproducibility.

Model outputs require judgement

FAILURE MODE

A visually convincing chart can hide leakage, bias, or a weak sample.

CONTROL

Pair every result with a baseline, error metric, data scope, and human review.

VERIFICATION

A second learner can reproduce the output from the same file and parameters.

Two-day learning arc



Trace the named state at each boundary; a missing state breaks reproducibility.

LMS evidence flow



Trace the named state at each boundary; a missing state breaks reproducibility.

Assessment briefing

FAILURE MODE

Unapproved collaboration or copied work invalidates individual evidence.

CONTROL

Use only approved course materials; complete every item and keep evidence traceable.

VERIFICATION

WA answers and PP outputs are uploaded to the LMS and signed off by the assessor.

How to read the technical slides

1

Mechanism

2

Contract

3

Calculation

4

Evidence

The final control panel on each concept names a failure, a control, and observable proof.

TOPIC 01

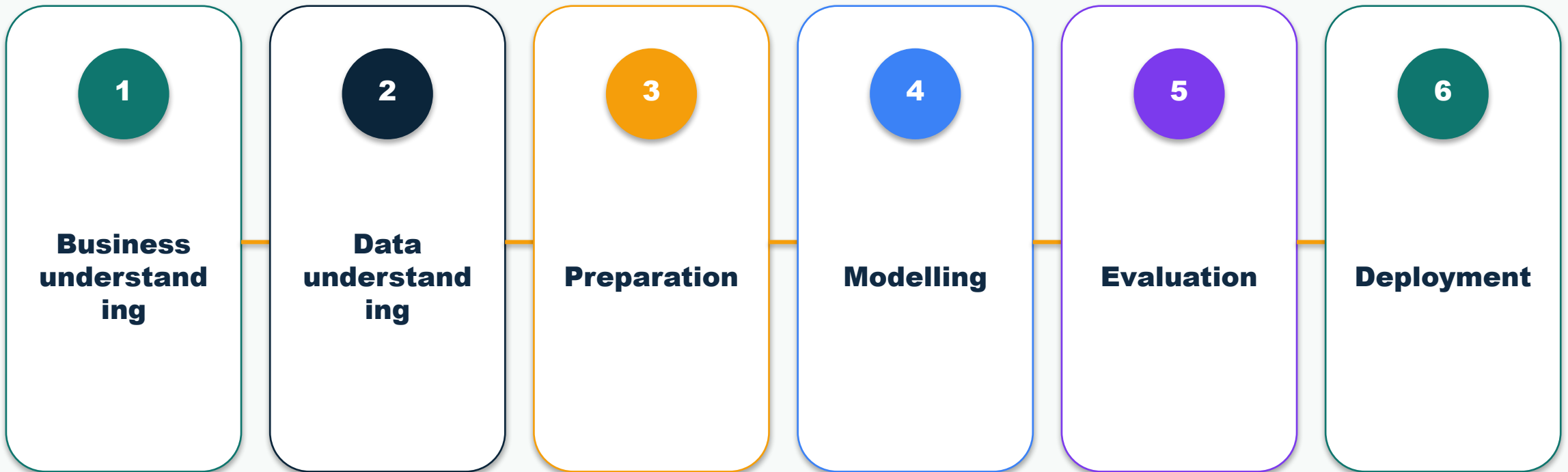
Overview of Data Mining and Machine Learning

LO1 · A8 · K4 · From business question to monitored evidence

Mechanism → contract → calculation → evidence → control



CRISP-DM evidence loop · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

CRISP-DM evidence loop · concrete contract



objective

measurable decision



unit_of_analysis

one row



target

observed outcome



acceptance

metric + threshold

CRISP-DM evidence loop · worked calculation

decision value = benefit of correct action – cost of errors

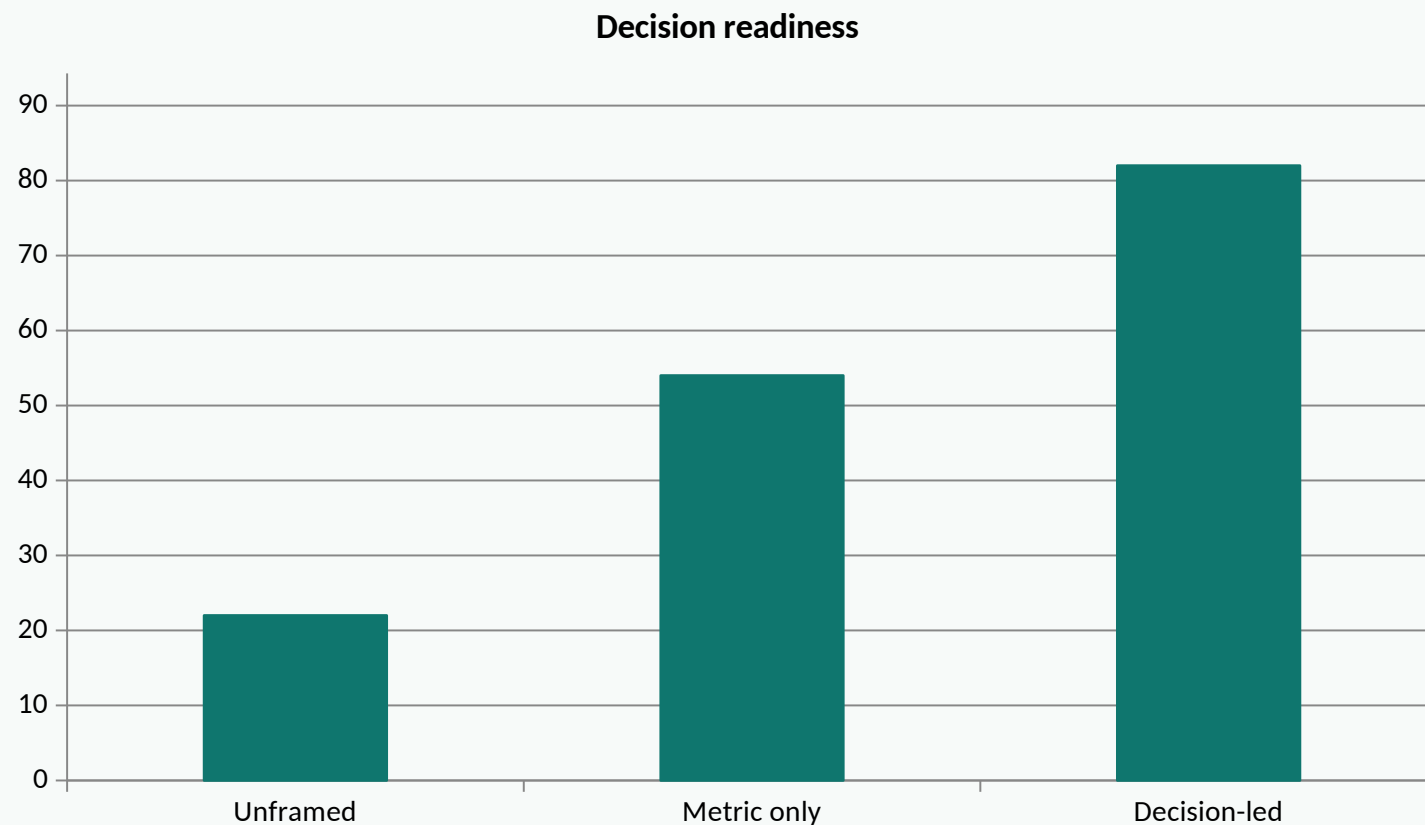
WORKED EXAMPLE

\$80 benefit – \$18 expected error cost = \$62 net value

READOUT
expected decision value

Show units, denominator, and the comparison baseline.

CRISP-DM evidence loop · evidence



WHAT THE DATA SHOWS

A reviewer can trace every model output to a decision and acceptance rule.

Evidence becomes a decision only after scope and uncertainty are stated.

CRISP-DM evidence loop · failure and control

FAILURE MODE

Optimising a convenient model metric that does not change the business decision.

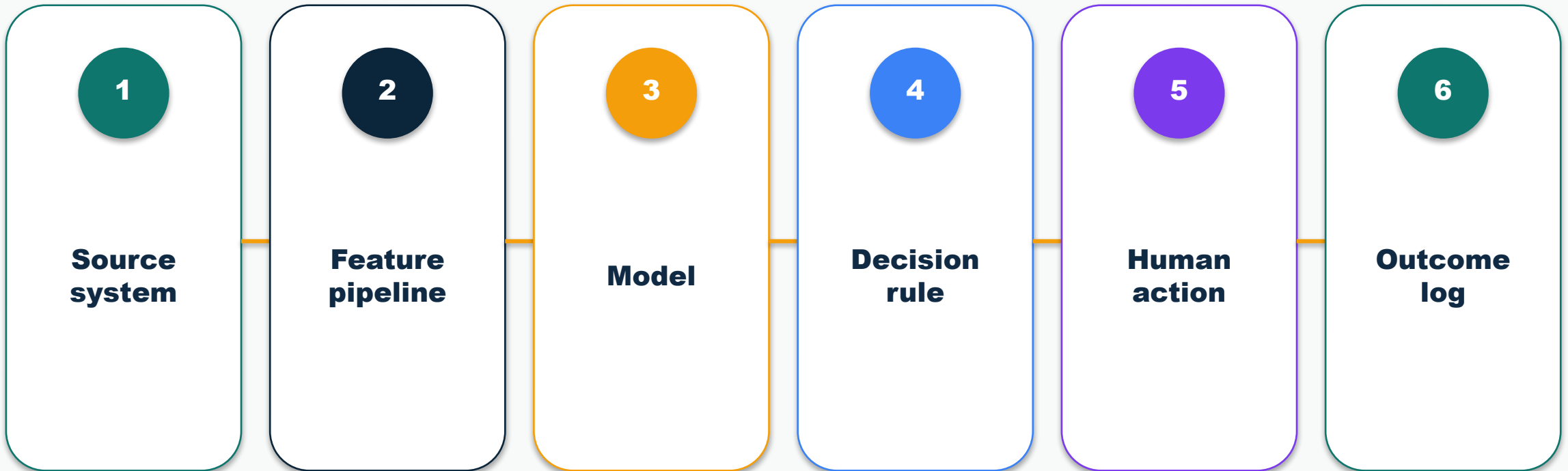
CONTROL

Write the decision, owner, action, and cost matrix before modelling.

VERIFICATION

A reviewer can trace every model output to a decision and acceptance rule.

Machine-learning system boundary · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Machine-learning system boundary · concrete contract



inputs

versioned fields



model_artifact

fitted parameters



threshold

operating point



outcome

delayed ground truth

Machine-learning system boundary · worked calculation

prediction $\neq$ decision; decision = policy(prediction, context)

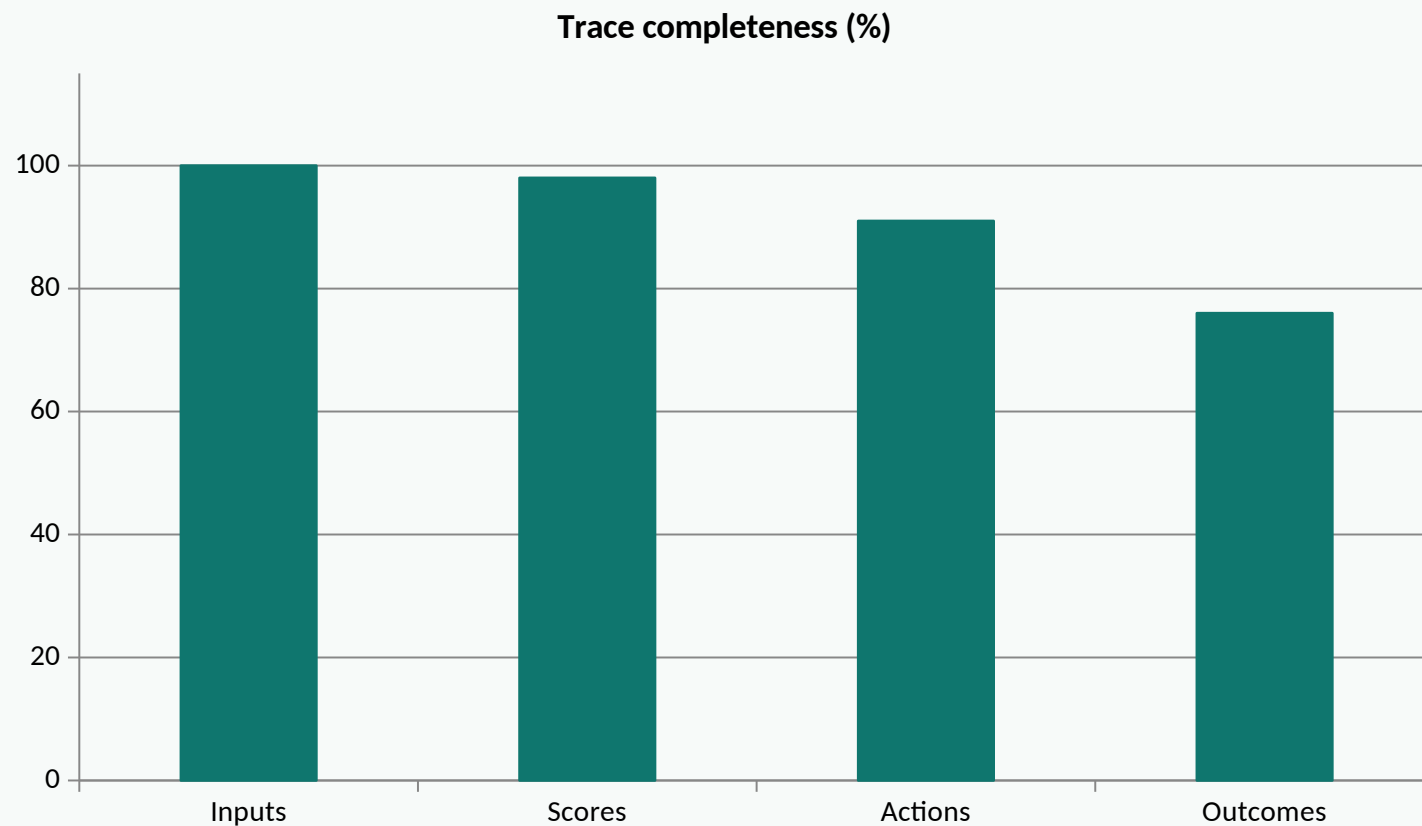
WORKED EXAMPLE

**0.73 churn probability + threshold 0.65 →
retention review**

READOUT
**decision trace
coverage**

Show units, denominator, and the comparison baseline.

Machine-learning system boundary · evidence



WHAT THE DATA SHOWS

Logs contain input version, score, threshold, action, and later outcome.

Evidence becomes a decision only after scope and uncertainty are stated.

Machine-learning system boundary · failure and control

FAILURE MODE

Treating a prediction as an instruction and hiding ownership of the final action.

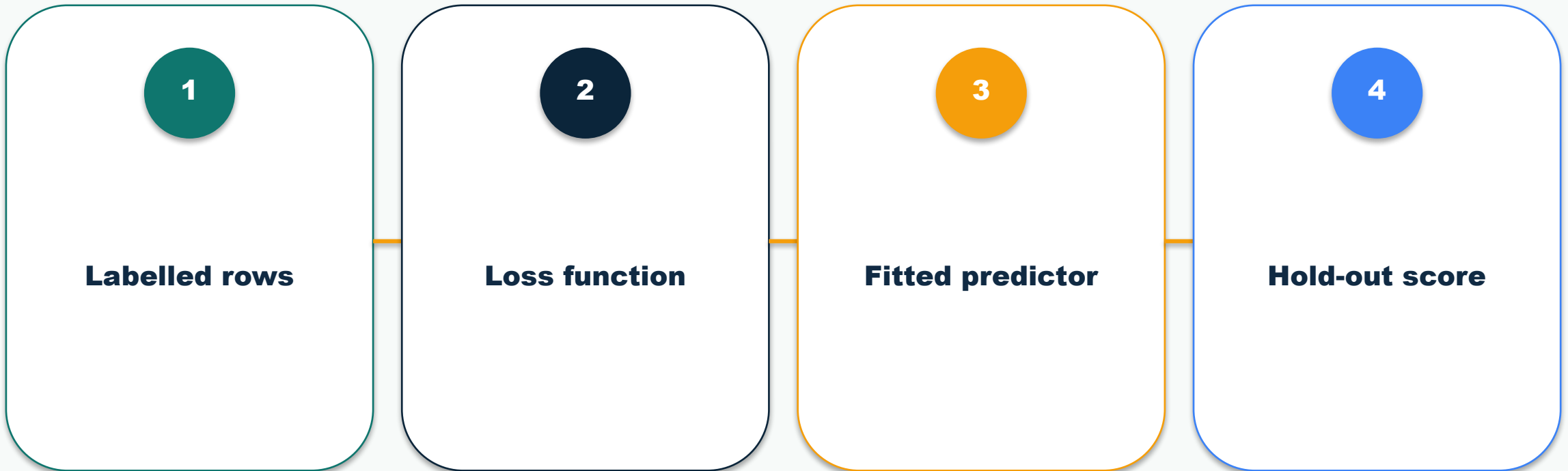
CONTROL

Separate inference, policy, and human approval in the architecture.

VERIFICATION

Logs contain input version, score, threshold, action, and later outcome.

Supervised versus unsupervised evidence · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Supervised versus unsupervised evidence - concrete contract



supervised

X plus y



unsupervised

X only



classification

discrete y



regression

continuous y

Supervised versus unsupervised evidence - worked calculation

supervised minimises error to y; clustering minimises within-group distance

WORKED EXAMPLE

churn label trains a classifier; spend patterns form clusters without labels

READOUT

external or internal validation

Show units, denominator, and the comparison baseline.

Supervised versus unsupervised evidence · evidence

SUPERVISED

Labelled target

Predict held-out y

Loss / class metric

UNSUPERVISED

No target label

Discover structure

Stability / utility

The evidence contract changes with the learning question; the scores are not interchangeable.

Supervised versus unsupervised evidence · failure and control

FAILURE MODE

Calling clusters 'true customer types' without external validation.

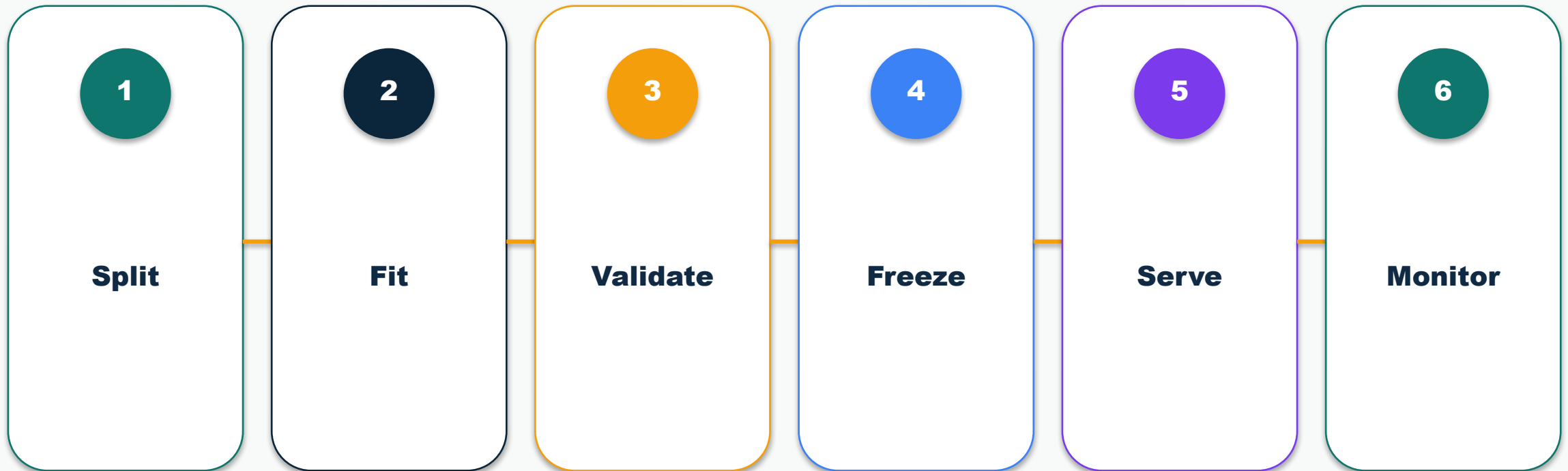
CONTROL

Use business review, stability tests, and later outcomes to validate segments.

VERIFICATION

The evaluation method matches the presence or absence of ground truth.

Train-evaluate-deploy lifecycle · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Train-evaluate-deploy lifecycle · concrete contract

**train**

parameter learning

**validation**

selection

**test**

final estimate

**production**

unseen stream

Train-evaluate-deploy lifecycle · worked calculation

generalisation gap = training score – test score

WORKED EXAMPLE

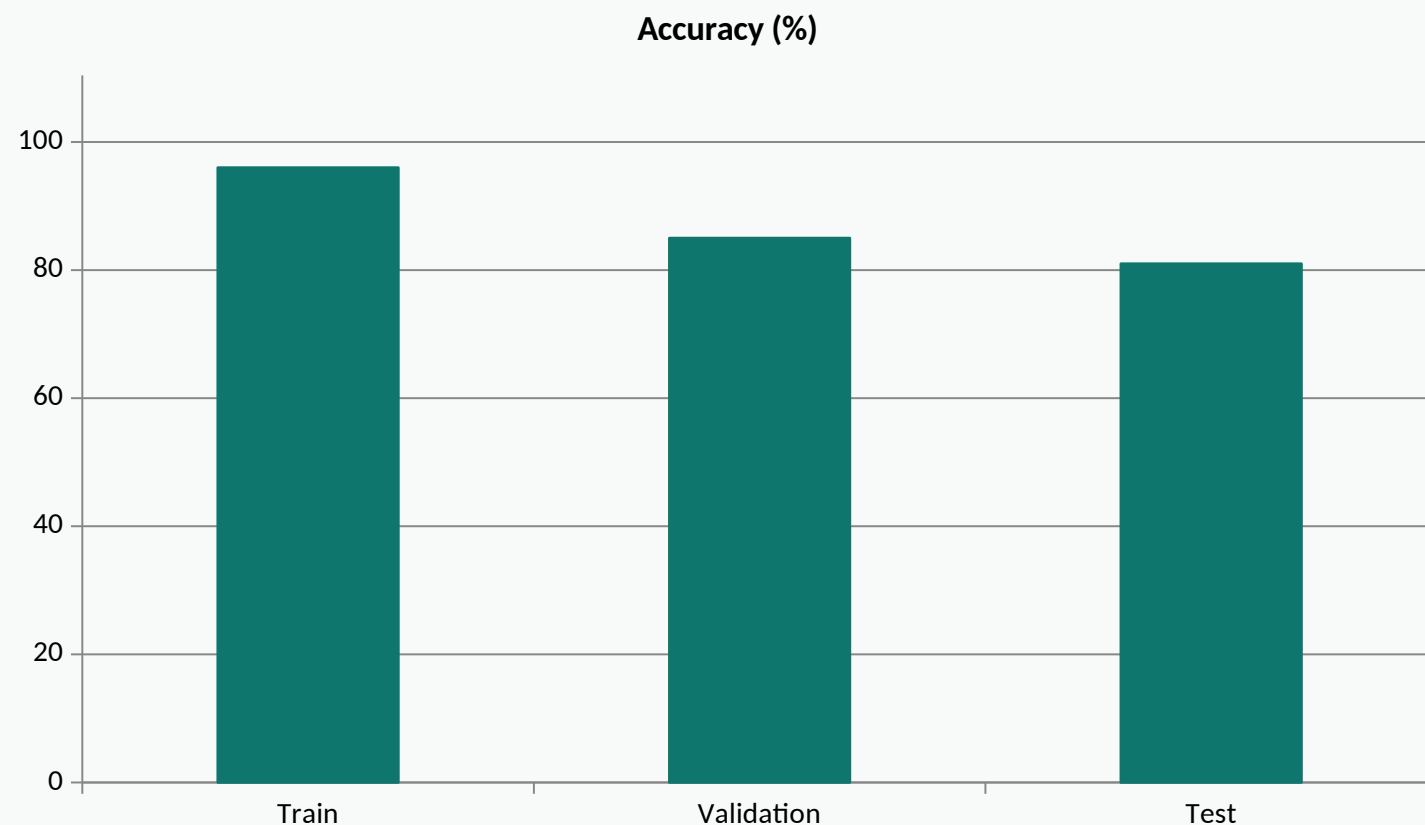
0.96 train – 0.81 test = 0.15 gap

READOUT

generalisation gap

Show units, denominator, and the comparison baseline.

Train-evaluate-deploy lifecycle · evidence



WHAT THE DATA SHOWS

Re-running the experiment with the same seed reproduces the split and score.

Evidence becomes a decision only after scope and uncertainty are stated.

Train–evaluate–deploy lifecycle · failure and control

FAILURE MODE

Repeatedly tuning against the test set leaks test information into model selection.

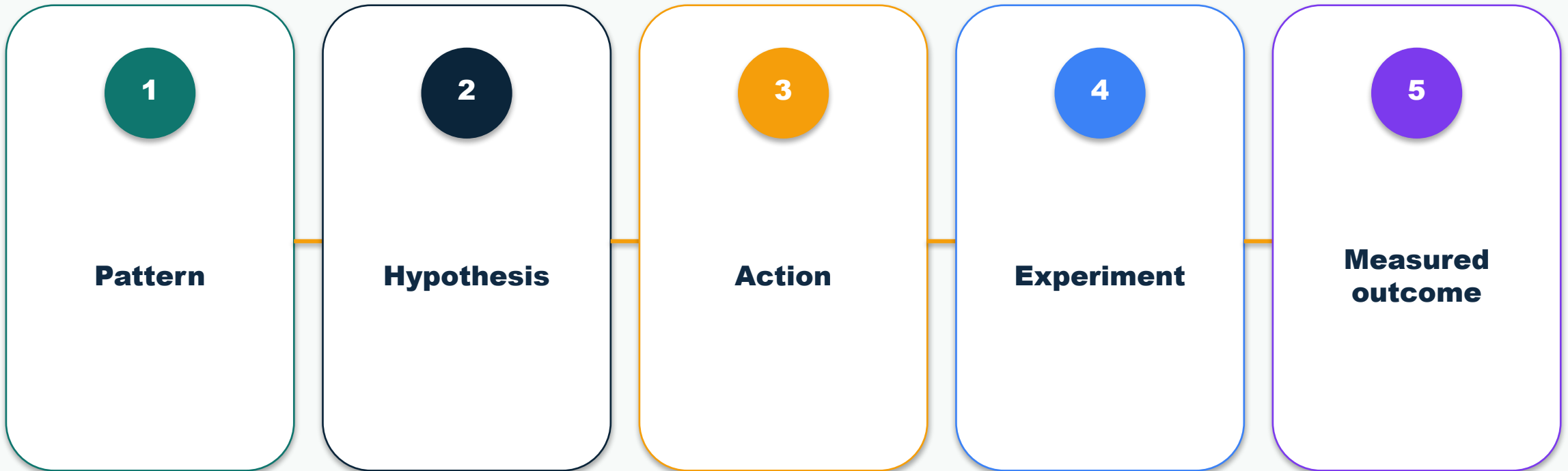
CONTROL

Lock a final test set and record experiment parameters separately.

VERIFICATION

Re-running the experiment with the same seed reproduces the split and score.

Business insight impact · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Business insight impact · concrete contract



segment

affected population



action

operational change



guardrail

harm limit



outcome

causal measure

Business insight impact · worked calculation

$$\text{uplift} = \text{outcome_treatment} - \text{outcome_control}$$

WORKED EXAMPLE

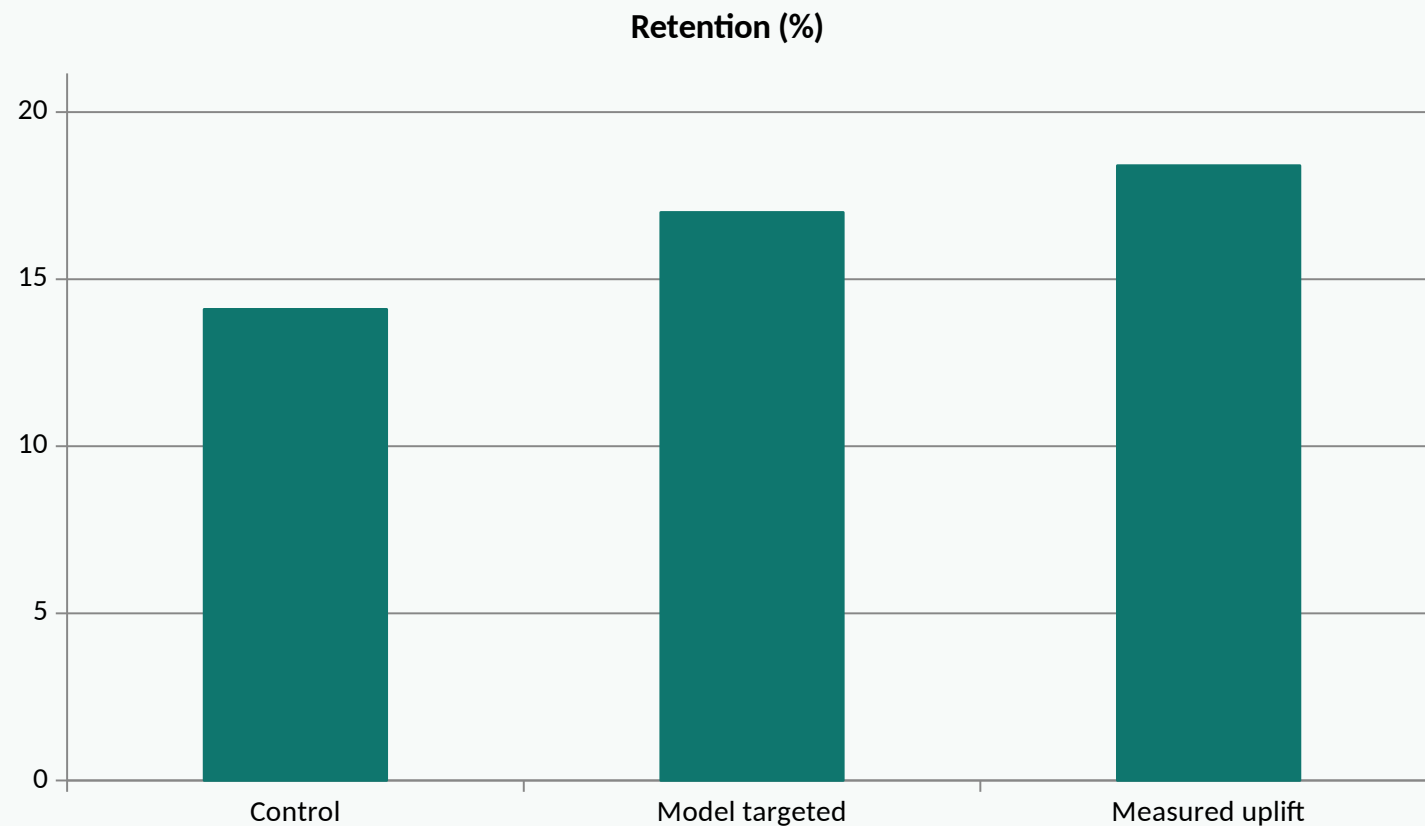
18.4% retention – 14.1% control = 4.3 pp uplift

READOUT

uplift and guardrails

Show units, denominator, and the comparison baseline.

Business insight impact · evidence



WHAT THE DATA SHOWS

The insight statement includes scope, uncertainty, expected action, and limitation.

Evidence becomes a decision only after scope and uncertainty are stated.

Business insight impact - failure and control

FAILURE MODE

Reporting correlation as causal impact.

CONTROL

Use controlled trials or explicitly label observational evidence.

VERIFICATION

The insight statement includes scope, uncertainty, expected action, and limitation.

Lab 01 · Data audit and business question

CENTRAL APP

**Data audit &
preparation**

<http://localhost:8501>

INPUT

Lab 01 mock CSV

MECHANISM

Data audit & preparation

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the
Learner Guide and lab folder.

Lab 01 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Data audit and business question produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 15 · Detect and explain anomalies

CENTRAL APP

**Anomaly
detection**

<http://localhost:8501>

INPUT

Lab 15 mock CSV

MECHANISM

Anomaly detection

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 15 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

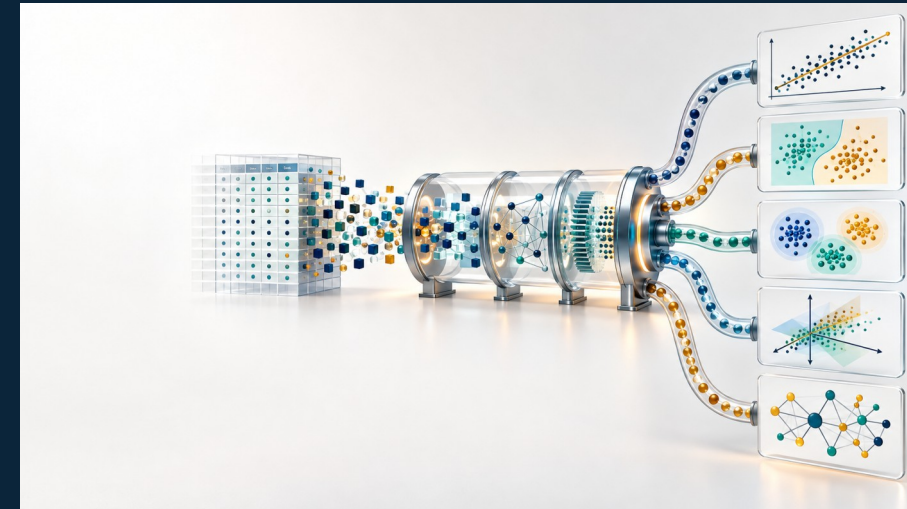
PASS: Detect and explain anomalies produces a reproducible artifact and a defensible interpretation—not only a screenshot.

TOPIC 02

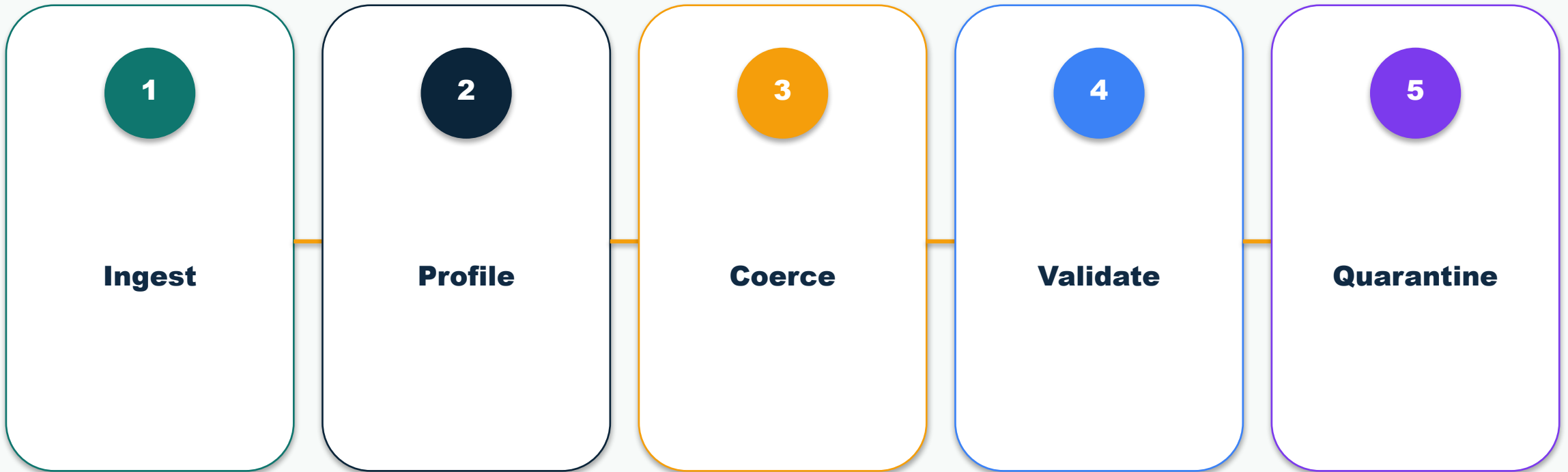
Data Preparation

LO2 · A1/A6 · K1/K7 · Reliable tables before reliable models

Mechanism → contract → calculation → evidence → control



Schema and data types · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Schema and data types · concrete contract

**customer_id**

string key

**event_time**

datetime

**amount**float ≥ 0 **target**

explicit role

Schema and data types · worked calculation

validity rate = valid rows / received rows

WORKED EXAMPLE

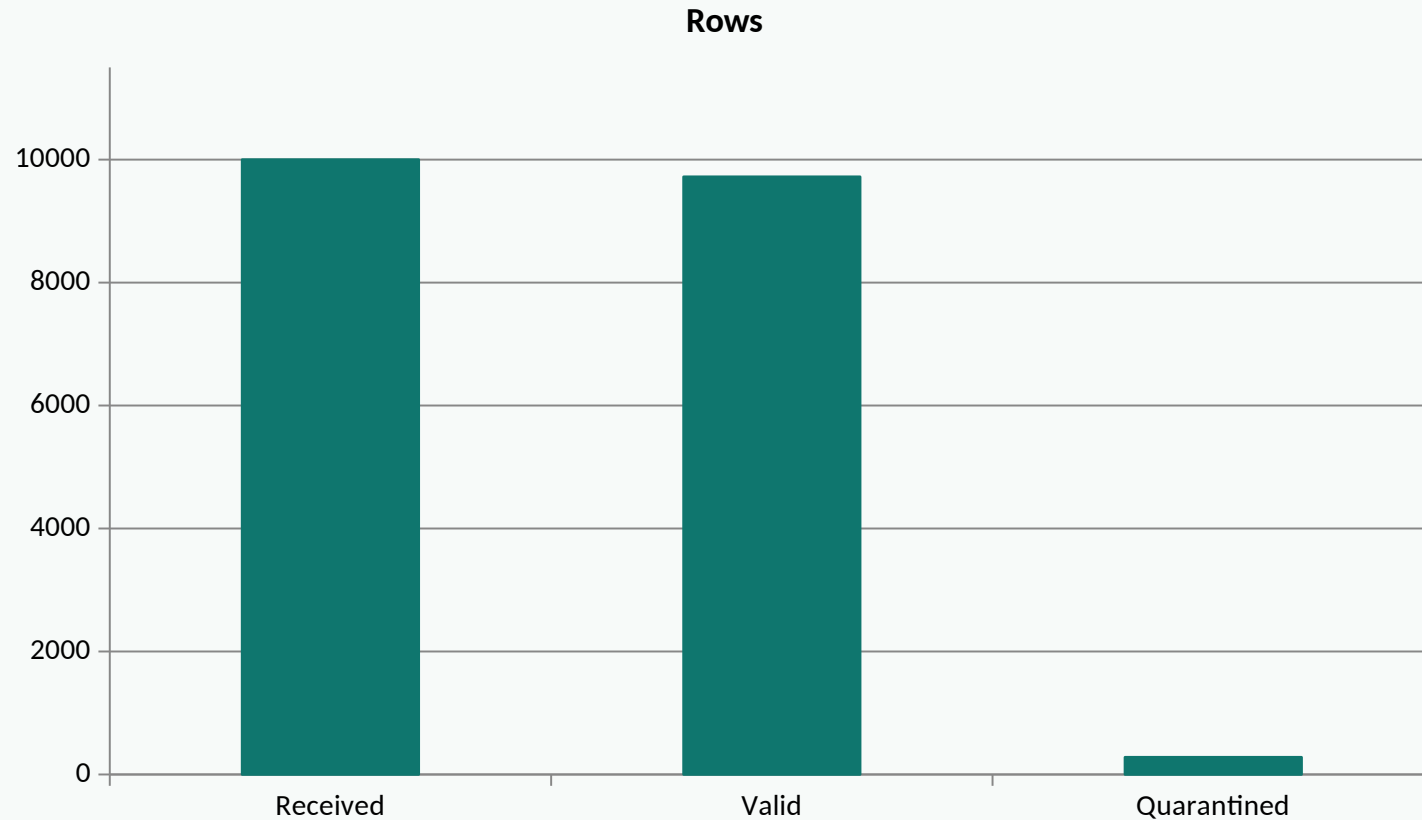
9,720 valid / 10,000 received = 97.2%

READOUT

validity rate

Show units, denominator, and the comparison baseline.

Schema and data types · evidence



WHAT THE DATA SHOWS

Row counts reconcile across received, accepted, and quarantined records.

Evidence becomes a decision only after scope and uncertainty are stated.

Schema and data types · failure and control

FAILURE MODE

**Silent type coercion
turns invalid values into
missing values without
an audit trail.**

CONTROL

**Validate ranges and log
rejected rows before
conversion.**

VERIFICATION

**Row counts reconcile
across received,
accepted, and
quarantined records.**

Filtering and row selection · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Filtering and row selection · concrete contract



predicate

amount > 0



scope

date window



null policy

explicit



index

preserved identifier

Filtering and row selection · worked calculation

$$\text{selected} = \sum I(\text{condition is true})$$

WORKED EXAMPLE

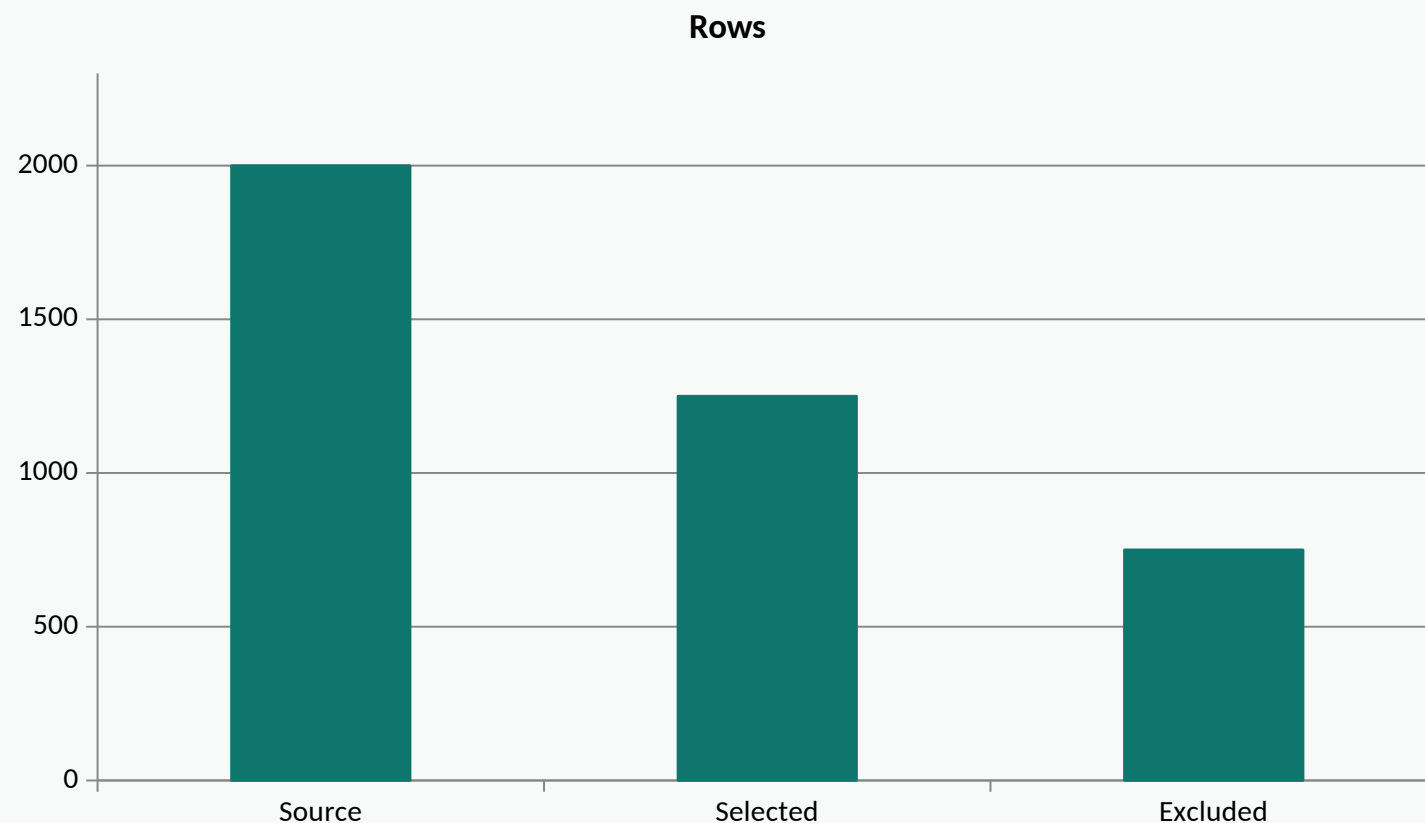
1,250 rows satisfy region='North' and amount>0

READOUT

selection rate

Show units, denominator, and the comparison baseline.

Filtering and row selection · evidence



WHAT THE DATA SHOWS

Selected + excluded + null-policy rows equals the source row count.

Evidence becomes a decision only after scope and uncertainty are stated.

Filtering and row selection · failure and control

FAILURE MODE

A filter unintentionally removes nulls because null comparisons evaluate false.

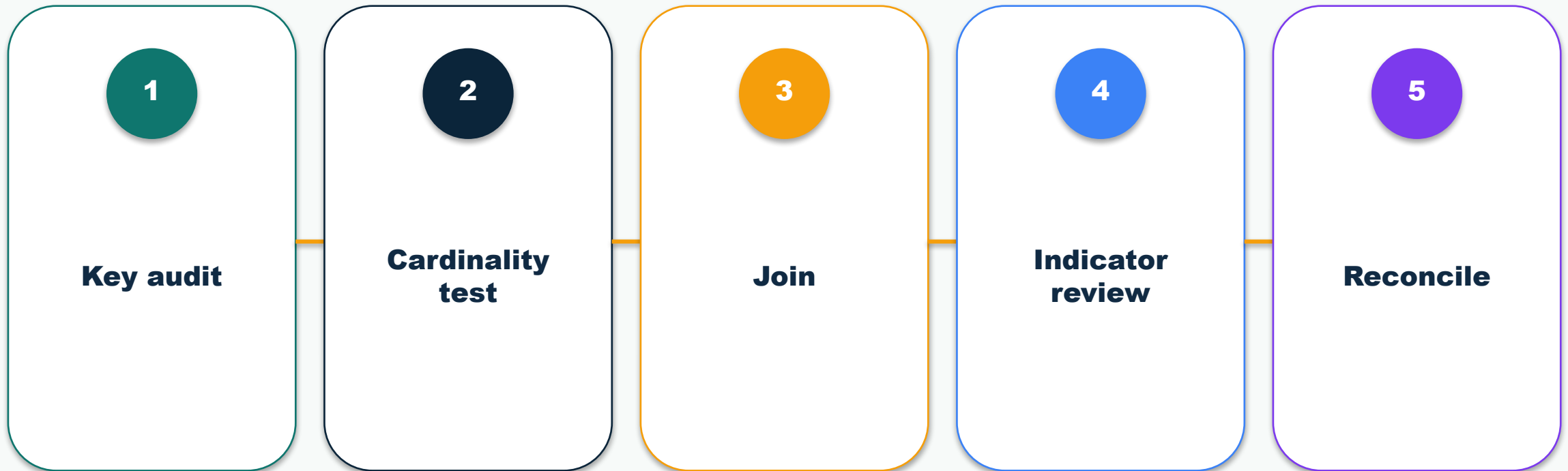
CONTROL

Profile null counts before and after every predicate.

VERIFICATION

Selected + excluded + null-policy rows equals the source row count.

Join and concatenate · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Join and concatenate · concrete contract



key

customer_id



left cardinality

many



right cardinality

one



join type

inner/left

Join and concatenate · worked calculation

join expansion = output rows / left rows

WORKED EXAMPLE

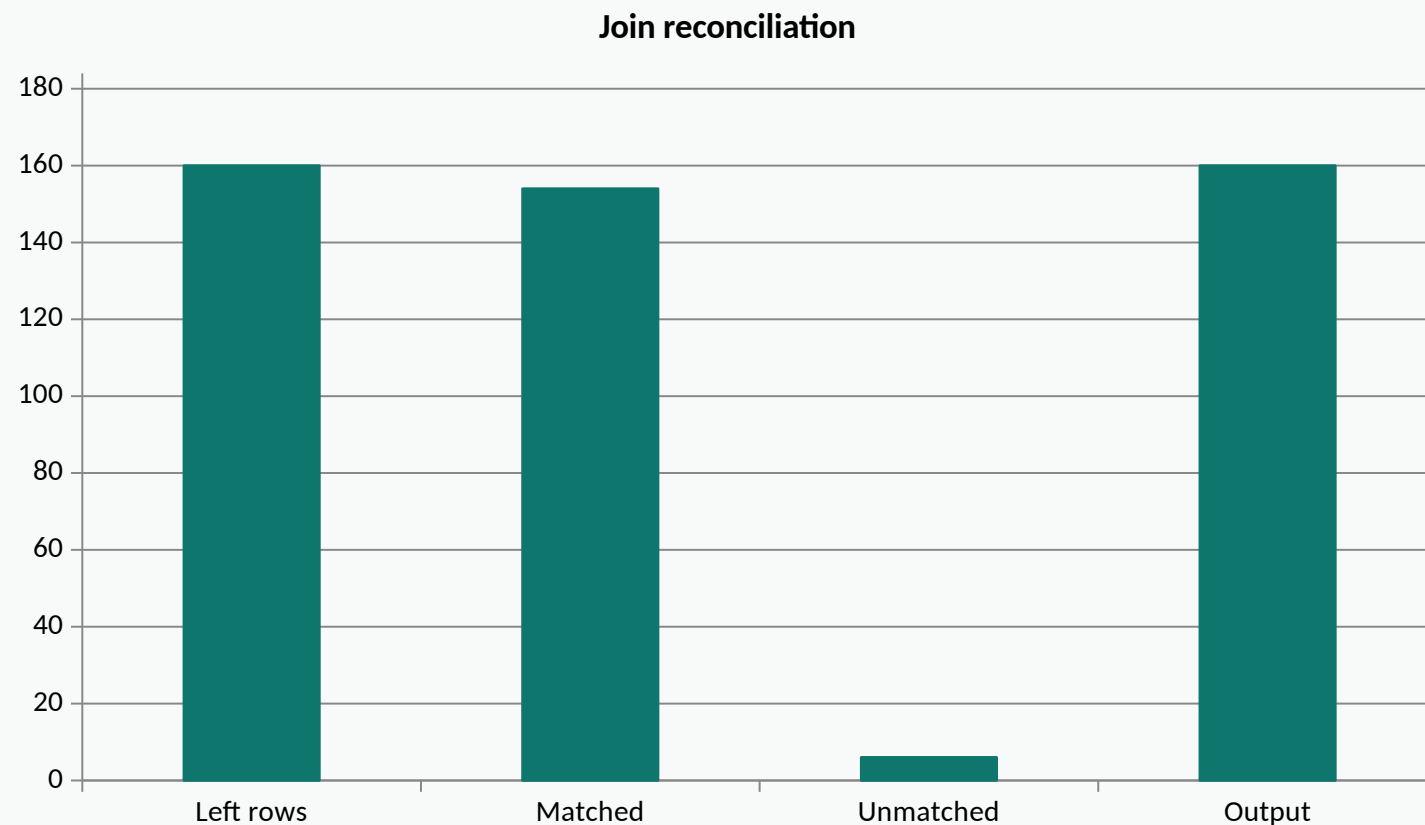
168 output / 160 orders = 1.05× warns of duplicate customer keys

READOUT

match rate and expansion

Show units, denominator, and the comparison baseline.

Join and concatenate · evidence



WHAT THE DATA SHOWS

Revenue before and after the intended join reconciles within the documented scope.

Evidence becomes a decision only after scope and uncertainty are stated.

Join and concatenate · failure and control

FAILURE MODE

**Many-to-many keys
duplicate facts and
inflate revenue.**

CONTROL

**Assert key uniqueness
on the dimension table
and inspect merge
indicators.**

VERIFICATION

**Revenue before and after
the intended join
reconciles within the
documented scope.**

Missing-value treatment · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Missing-value treatment · concrete contract

#

MCAR/MAR/MNAR hypothesis

{

numeric

median/mean

{

category

mode/Unknown

{

indicator

was_missing

Missing-value treatment · worked calculation

median = middle ordered value; robust to extreme values

WORKED EXAMPLE

[20, 22, 24, 95] → median 23, mean 40.25

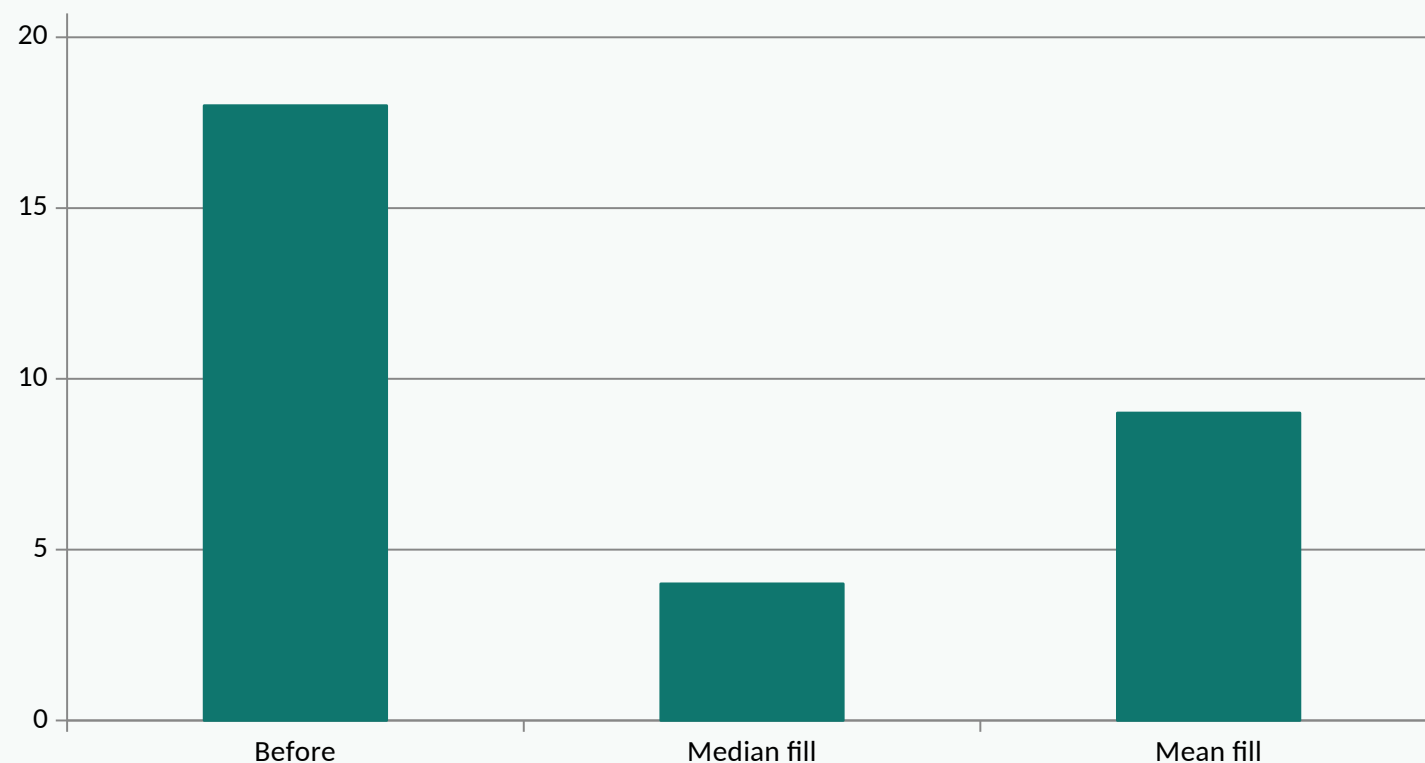
READOUT

distribution shift

Show units, denominator, and the comparison baseline.

Missing-value treatment · evidence

Absolute distribution shift



WHAT THE DATA SHOWS

No unintended nulls remain and train/test preprocessing is identical.

Evidence becomes a decision only after scope and uncertainty are stated.

Missing-value treatment · failure and control

FAILURE MODE

Computing imputation values on the full dataset leaks test distribution.

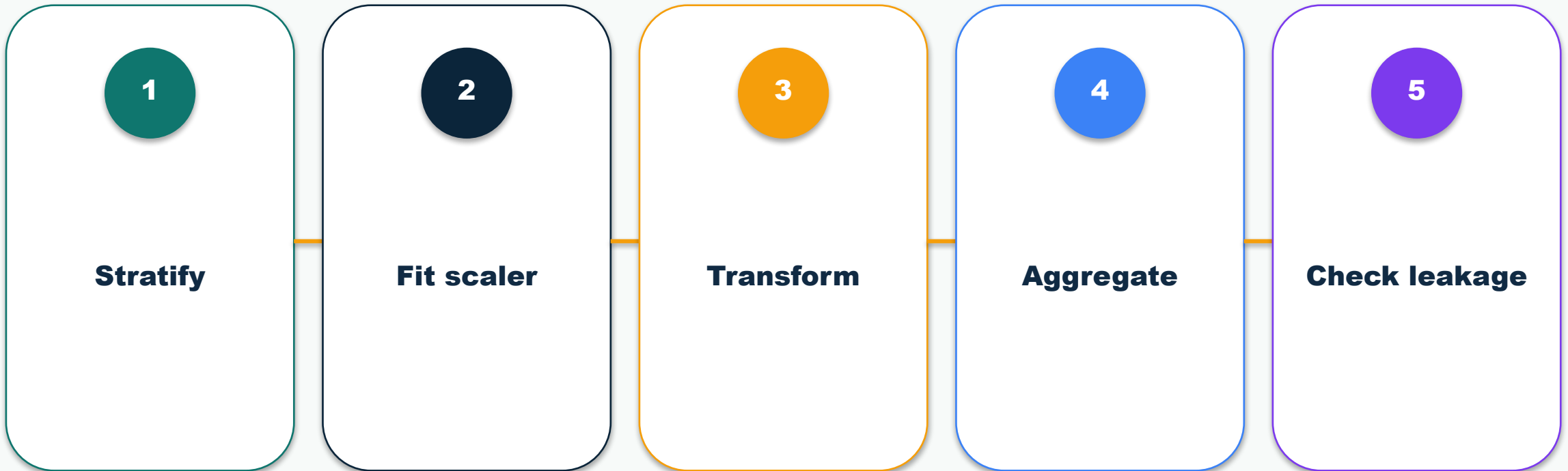
CONTROL

Fit imputers only on training data inside a pipeline.

VERIFICATION

No unintended nulls remain and train/test preprocessing is identical.

Scaling, sampling, and aggregation - runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Scaling, sampling, and aggregation - concrete contract

#

$$z = (x - \mu) / \sigma$$

#

$$\text{min-max} = (x - \text{min}) / (\text{max} - \text{min})$$

{ }

group key

declared

{ }

measure

count/sum/mean

Scaling, sampling, and aggregation - worked calculation

$$z = (850 - 500) / 140 = 2.50$$

WORKED EXAMPLE

Revenue 850 is 2.5 standard deviations above training mean

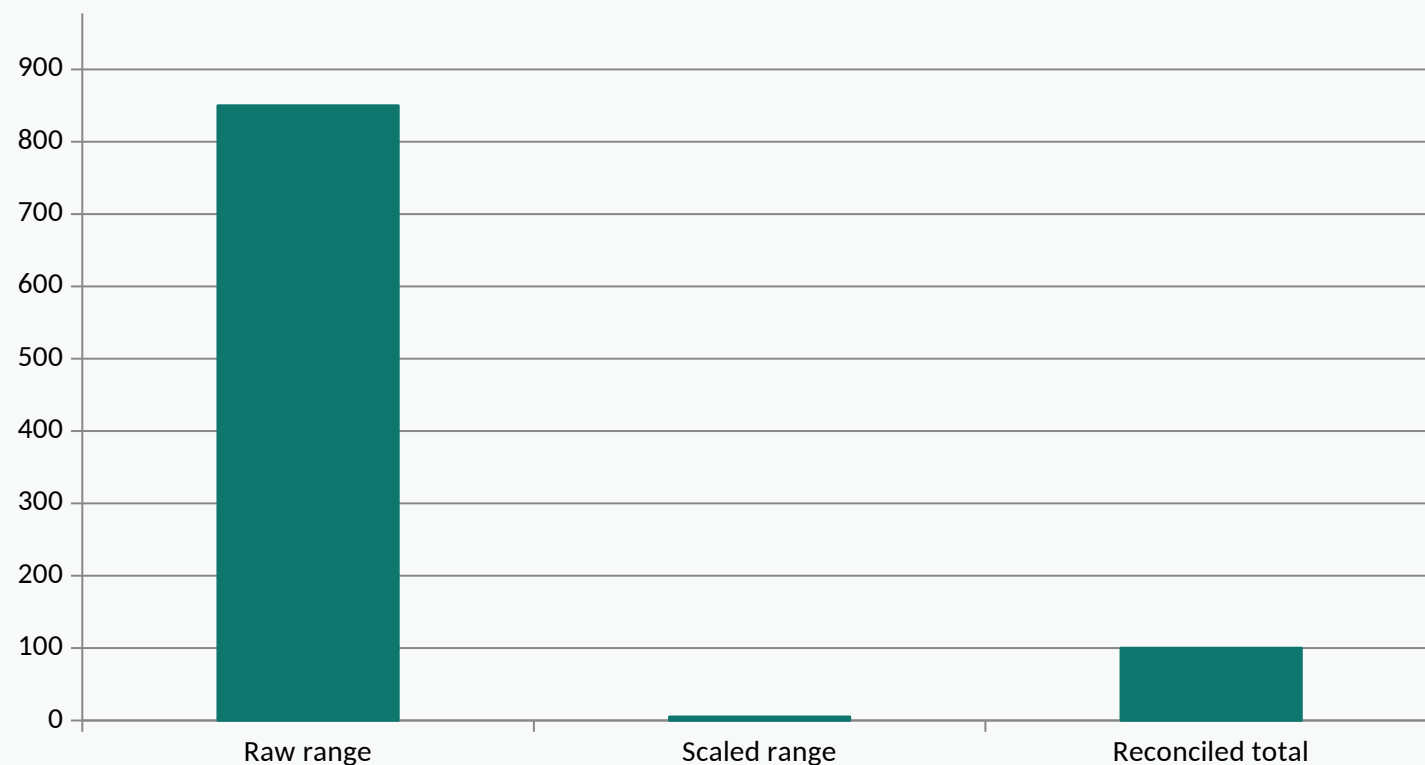
READOUT

standardised range and group totals

Show units, denominator, and the comparison baseline.

Scaling, sampling, and aggregation · evidence

Illustrative magnitude



WHAT THE DATA SHOWS

**Training features
centre near zero and
group totals reconcile
to source totals.**

Evidence becomes a decision only after scope
and uncertainty are stated.

Scaling, sampling, and aggregation - failure and control

FAILURE MODE

**Scaling before splitting
lets test statistics
influence training
features.**

CONTROL

**Split first; fit the scaler
within the training
pipeline.**

VERIFICATION

**Training features centre
near zero and group
totals reconcile to
source totals.**

Lab 02 · Clean, impute, and export

CENTRAL APP

Data audit & preparation

<http://localhost:8501>

INPUT

Lab 02 mock CSV

MECHANISM

Data audit & preparation

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 02 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Clean, impute, and export produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 03 · Filter, join, and aggregate

CENTRAL APP

**Join &
aggregate**

<http://localhost:8501>

INPUT

Lab 03 mock CSV

MECHANISM

Join & aggregate

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 03 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Filter, join, and aggregate produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 04 · Scale features before modelling

CENTRAL APP

Regression

<http://localhost:8501>

INPUT

Lab 04 mock CSV

MECHANISM

Regression

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 04 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Scale features before modelling produces a reproducible artifact and a defensible interpretation—not only a screenshot.

TOPIC 03

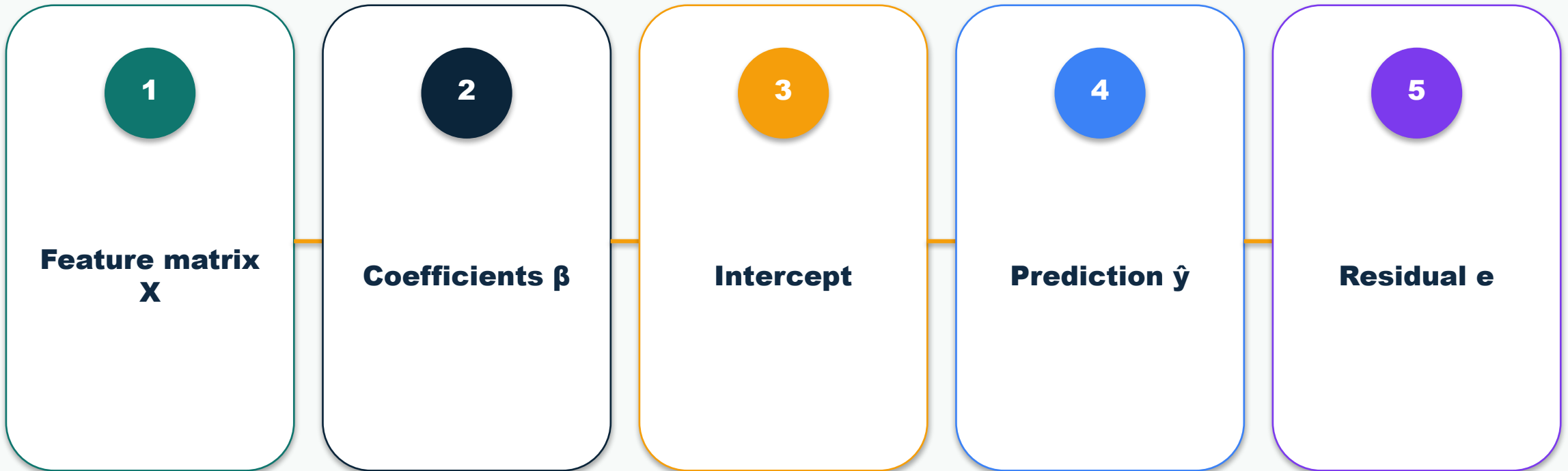
Regression

LO3 · A2 · K2/K10 · Predicting continuous outcomes with controlled error

Mechanism → contract → calculation → evidence → control



Linear model mechanism · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Linear model mechanism · concrete contract



row

observation



column

feature



β

marginal slope



$\hat{y}$

numeric outcome

Linear model mechanism · worked calculation

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

WORKED EXAMPLE

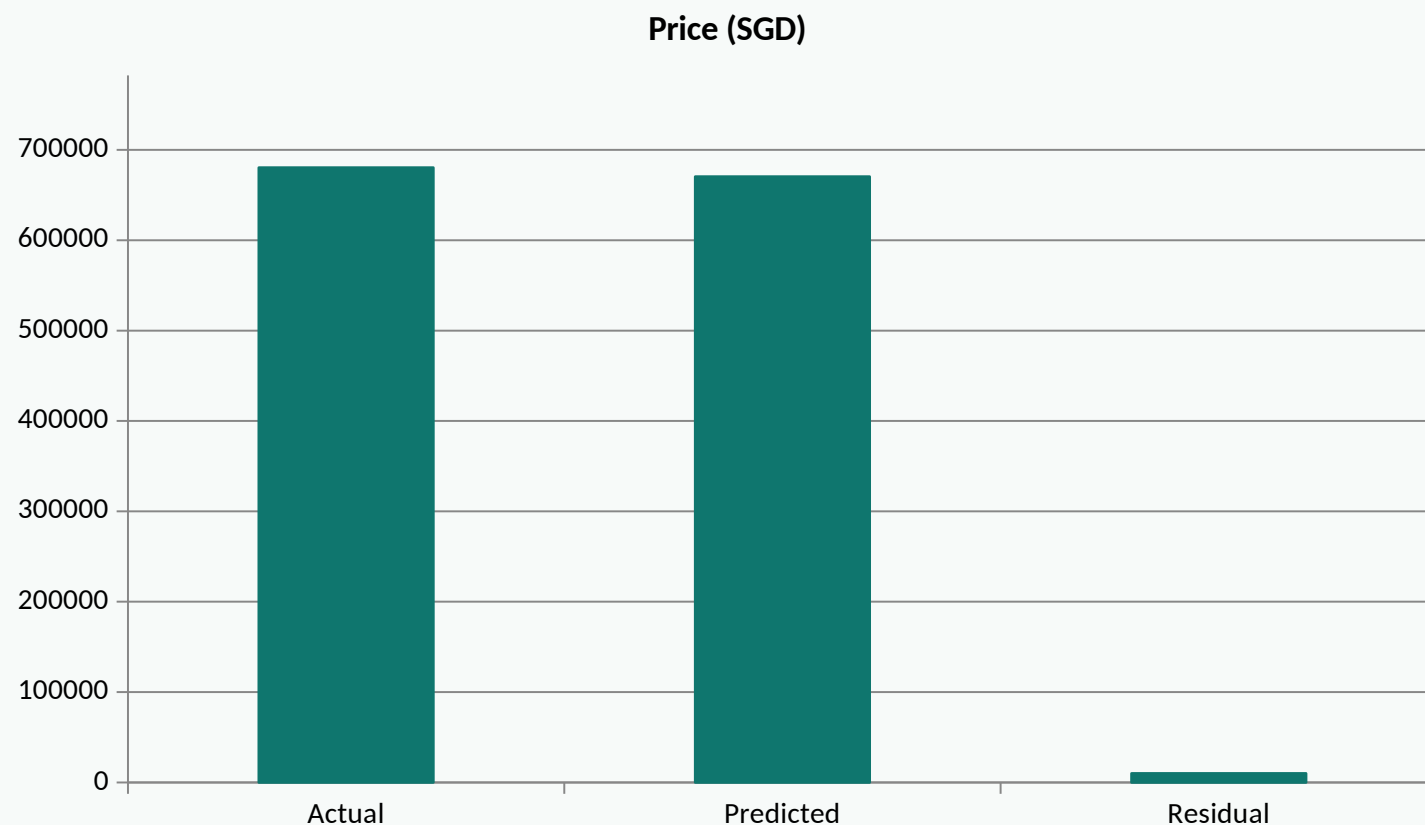
$$\hat{y} = 120,000 + 5,200 \times 92 + 7,200 \times 10 = \$670,400$$

READOUT

coefficient and residual

Show units, denominator, and the comparison baseline.

Linear model mechanism · evidence



WHAT THE DATA SHOWS

Predictions, coefficients, and residuals can be reproduced from the frozen pipeline.

Evidence becomes a decision only after scope and uncertainty are stated.

Linear model mechanism · failure and control

FAILURE MODE

Interpreting a coefficient causally when correlated variables or selection bias are present.

CONTROL

Use coefficients as conditional associations unless the design supports causality.

VERIFICATION

Predictions, coefficients, and residuals can be reproduced from the frozen pipeline.

Residual diagnostics · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Residual diagnostics · concrete contract

#

residual $e=y-\hat{y}$

#

zero-centred cloud

#

constant variance

#

independent errors

Residual diagnostics · worked calculation

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2$$

WORKED EXAMPLE

errors [10, -5, 15] → SSE = 100+25+225 = 350

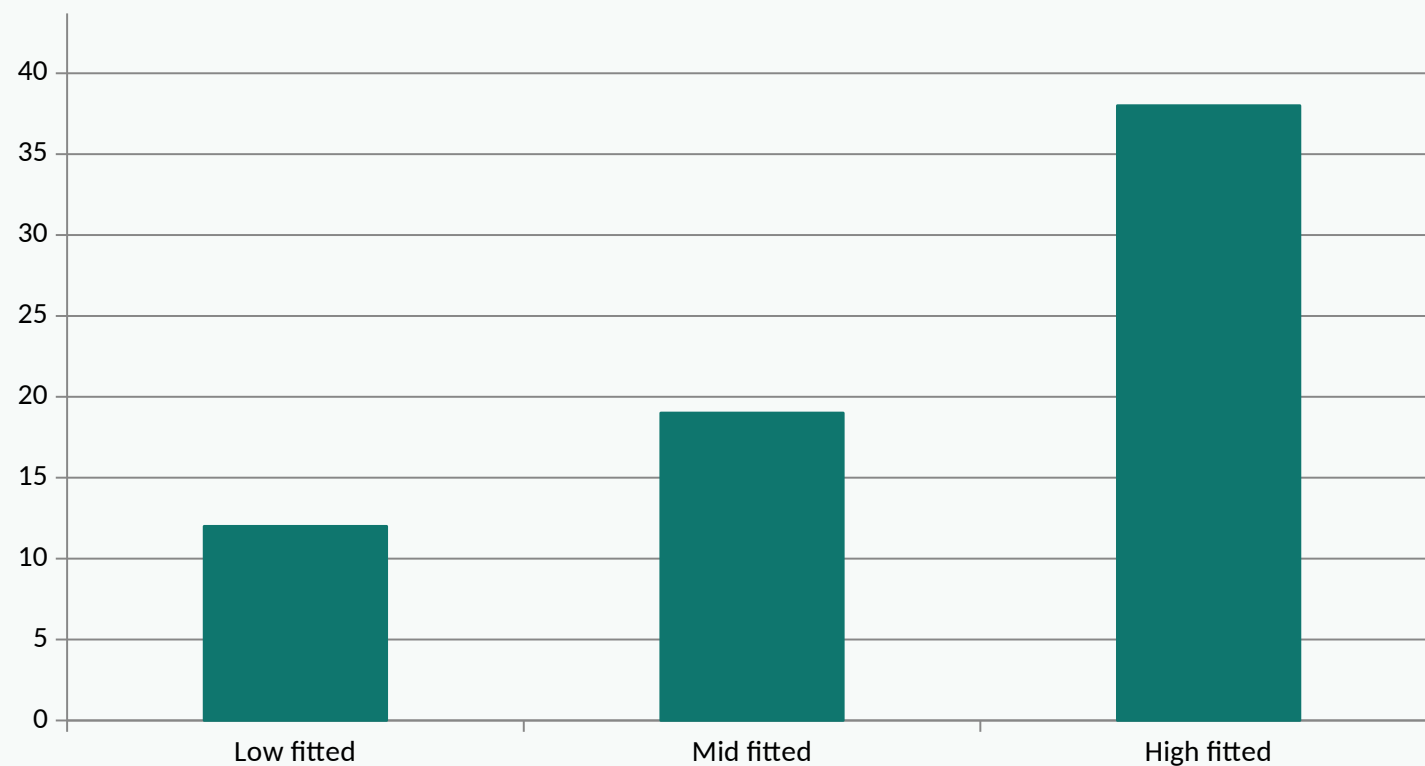
READOUT

residual pattern

Show units, denominator, and the comparison baseline.

Residual diagnostics · evidence

Residual spread



WHAT THE DATA SHOWS

Residuals show no systematic curve, drift, or group-specific bias.

Evidence becomes a decision only after scope and uncertainty are stated.

Residual diagnostics · failure and control

FAILURE MODE

A funnel shape means error variance grows with the prediction.

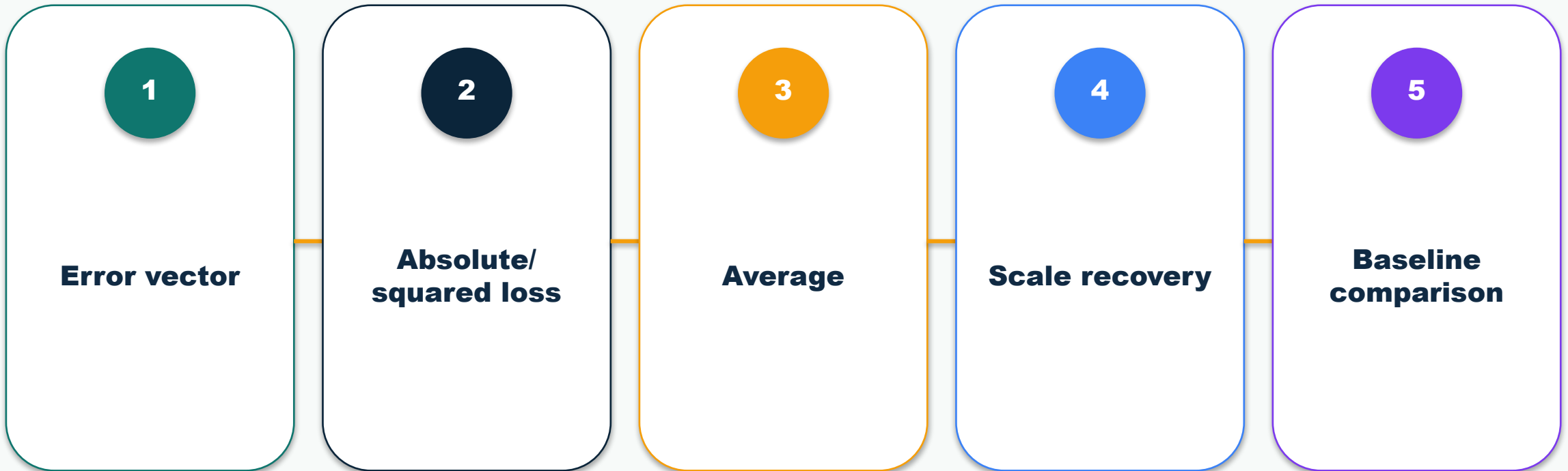
CONTROL

Transform the target, add relevant features, or use robust intervals.

VERIFICATION

Residuals show no systematic curve, drift, or group-specific bias.

MAE, MSE, RMSE, and R^2 · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

MAE, MSE, RMSE, and R^2 · concrete contract

**MAE**

linear penalty

**MSE**

squared penalty

**RMSE**

target units

 **R^2**

versus mean baseline

MAE, MSE, RMSE, and R^2 · worked calculation

$$R^2 = 1 - \text{SSE}/\text{SST}$$

WORKED EXAMPLE

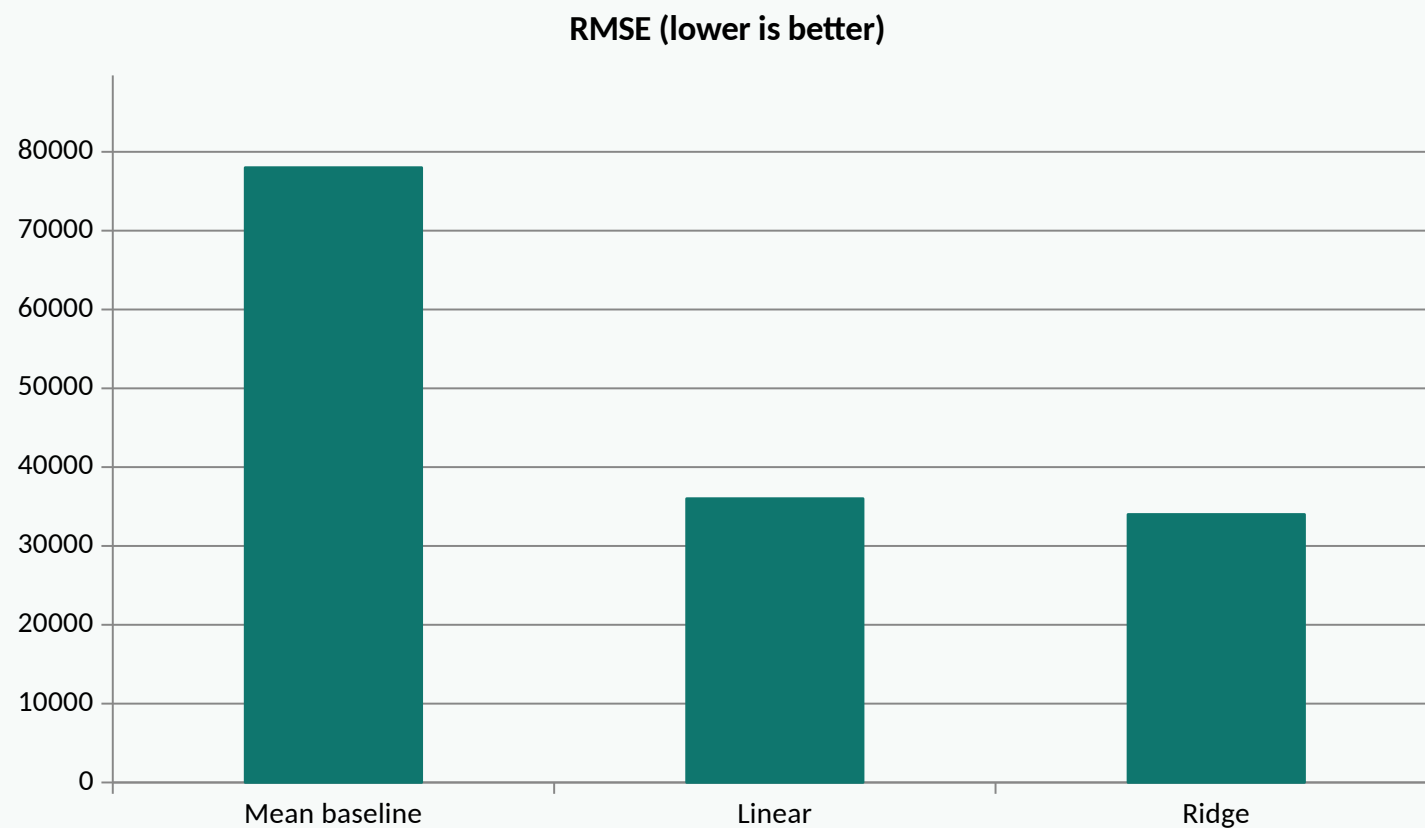
SSE 1,800; SST 9,000 $\rightarrow R^2 = 0.80$

READOUT

hold-out MAE/RMSE/ R^2

Show units, denominator, and the comparison baseline.

MAE, MSE, RMSE, and R^2 · evidence



WHAT THE DATA SHOWS

Metrics are computed on unseen rows and compared with a simple baseline.

Evidence becomes a decision only after scope and uncertainty are stated.

MAE, MSE, RMSE, and R^2 · failure and control

FAILURE MODE

High R^2 can coexist with large business-critical errors in a subgroup.

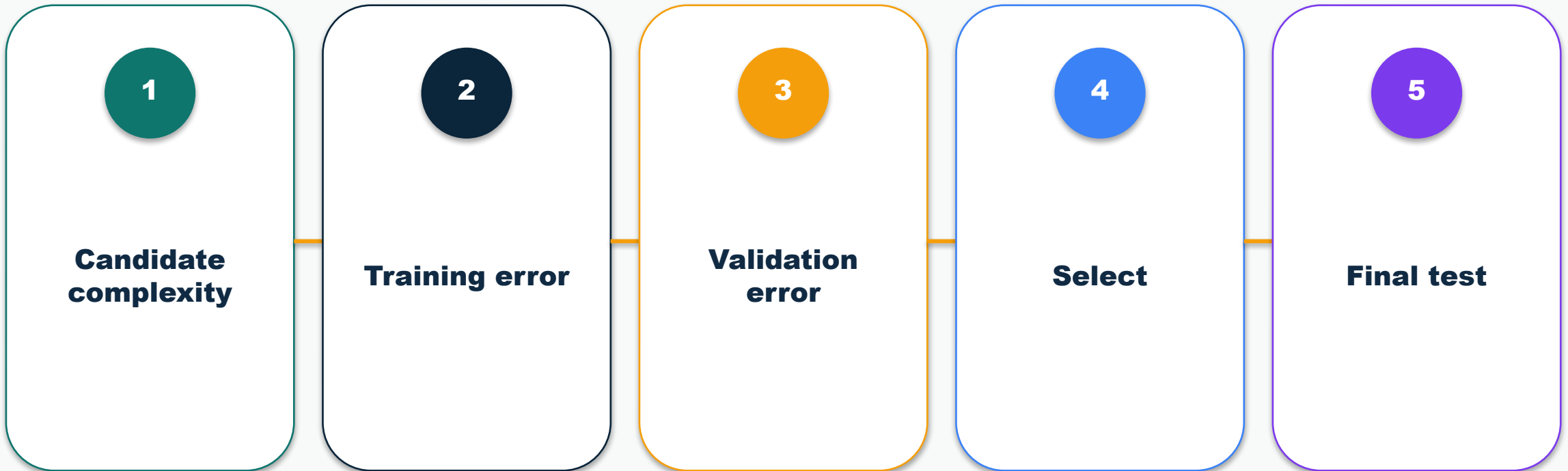
CONTROL

Report absolute error and segment diagnostics alongside R^2 .

VERIFICATION

Metrics are computed on unseen rows and compared with a simple baseline.

Underfitting, overfitting, and validation - runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Underfitting, overfitting, and validation - concrete contract



underfit

both errors high



fit

low validation error



overfit

widening gap



test

used once

Underfitting, overfitting, and validation - worked calculation

$$\text{gap} = \text{train RMSE} - \text{test RMSE}$$

WORKED EXAMPLE

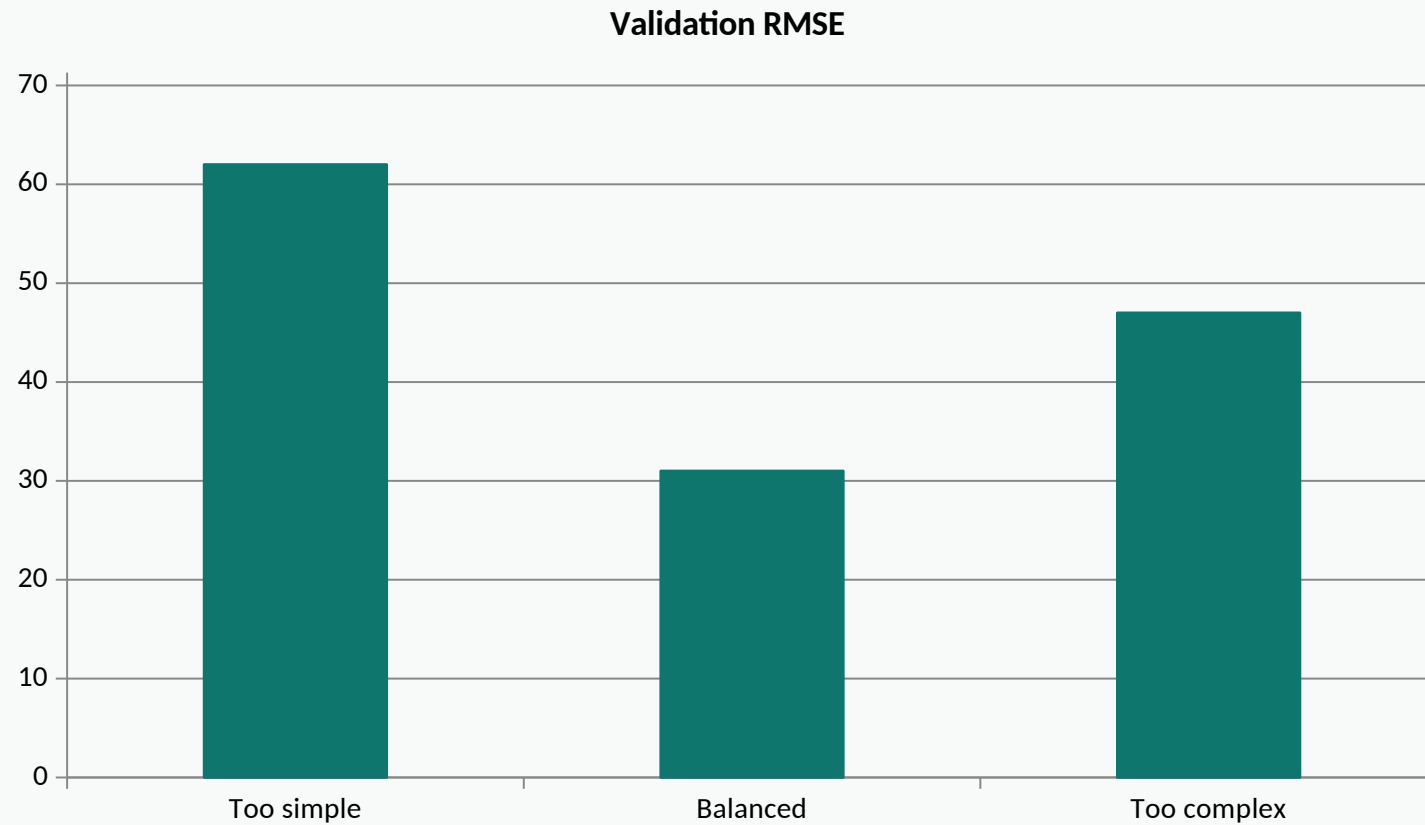
train 12k vs test 47k → -35k gap indicates overfit

READOUT

validation curve

Show units, denominator, and the comparison baseline.

Underfitting, overfitting, and validation · evidence



WHAT THE DATA SHOWS

Learning curves show whether more data or a simpler model is likely to help.

Evidence becomes a decision only after scope and uncertainty are stated.

Underfitting, overfitting, and validation - failure and control

FAILURE MODE

Selecting the deepest model because it minimises training error.

CONTROL

Choose complexity at the validation minimum and retain a final test set.

VERIFICATION

Learning curves show whether more data or a simpler model is likely to help.

Ridge and Lasso regularisation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Ridge and Lasso regularisation · concrete contract



Ridge

L2 penalty



Lasso

L1 penalty



α

penalty strength



intercept

usually unpenalised

Ridge and Lasso regularisation · worked calculation

Ridge: minimise $SSE + \alpha \sum \beta^2$; Lasso: $SSE + \alpha \sum |\beta|$

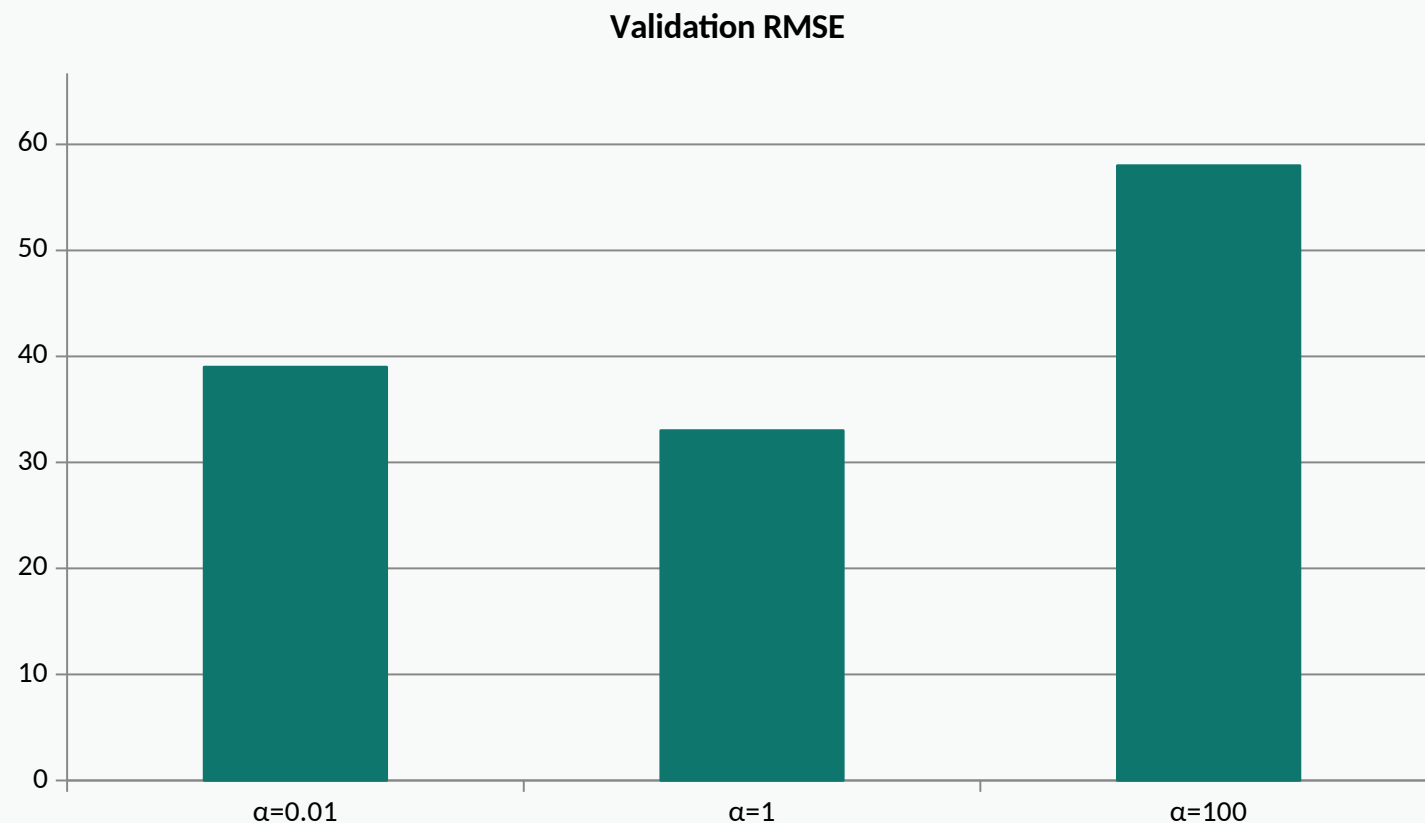
WORKED EXAMPLE

$\alpha=1$ shrinks unstable coefficients while retaining all Ridge features

READOUT
cross-validated error
versus α

Show units, denominator, and the comparison baseline.

Ridge and Lasso regularisation · evidence



WHAT THE DATA SHOWS

Selected α improves or preserves validation error without collapsing useful signal.

Evidence becomes a decision only after scope and uncertainty are stated.

Ridge and Lasso regularisation · failure and control

FAILURE MODE

Comparing penalty strengths on unscaled features makes the penalty uneven.

CONTROL

Standardise within the pipeline before Ridge or Lasso.

VERIFICATION

Selected α improves or preserves validation error without collapsing useful signal.

Lab 05 · Build a regression baseline

CENTRAL APP

Regression

<http://localhost:8501>

INPUT

Lab 05 mock CSV

MECHANISM

Regression

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 05 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Build a regression baseline produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 06 · Control overfitting with Ridge and Lasso

CENTRAL APP

Regression

<http://localhost:8501>

INPUT

Lab 06 mock CSV

MECHANISM

Regression

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 06 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

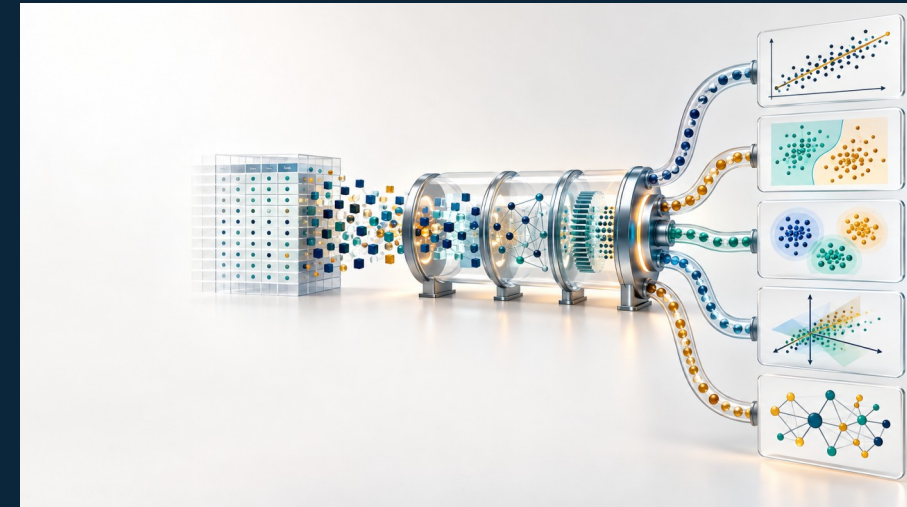
PASS: Control overfitting with Ridge and Lasso produces a reproducible artifact and a defensible interpretation—not only a screenshot.

TOPIC 04

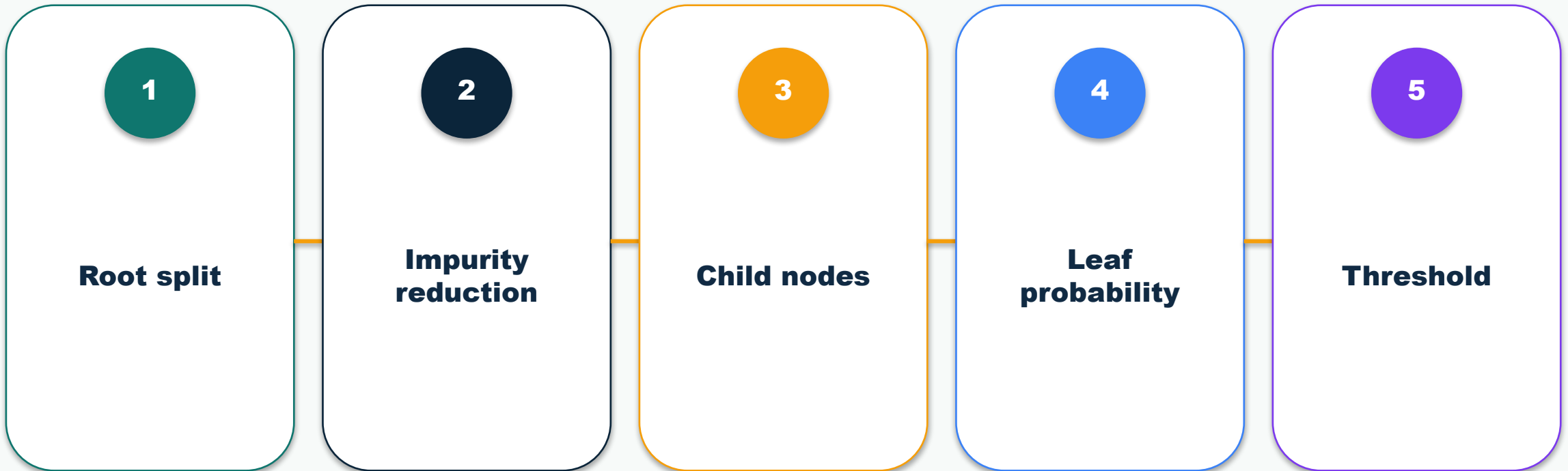
Classification

LO4 · A7 · K6 · Separating classes and choosing an operating threshold

Mechanism → contract → calculation → evidence → control



Decision tree and ensemble path · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Decision tree and ensemble path · concrete contract

#

feature threshold

#

Gini/entropy

#

max_depth

#

min_samples_leaf

Decision tree and ensemble path · worked calculation

$$\text{Gini} = 1 - \sum p_k^2$$

WORKED EXAMPLE

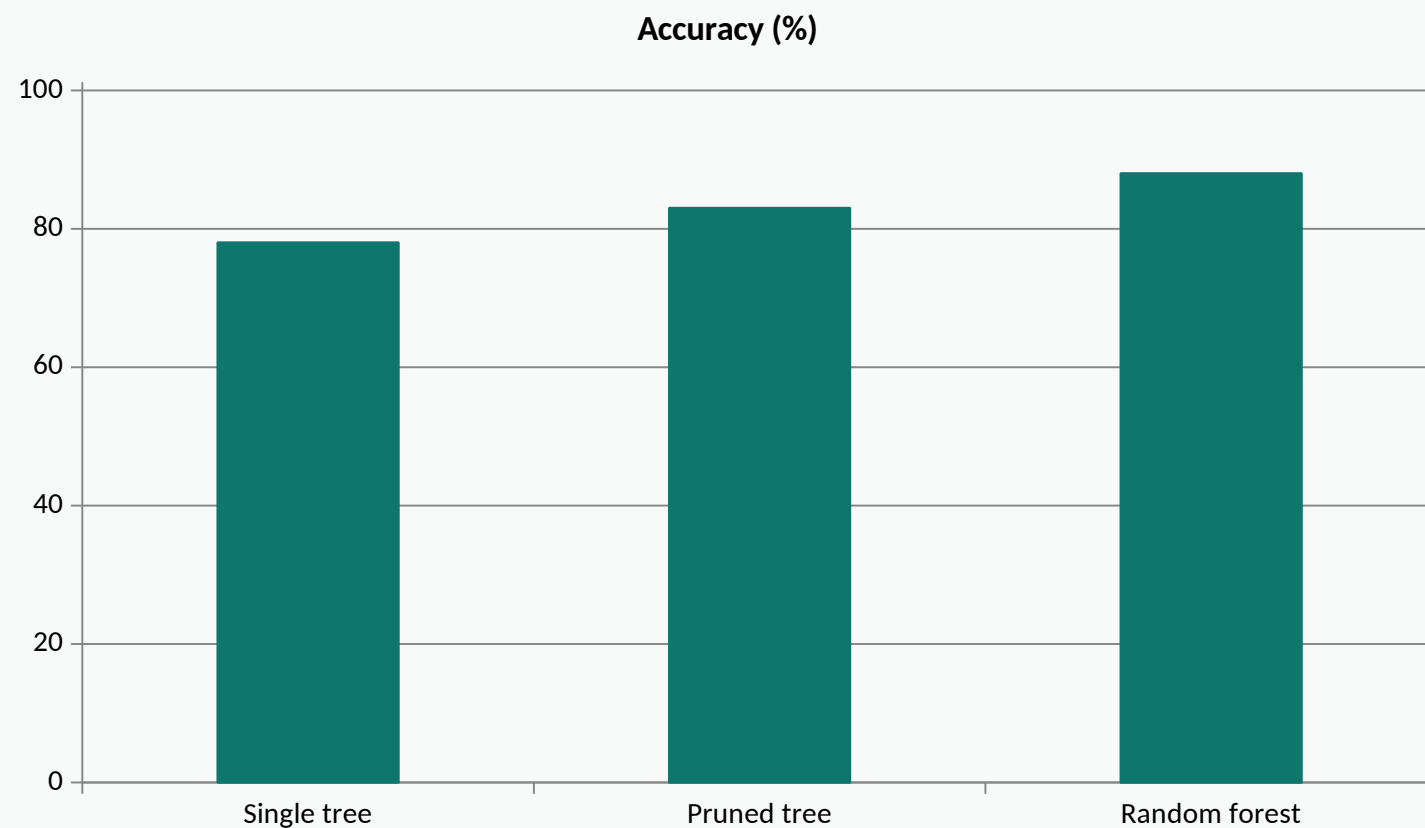
$$p(\text{churn})=0.7 \rightarrow \text{Gini}=1-0.7^2-0.3^2=0.42$$

READOUT

**depth, leaf support,
validation score**

Show units, denominator, and the comparison baseline.

Decision tree and ensemble path · evidence



WHAT THE DATA SHOWS

Leaf counts, feature importance, and hold-out scores remain plausible.

Evidence becomes a decision only after scope and uncertainty are stated.

Decision tree and ensemble path · failure and control

FAILURE MODE

Deep leaves memorise rare combinations and become unstable.

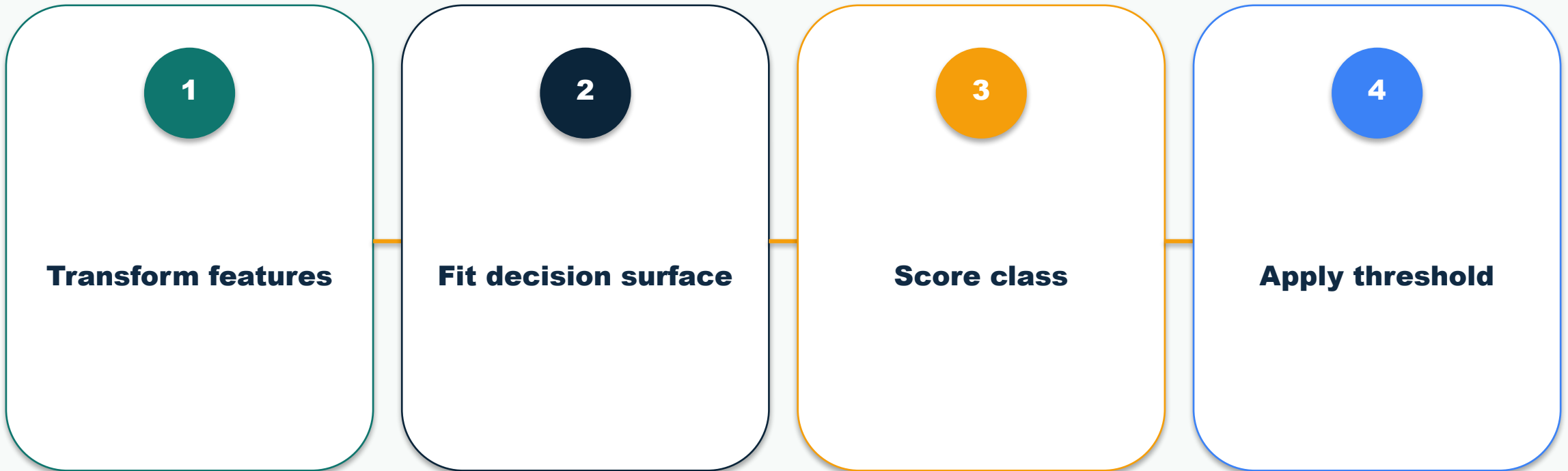
CONTROL

Limit depth/leaf size and compare with bagged Random Forest estimates.

VERIFICATION

Leaf counts, feature importance, and hold-out scores remain plausible.

Logistic, KNN, SVM, and Naive Bayes · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Logistic, KNN, SVM, and Naive Bayes - concrete contract



logistic

log-odds



KNN

neighbourhood vote



SVM

margin/kernel



NB

conditional likelihood

Logistic, KNN, SVM, and Naive Bayes · worked calculation

$$\text{logit}(p) = \beta_0 + \beta x; p = 1 / (1 + \exp(-z))$$

WORKED EXAMPLE

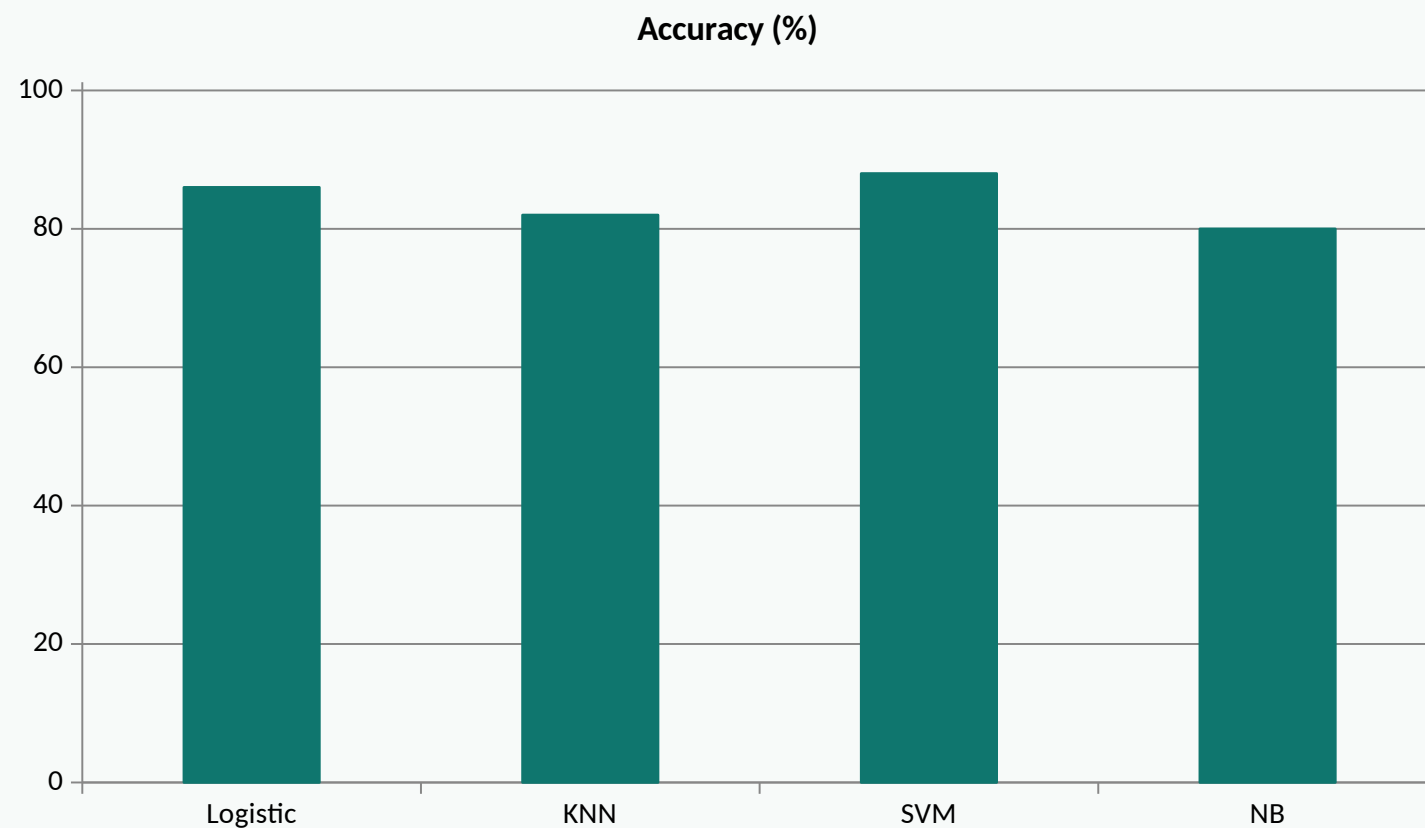
$$z = 1.386 \rightarrow p \approx 0.80$$

READOUT

**probability calibration
and discrimination**

Show units, denominator, and the comparison baseline.

Logistic, KNN, SVM, and Naive Bayes - evidence



WHAT THE DATA SHOWS

Each classifier is evaluated on the same rows, labels, and metrics.

Evidence becomes a decision only after scope and uncertainty are stated.

Logistic, KNN, SVM, and Naive Bayes - failure and control

FAILURE MODE

KNN and SVM are distorted when features have incompatible scales.

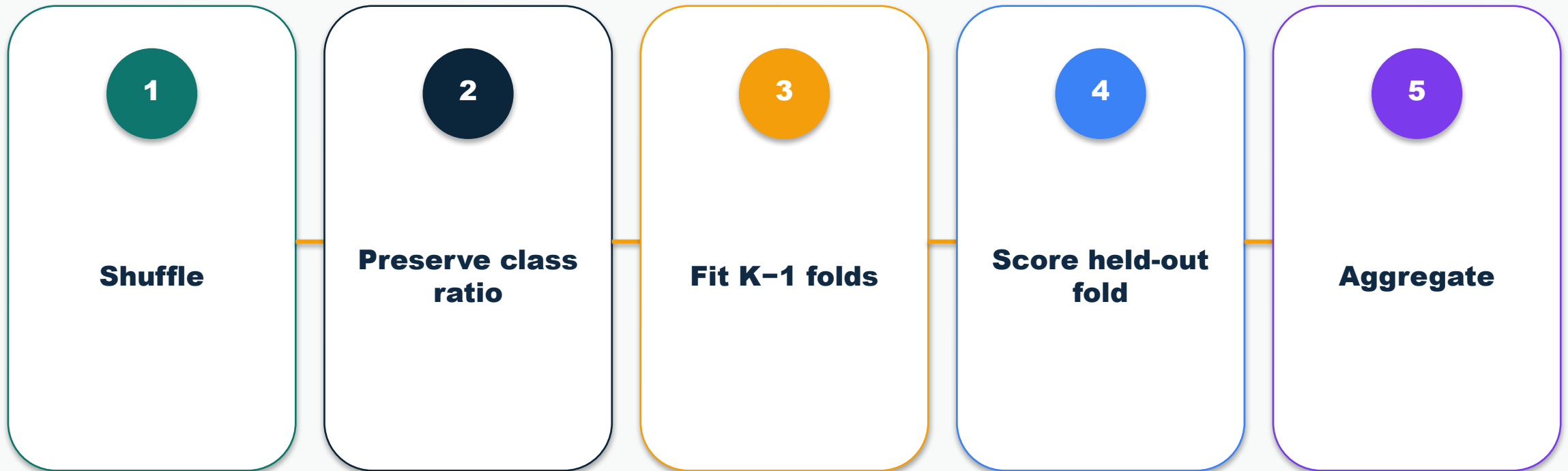
CONTROL

Use a shared preprocessing pipeline and compare models under identical folds.

VERIFICATION

Each classifier is evaluated on the same rows, labels, and metrics.

Stratified K-fold cross-validation · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Stratified K-fold cross-validation · concrete contract



k

number of folds



stratify

class balance



seed

reproducibility



mean ± standard deviation

Stratified K-fold cross-validation · worked calculation

$$\text{CV mean} = \sum \text{score}_i / k$$

WORKED EXAMPLE

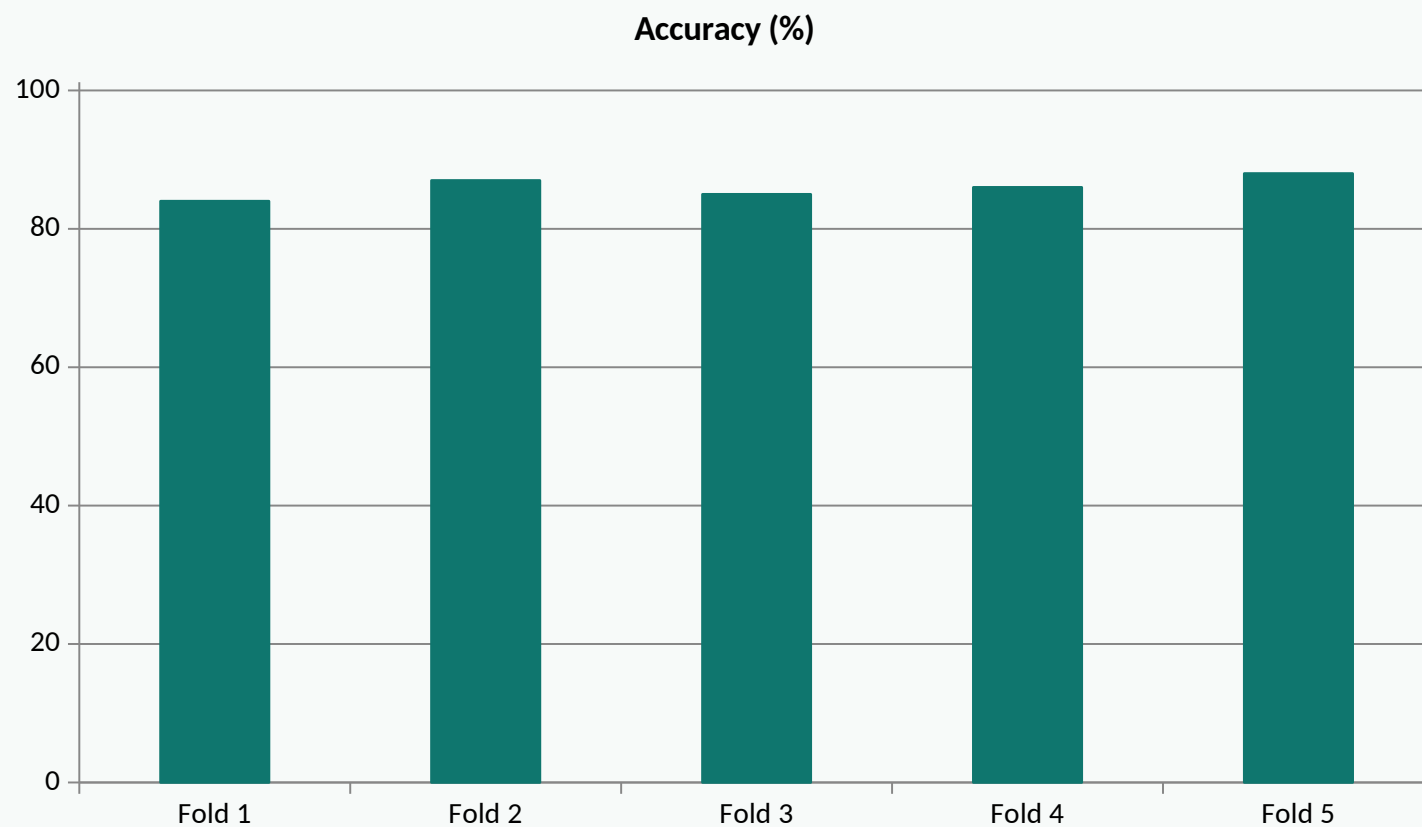
[.84,.87,.85,.86,.88] → mean .86

READOUT

**mean and fold
variability**

Show units, denominator, and the comparison baseline.

Stratified K-fold cross-validation · evidence



WHAT THE DATA SHOWS

Every row is held out once and class proportions are stable across folds.

Evidence becomes a decision only after scope and uncertainty are stated.

Stratified K-fold cross-validation · failure and control

FAILURE MODE

Preprocessing outside the cross-validation loop leaks fold information.

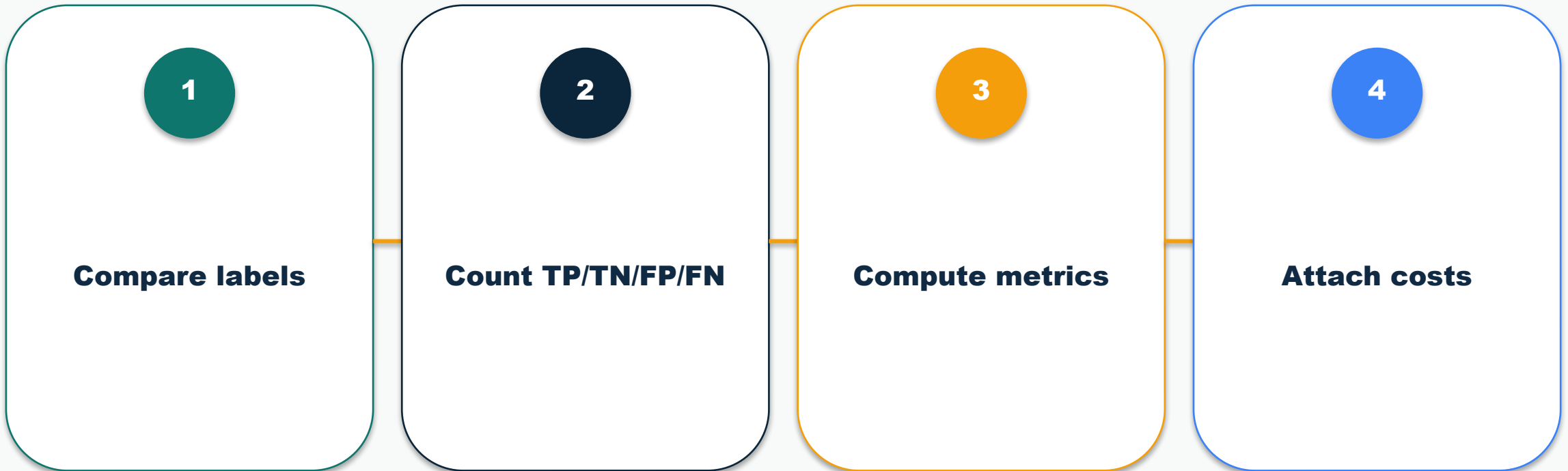
CONTROL

Put imputation, encoding, scaling, and model in one pipeline.

VERIFICATION

Every row is held out once and class proportions are stable across folds.

Confusion matrix and class metrics · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Confusion matrix and class metrics · concrete contract

#

precision=TP/(TP+FP)

#

recall=TP/(TP+FN)

#

specificity=TN/(TN+FP)

#

F1 harmonic mean

Confusion matrix and class metrics · worked calculation

$$F1 = 2PR / (P + R)$$

WORKED EXAMPLE

$$P=.75, R=.60 \rightarrow F1=.667$$

READOUT

**precision, recall, F1,
specificity**

Show units, denominator, and the comparison baseline.

Confusion matrix and class metrics · evidence

		PREDICTED	
		POSITIVE	NEGATIVE
ACTUAL	POSITIVE	TP 45	FN 30
	NEGATIVE	FP 15	TN 210

METRIC READOUT

Precision 75%
Recall 60%
F1 66.7%
Cases 300

Confusion matrix and class metrics · failure and control

FAILURE MODE

**Accuracy hides failure
when the positive class
is rare.**

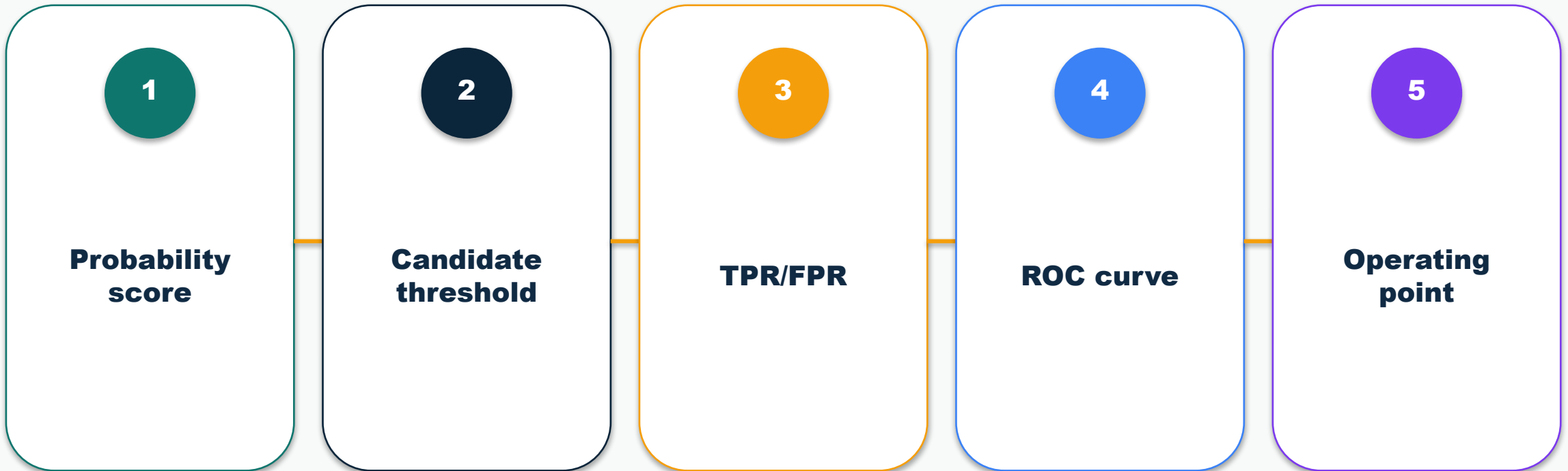
CONTROL

**Report the full matrix
and select metrics
based on error cost.**

VERIFICATION

**Matrix cell counts sum
to the evaluated sample
size.**

ROC, AUC, and threshold policy · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

ROC, AUC, and threshold policy · concrete contract

#

TPR=recall

#

FPR=FP/(FP+TN)

{

AUC

ranking probability

{

threshold

policy choice

ROC, AUC, and threshold policy · worked calculation

$$\text{expected cost} = \text{FP} \times \text{C_FP} + \text{FN} \times \text{C_FN}$$

WORKED EXAMPLE

$$15 \text{ FP} \times \$20 + 30 \text{ FN} \times \$120 = \$3,900$$

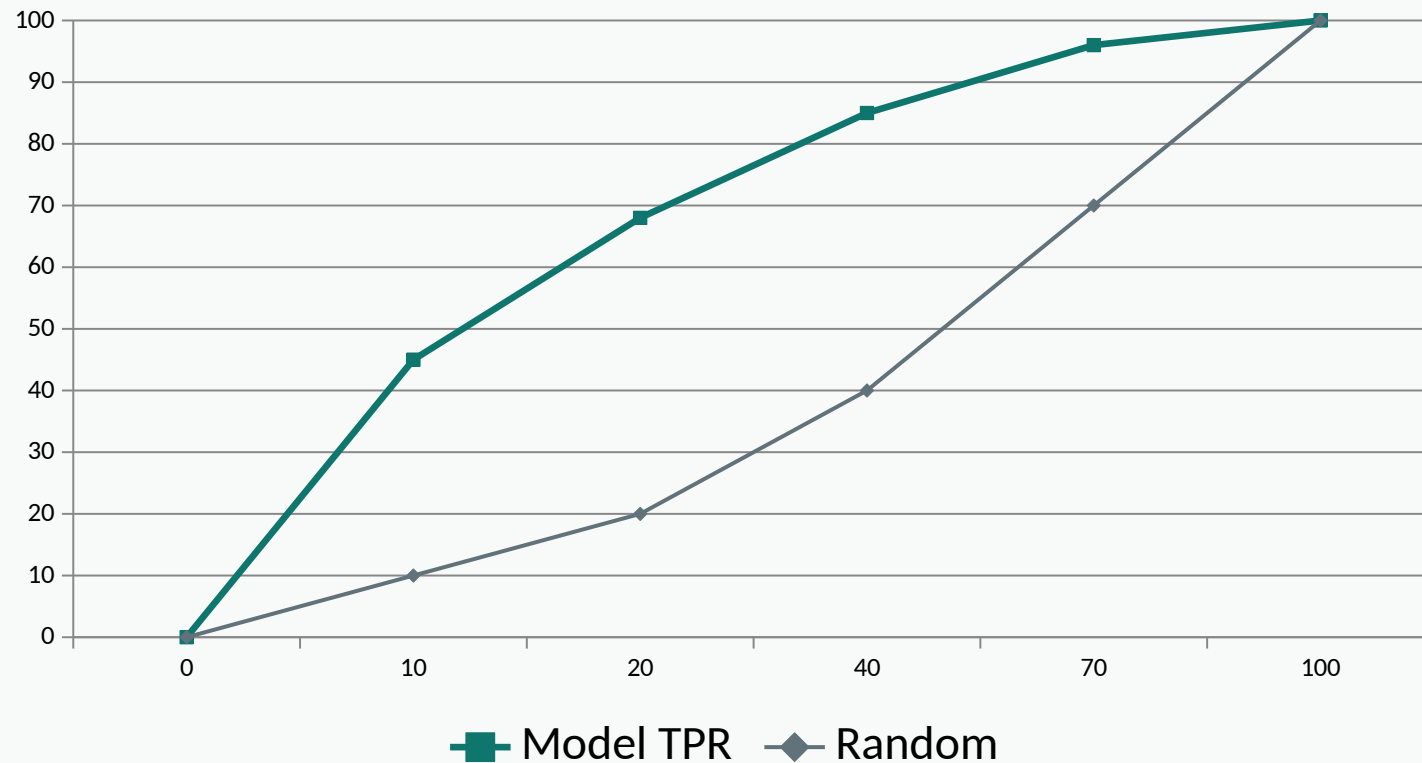
READOUT

**AUC plus operating
cost**

Show units, denominator, and the comparison baseline.

ROC, AUC, and threshold policy · evidence

ROC: true-positive rate versus false-positive rate (%)



OPERATING POINT

AUC summarises ranking.

The threshold still controls TPR, FPR, workload, and cost.

ROC, AUC, and threshold policy · failure and control

FAILURE MODE

Choosing 0.5 by convention when false negatives are much costlier.

CONTROL

Select a threshold from business costs and capacity constraints.

VERIFICATION

The chosen point states threshold, TPR, FPR, volume, and expected cost.

Lab 07 · Compare classification algorithms

CENTRAL APP

Classification

<http://localhost:8501>

INPUT

Lab 07 mock CSV

MECHANISM

Classification

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 07 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Compare classification algorithms produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 08 · Use stratified K-fold validation

CENTRAL APP

Cross-validation

<http://localhost:8501>

INPUT

Lab 08 mock CSV

MECHANISM

Cross-validation

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 08 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

**PASS: Use stratified K-fold validation produces a reproducible artifact and a defensible interpretation
—not only a screenshot.**

Lab 09 · Interpret confusion metrics and error costs

CENTRAL APP

Classification

<http://localhost:8501>

INPUT

Lab 09 mock CSV

MECHANISM

Classification

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 09 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Interpret confusion metrics and error costs produces a reproducible artifact and a defensible interpretation—not only a screenshot.

TOPIC 05

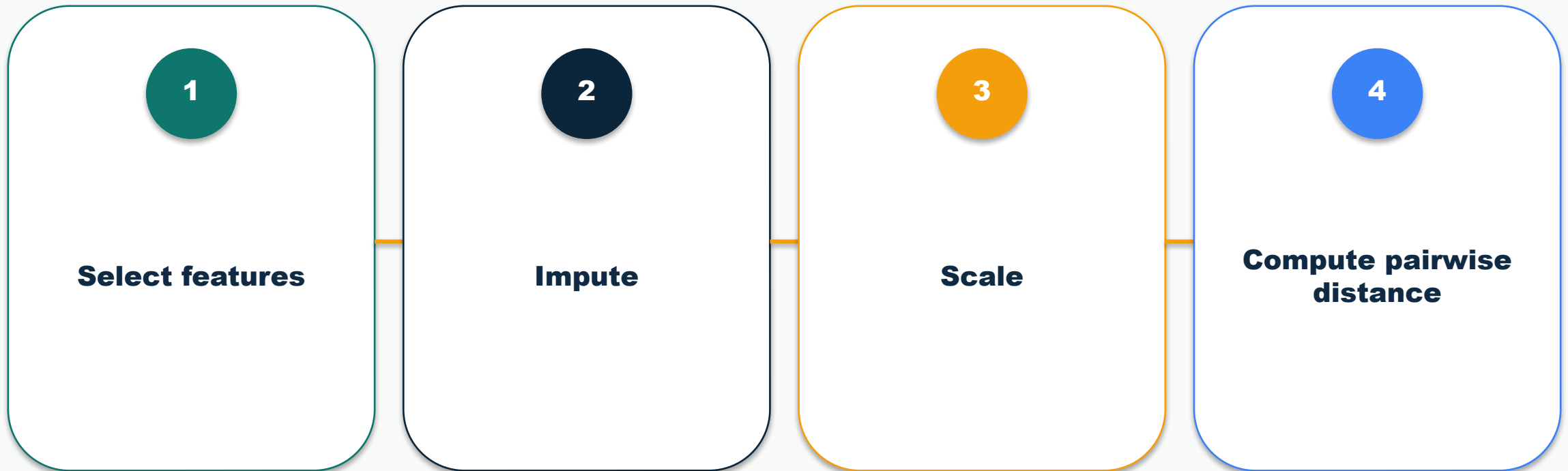
Clustering

LO5 · A5/A9 · K3/K5 · Discovering structure without labels

Mechanism → contract → calculation → evidence → control



Distance and scaling · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Distance and scaling · concrete contract

**Euclidean**

continuous

**Manhattan**

absolute steps

**cosine**

direction

**categorical**

requires encoding

Distance and scaling · worked calculation

$$d(x,y)=\sqrt{\sum (x_i-y_i)^2}$$

WORKED EXAMPLE

(2,3) to (5,7) → $\sqrt{(9+16)}=5$

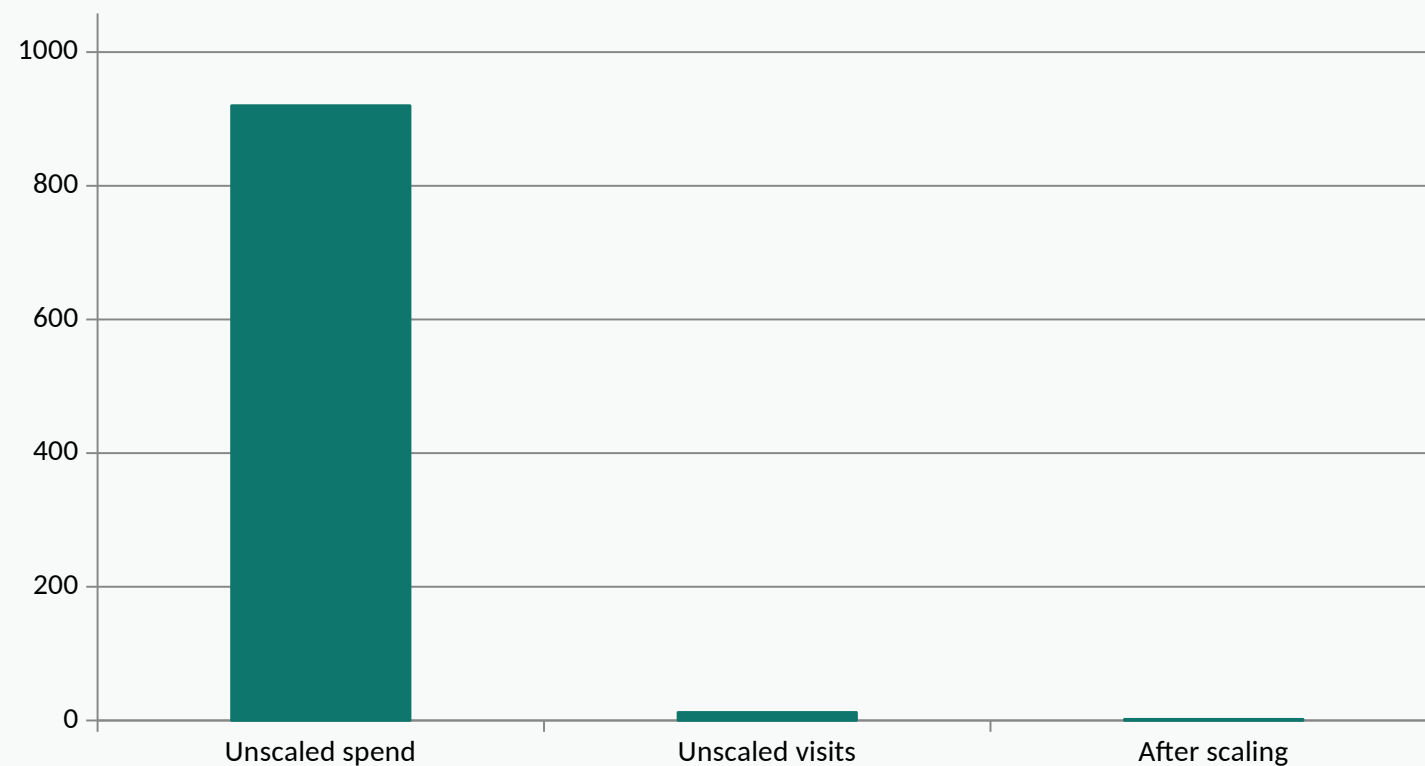
READOUT

distance distribution

Show units, denominator, and the comparison baseline.

Distance and scaling · evidence

Distance contribution



WHAT THE DATA SHOWS

**Nearest neighbours
are sensible under
domain review and
robust to scaling
choices.**

Evidence becomes a decision only after scope
and uncertainty are stated.

Distance and scaling · failure and control

FAILURE MODE

A high-magnitude spend field dominates an unscaled visit-frequency field.

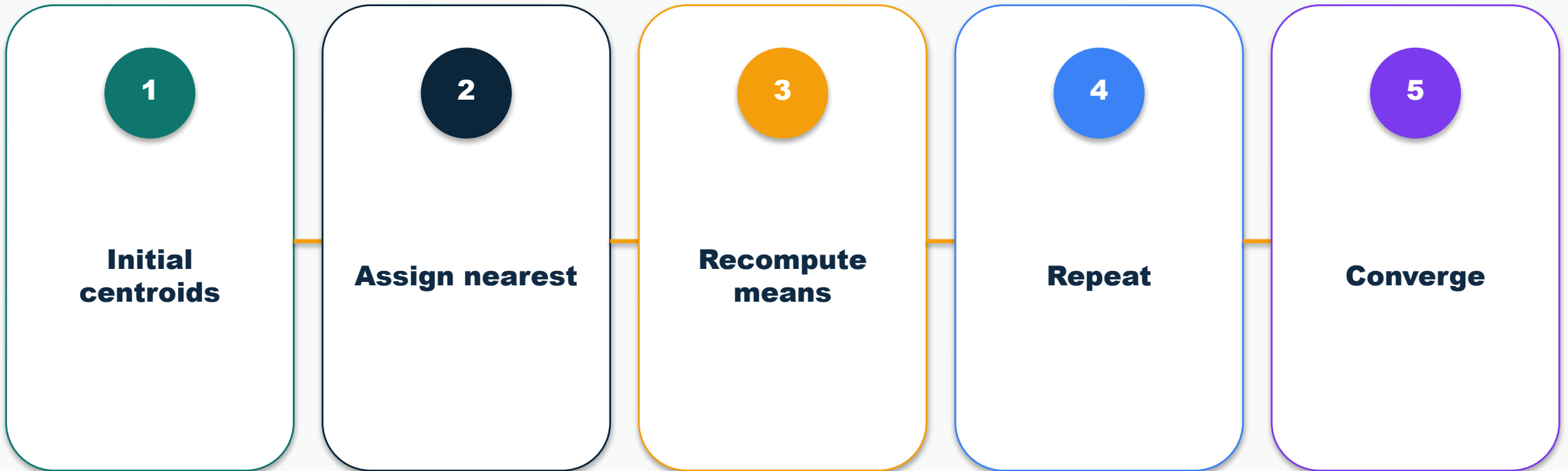
CONTROL

Standardise numeric variables and justify the distance metric.

VERIFICATION

Nearest neighbours are sensible under domain review and robust to scaling choices.

K-means assignment/update loop · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

K-means assignment/update loop · concrete contract

**k**

requested clusters

**centroid**

feature mean

**inertia**

within-cluster SSE

**n_init**

multiple starts

K-means assignment/update loop · worked calculation

$$\text{inertia} = \sum ||\mathbf{x}_i - \mu_{\text{cluster}(i)}||^2$$

WORKED EXAMPLE

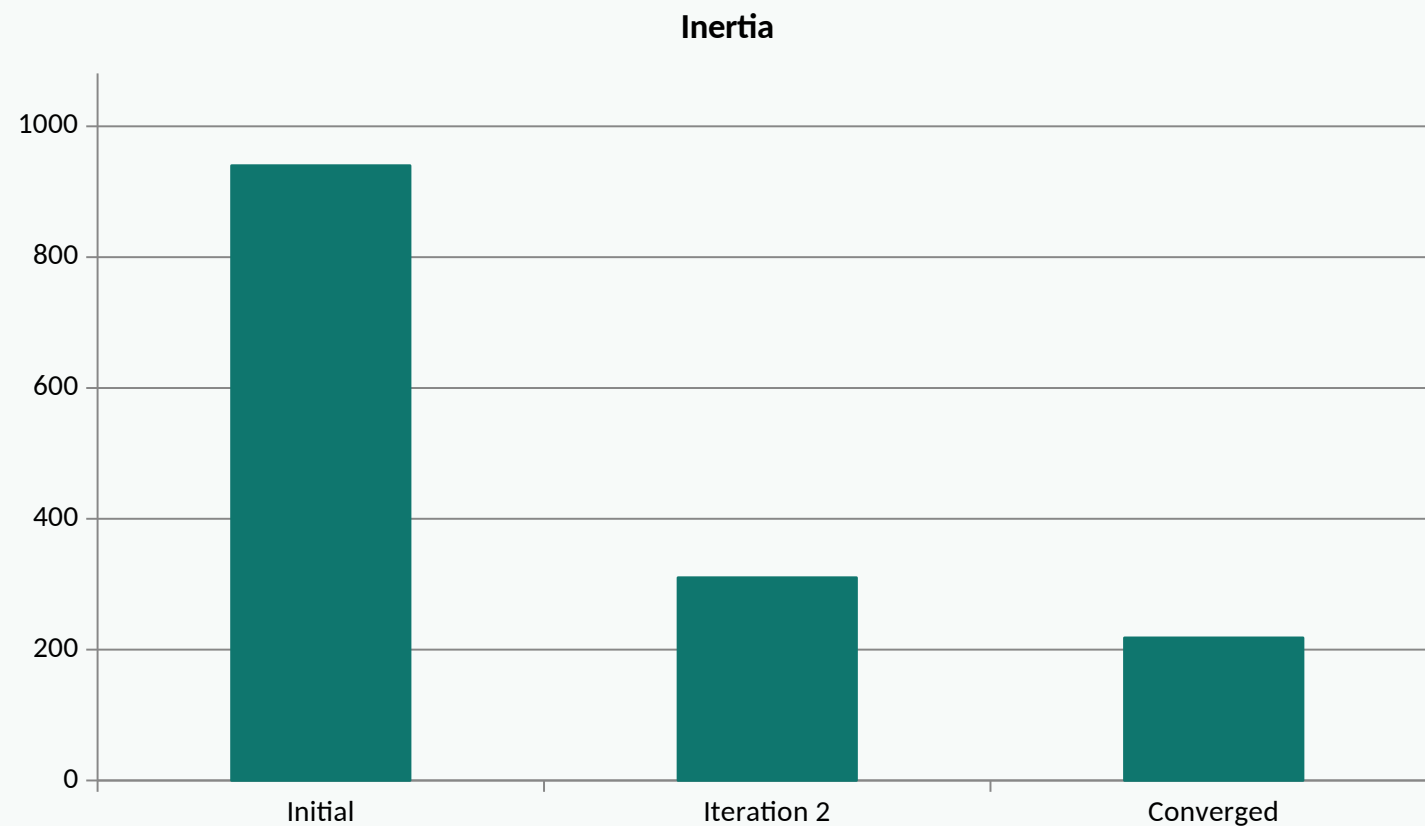
three compact groups reduce inertia from 940 to 218

READOUT

inertia and stability

Show units, denominator, and the comparison baseline.

K-means assignment/update loop · evidence



WHAT THE DATA SHOWS

Centroids, inertia, and labels reproduce under the recorded seed.

Evidence becomes a decision only after scope and uncertainty are stated.

K-means assignment/update loop · failure and control

FAILURE MODE

**One random start
reaches a poor local
minimum.**

CONTROL

**Use multiple
initialisations and
compare label stability.**

VERIFICATION

**Centroids, inertia, and
labels reproduce under
the recorded seed.**

Silhouette analysis · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Silhouette analysis · concrete contract



range

−1 to 1



near 1

separated



near 0

boundary



negative

likely misassigned

Silhouette analysis · worked calculation

$$s = (b - a) / \max(a, b)$$

WORKED EXAMPLE

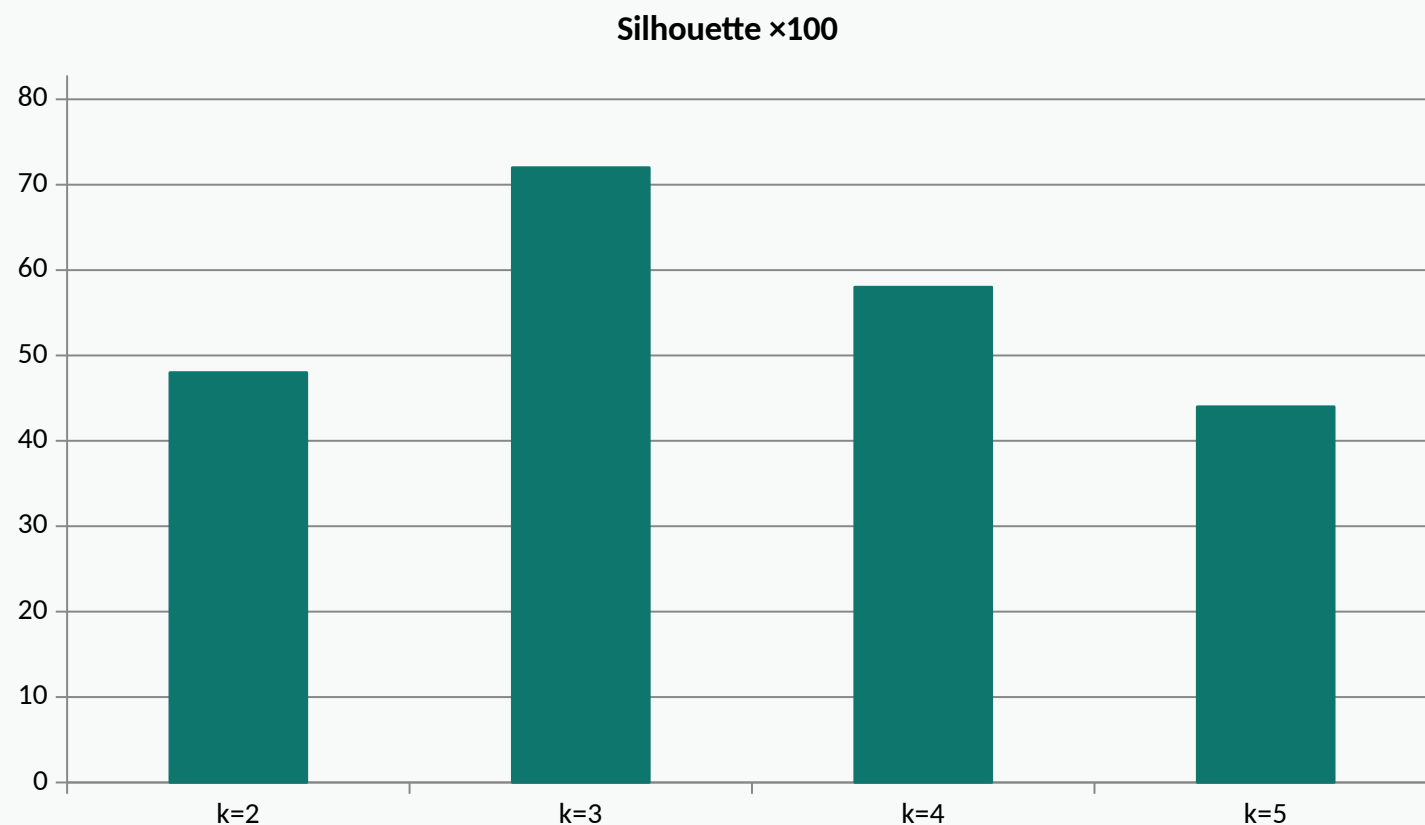
$$a=2, b=5 \rightarrow s=0.60$$

READOUT

**mean and per-cluster
silhouette**

Show units, denominator, and the comparison baseline.

Silhouette analysis · evidence



WHAT THE DATA SHOWS

The chosen k has acceptable separation and interpretable cluster sizes.

Evidence becomes a decision only after scope and uncertainty are stated.

Silhouette analysis · failure and control

FAILURE MODE

Selecting the highest average while ignoring a tiny or unstable cluster.

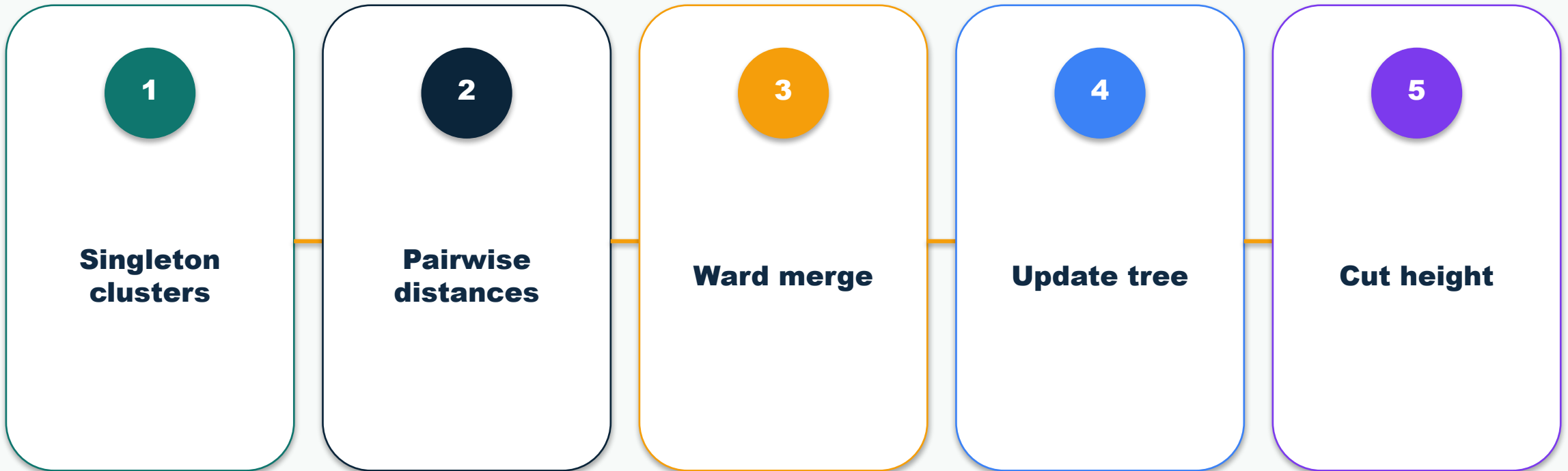
CONTROL

Review the distribution, cluster sizes, and business usefulness.

VERIFICATION

The chosen k has acceptable separation and interpretable cluster sizes.

Hierarchical clustering and dendrogram · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Hierarchical clustering and dendrogram · concrete contract



linkage

Ward



merge distance



dendrogram height



cut

cluster count

Hierarchical clustering and dendrogram · worked calculation

Ward merges the pair with the smallest increase in within-cluster variance

WORKED EXAMPLE

large jump from 6.2 to 14.8 suggests a cut before the jump

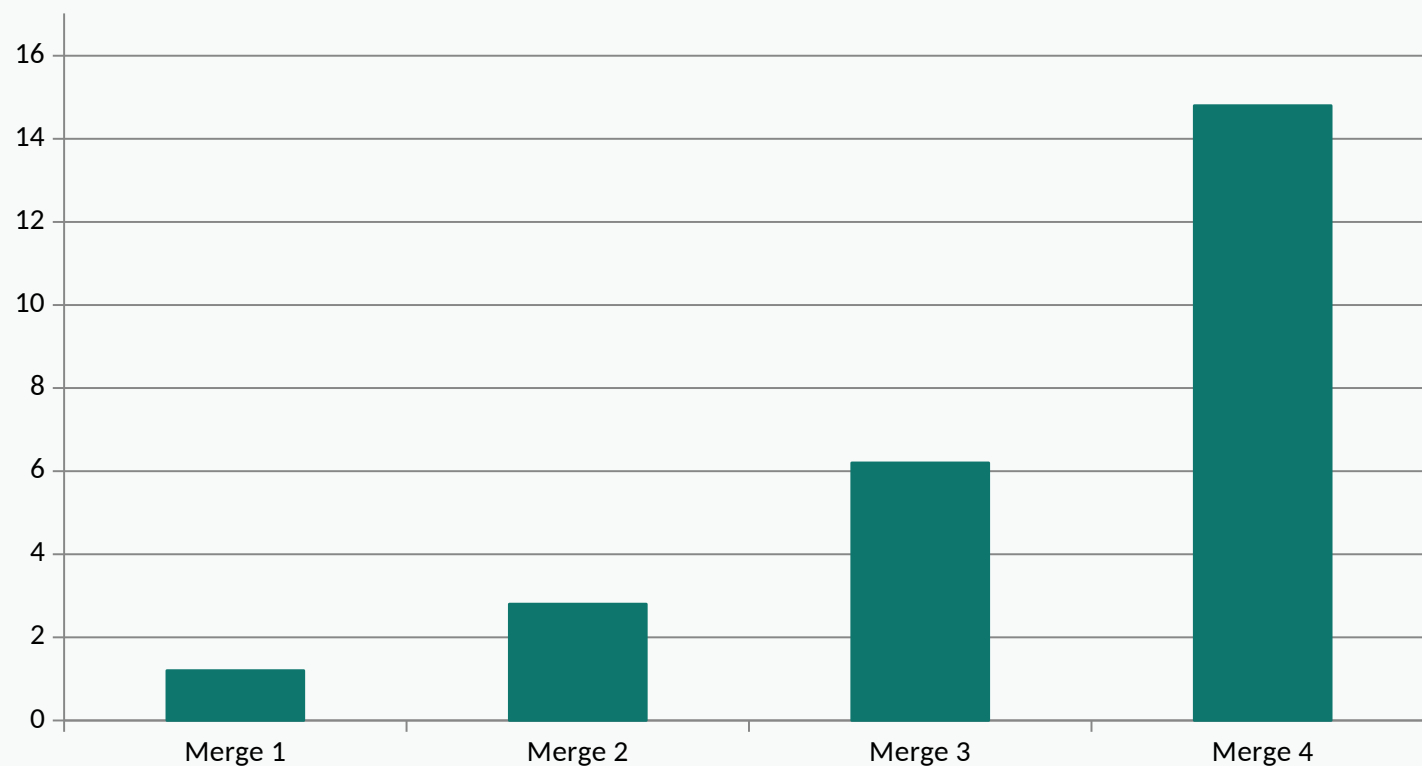
READOUT

merge-distance jump

Show units, denominator, and the comparison baseline.

Hierarchical clustering and dendrogram · evidence

Ward distance



WHAT THE DATA SHOWS

The cut produces the stated cluster count and matching scatter colours.

Evidence becomes a decision only after scope and uncertainty are stated.

Hierarchical clustering and dendrogram · failure and control

FAILURE MODE

Treating horizontal spacing in a dendrogram as a numeric distance.

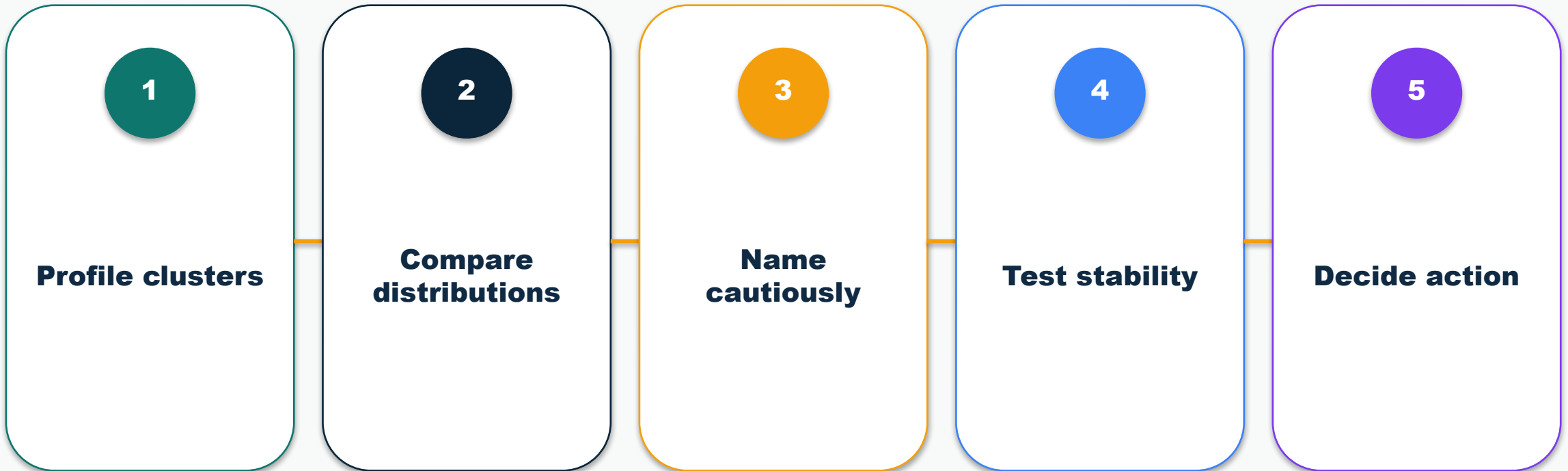
CONTROL

Read merge distance from the vertical axis and document the cut height.

VERIFICATION

The cut produces the stated cluster count and matching scatter colours.

Cluster interpretation and limitations - runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Cluster interpretation and limitations - concrete contract



size

n and %



centroid

scaled profile



stability

repeated samples



action

testable intervention

Cluster interpretation and limitations - worked calculation

standardised lift = cluster mean – overall mean

WORKED EXAMPLE

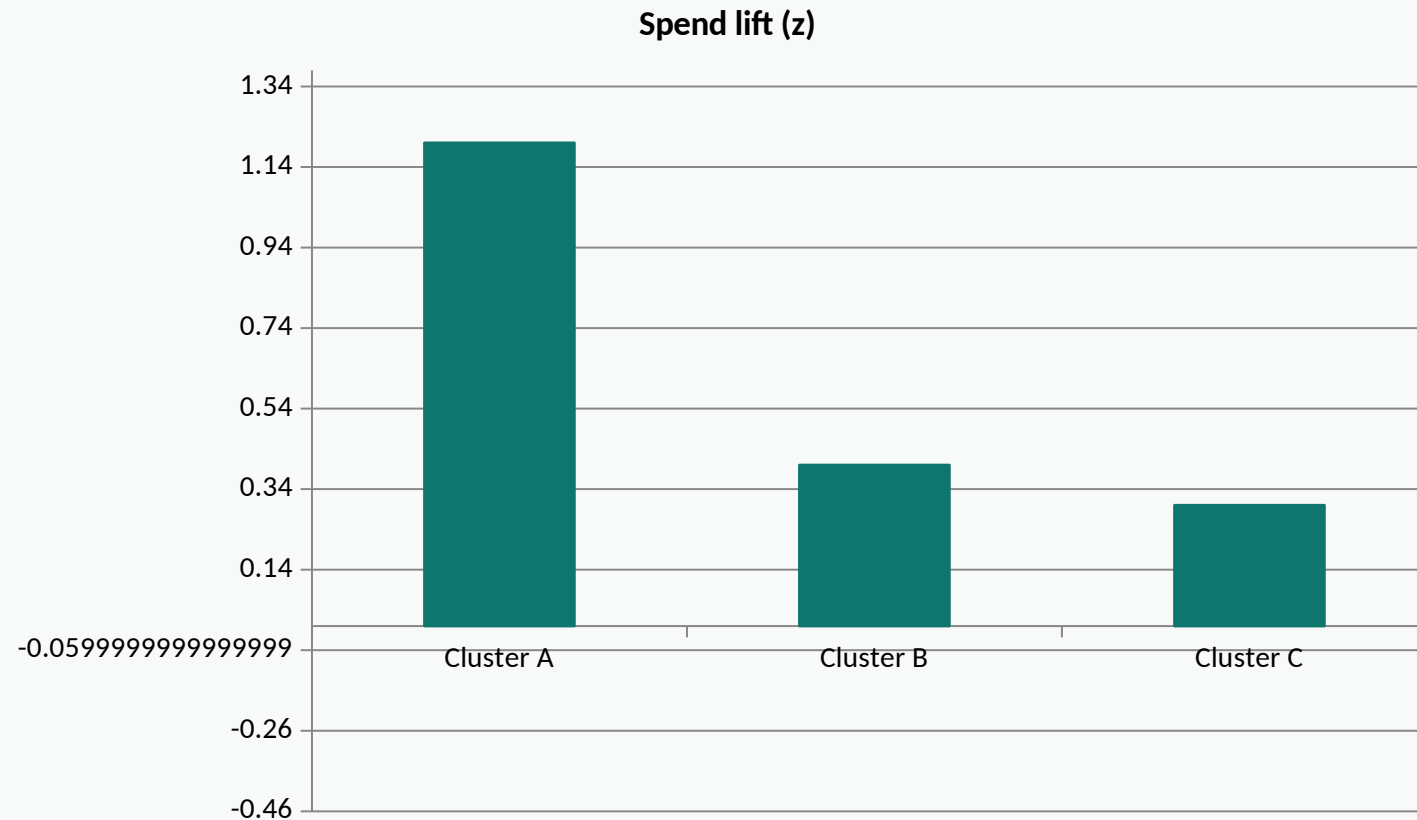
cluster A spend $z=+1.2$ and visits $z=+0.8$

READOUT

profile lift and stability

Show units, denominator, and the comparison baseline.

Cluster interpretation and limitations - evidence



WHAT THE DATA SHOWS

Cluster actions are tested and do not encode prohibited discrimination.

Evidence becomes a decision only after scope and uncertainty are stated.

Cluster interpretation and limitations - failure and control

FAILURE MODE

**Narrative labels
exaggerate weak
numerical differences.**

CONTROL

**Publish measured traits,
uncertainty, and
exclusions before
naming segments.**

VERIFICATION

**Cluster actions are
tested and do not
encode prohibited
discrimination.**

Lab 10 · Select K with silhouette evidence

CENTRAL APP

Clustering

<http://localhost:8501>

INPUT

Lab 10 mock CSV

MECHANISM

Clustering

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 10 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Select K with silhouette evidence produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 11 · Cut a hierarchical dendrogram

CENTRAL APP

**Hierarchical
clustering**

<http://localhost:8501>

INPUT

Lab 11 mock CSV

MECHANISM

Hierarchical clustering

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 11 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

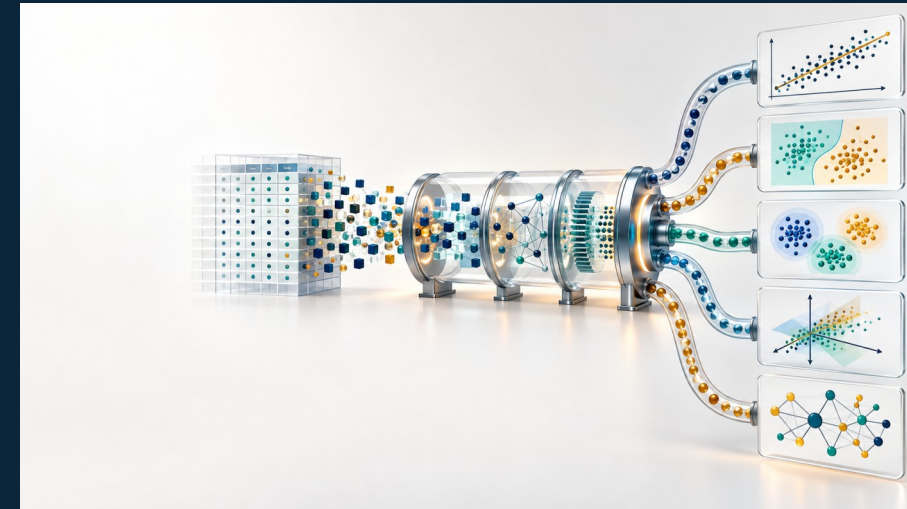
**PASS: Cut a hierarchical dendrogram produces a reproducible artifact and a defensible interpretation
—not only a screenshot.**

TOPIC 06

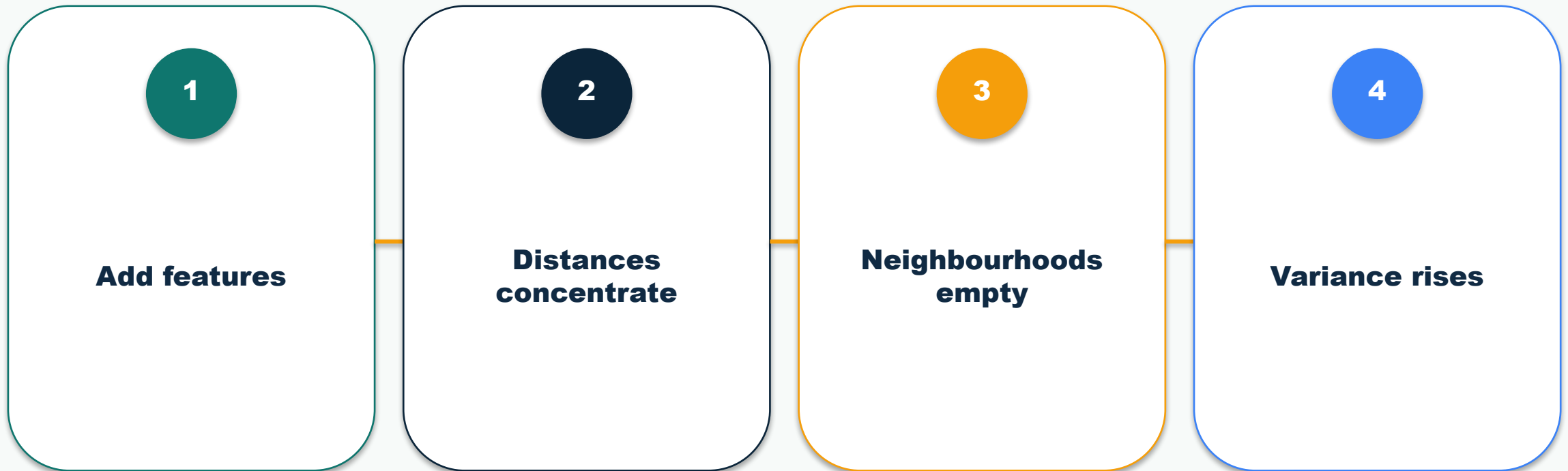
Dimension Reduction

LO6 · A4 · K8 · Compressing variation while retaining evidence

Mechanism → contract → calculation → evidence → control



Curse of dimensionality · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Curse of dimensionality · concrete contract



p

feature count



n

sample count



sparsity

volume effect



redundancy

correlated inputs

Curse of dimensionality · worked calculation

required coverage grows exponentially with p

WORKED EXAMPLE

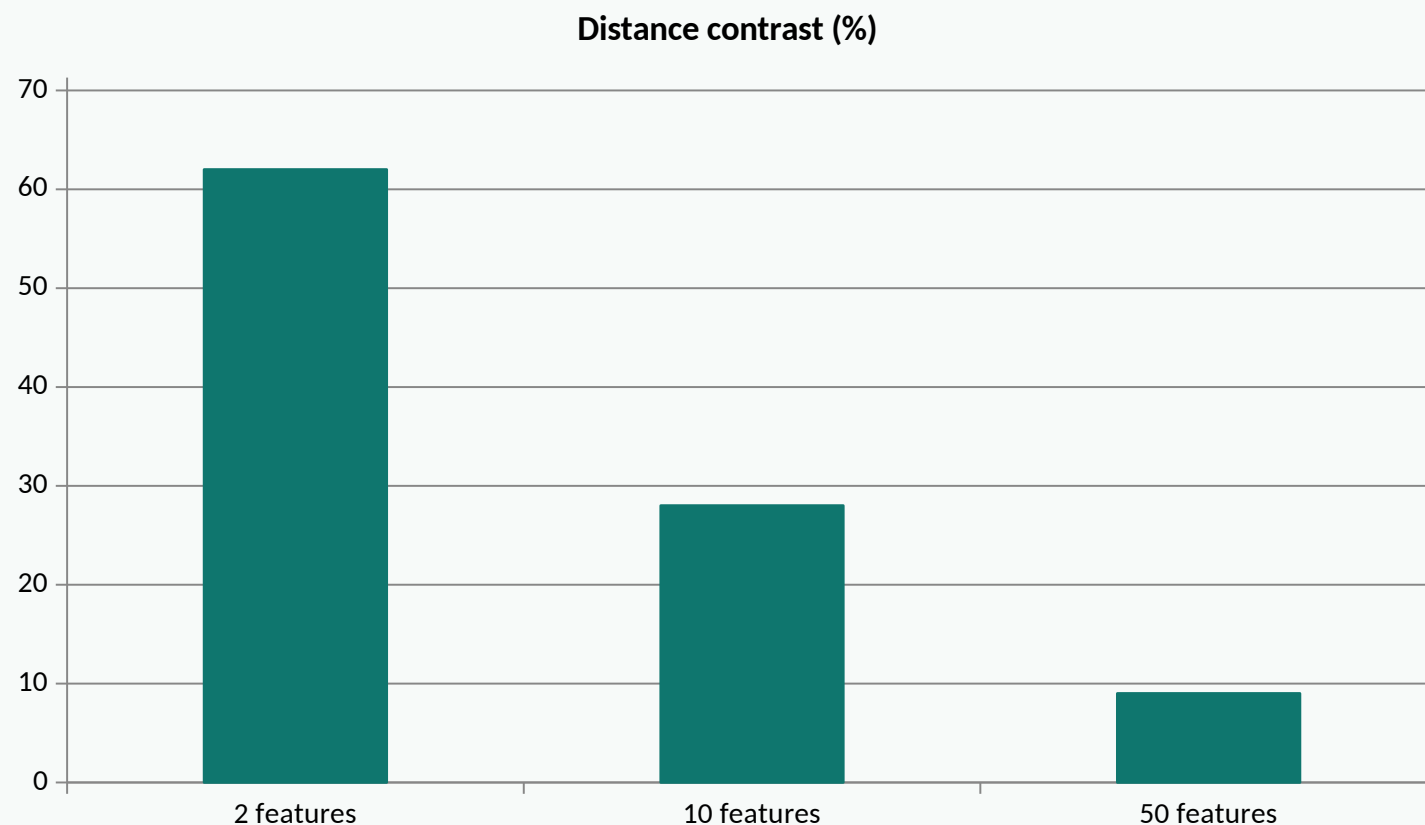
10 bins per feature $\rightarrow 10^6$ cells for 6 features

READOUT

distance contrast

Show units, denominator, and the comparison baseline.

Curse of dimensionality · evidence



WHAT THE DATA SHOWS

**Validation improves
or remains stable
when a feature is
retained.**

Evidence becomes a decision only after scope
and uncertainty are stated.

Curse of dimensionality · failure and control

FAILURE MODE

**Adding weak features
makes neighbours less
distinguishable.**

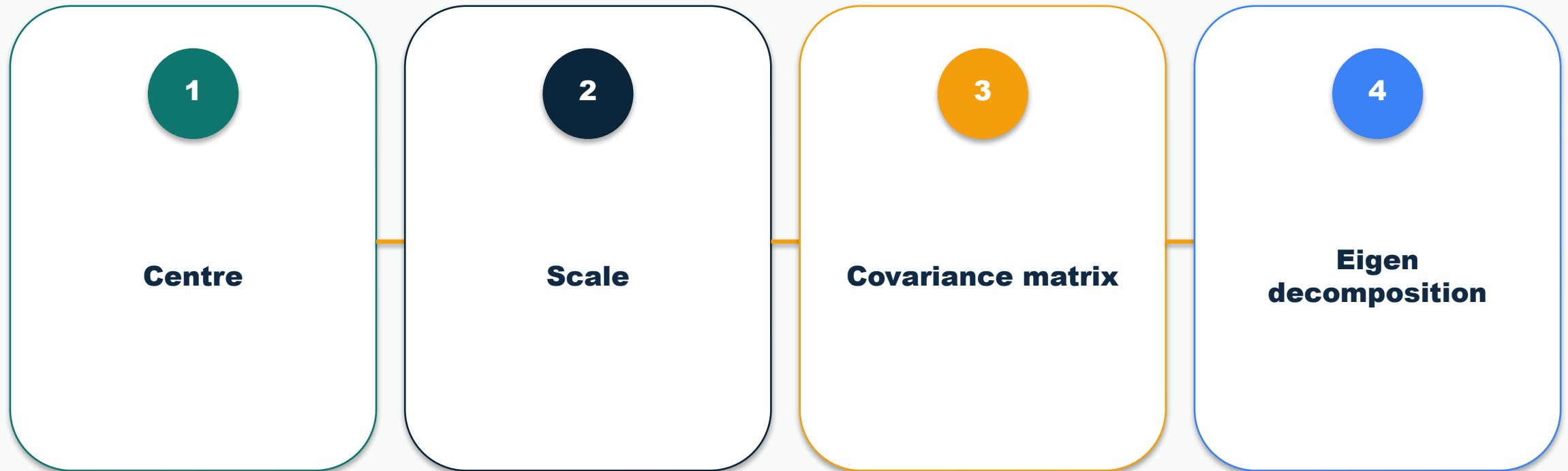
CONTROL

**Remove
leakage/redundancy and
validate feature
additions.**

VERIFICATION

**Validation improves or
remains stable when a
feature is retained.**

Standardisation and covariance · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Standardisation and covariance · concrete contract

#

mean vector μ

#

standard deviation σ

#

covariance Σ

#

symmetric matrix

Standardisation and covariance · worked calculation

$$\text{cov}(\mathbf{x}, \mathbf{y}) = \sum (\mathbf{x} - \bar{\mathbf{x}})(\mathbf{y} - \bar{\mathbf{y}}) / (n - 1)$$

WORKED EXAMPLE

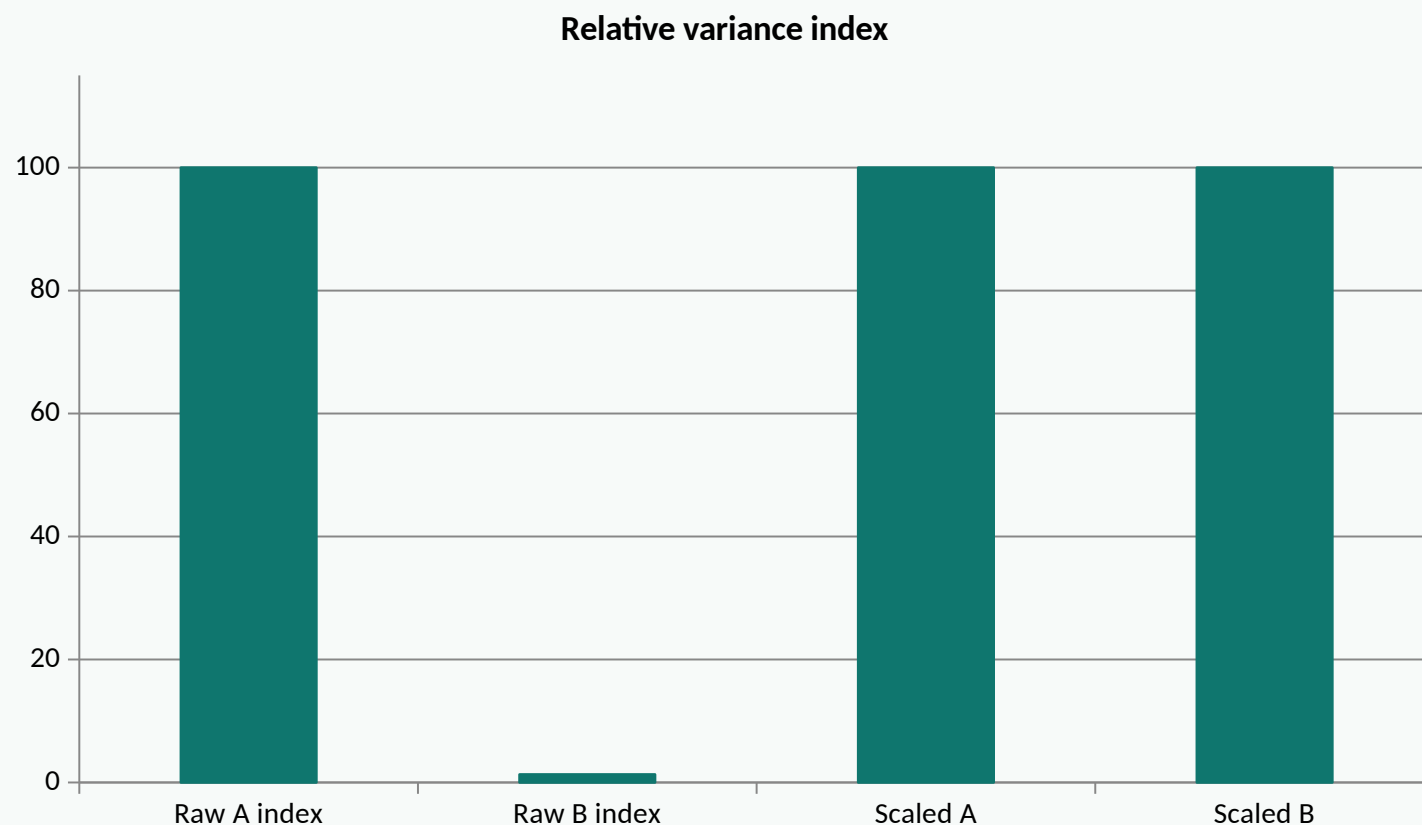
positive covariance means the two centred features move together

READOUT

covariance structure

Show units, denominator, and the comparison baseline.

Standardisation and covariance · evidence



WHAT THE DATA SHOWS

Each scaled feature has near-zero mean and unit variance on training data.

Evidence becomes a decision only after scope and uncertainty are stated.

Standardisation and covariance · failure and control

FAILURE MODE

PCA on raw units lets high-variance currency fields dominate ratios.

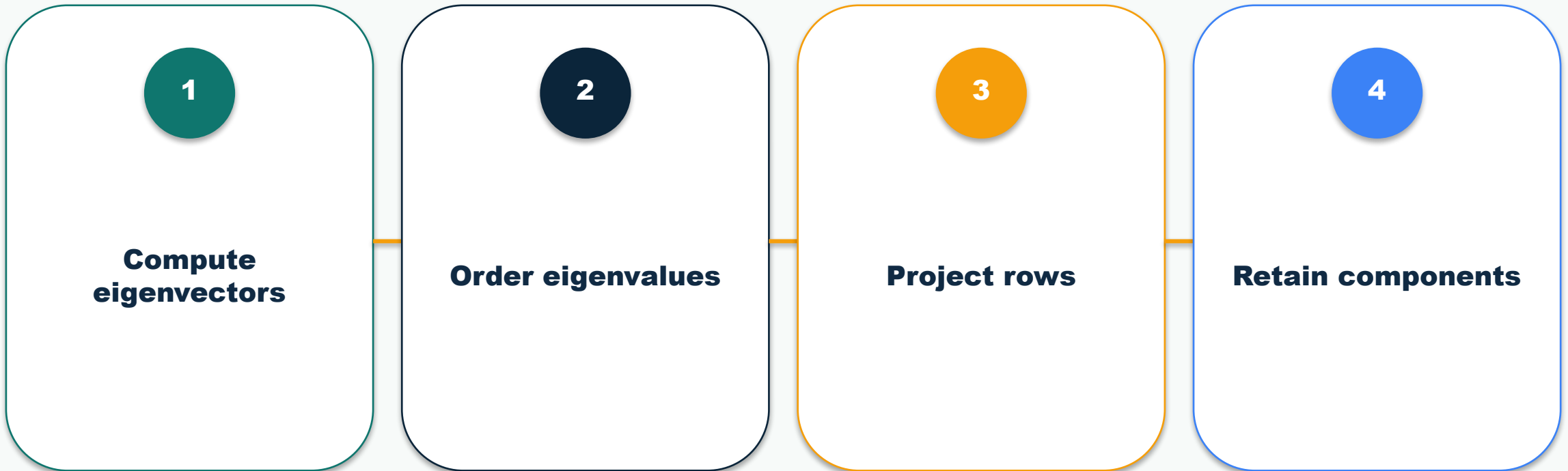
CONTROL

Standardise features when units and scales differ.

VERIFICATION

Each scaled feature has near-zero mean and unit variance on training data.

Principal components · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Principal components · concrete contract

**loading**

feature weight

**score**

projected row

**eigenvalue**

variance

**orthogonal axes**

Principal components · worked calculation

$$Z = XW$$

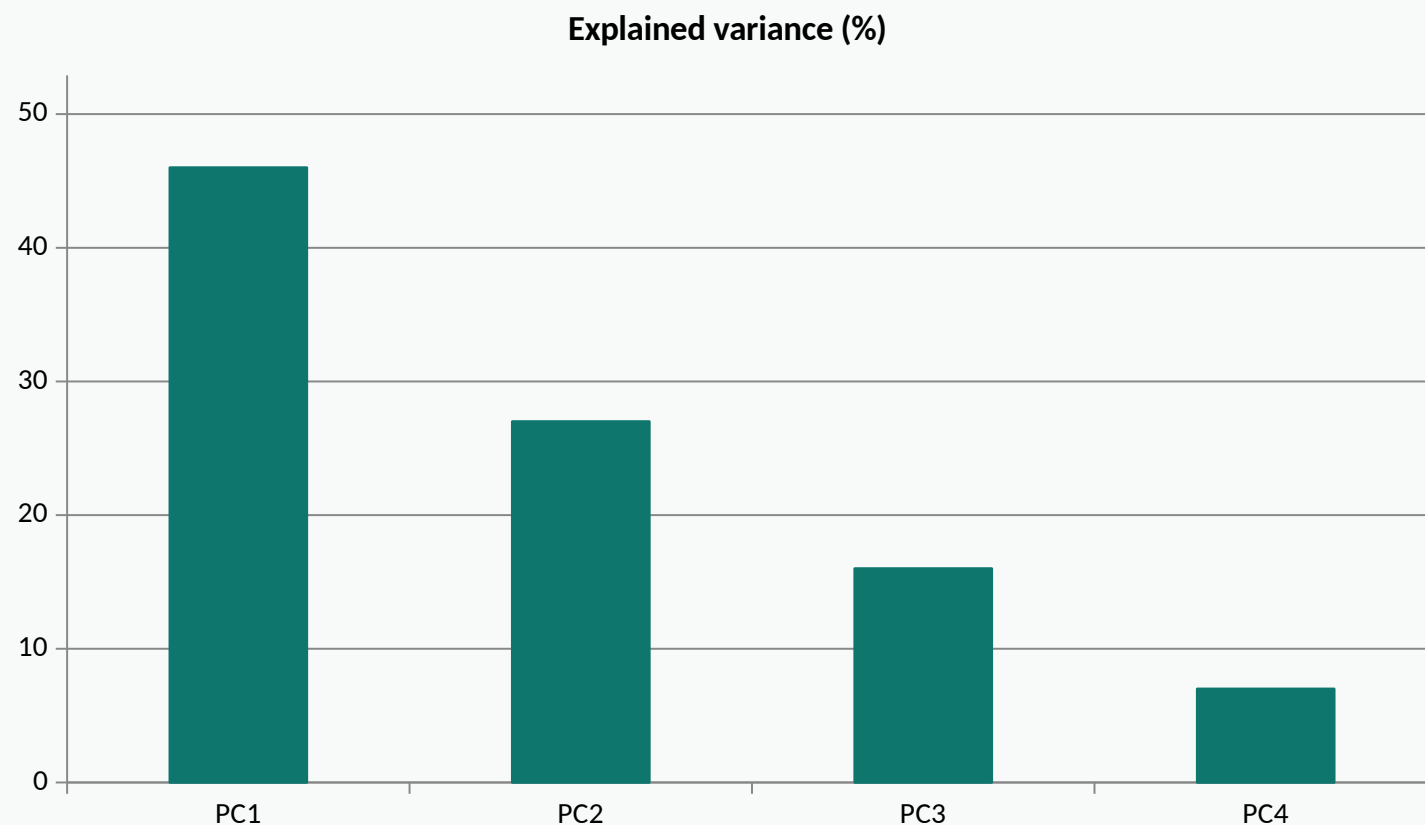
WORKED EXAMPLE

two correlated measures rotate into PC1 along shared variation

READOUT
explained variance ratio

Show units, denominator, and the comparison baseline.

Principal components · evidence



WHAT THE DATA SHOWS

Projected scores reproduce from saved scaler and PCA objects.

Evidence becomes a decision only after scope and uncertainty are stated.

Principal components · failure and control

FAILURE MODE

Interpreting component sign as inherently positive or negative; signs may flip.

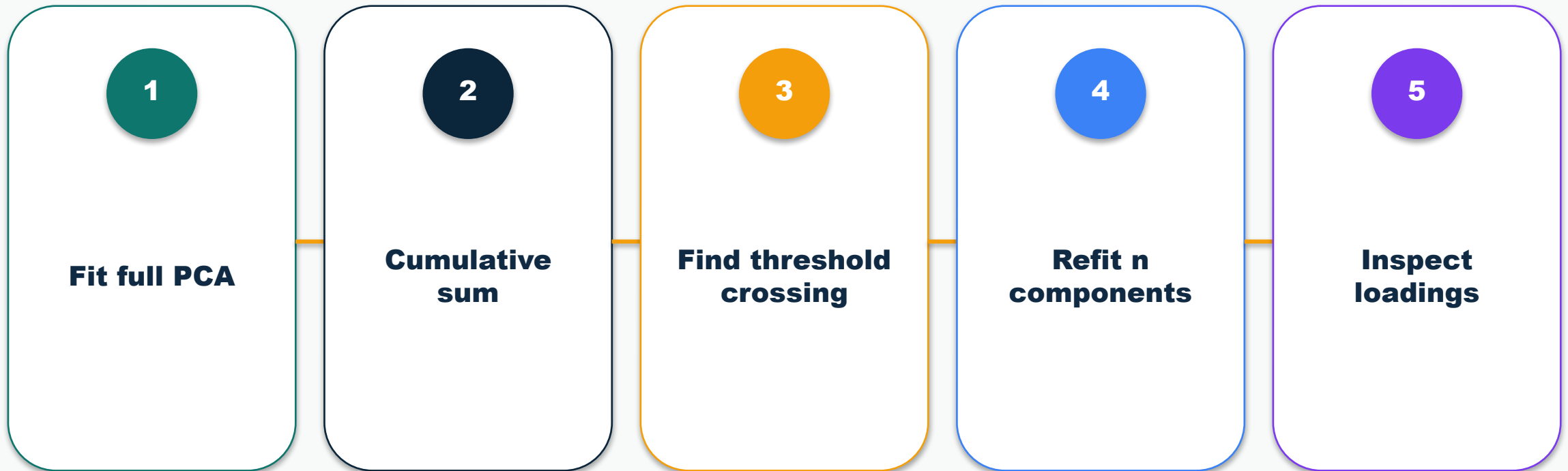
CONTROL

Interpret relative loading magnitude and relationships, not arbitrary sign alone.

VERIFICATION

Projected scores reproduce from saved scaler and PCA objects.

Retained variance and loadings · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Retained variance and loadings · concrete contract



variance target

0.70



n

first crossing



loading magnitude



reconstruction loss

Retained variance and loadings · worked calculation

$$n = \min\{k : \sum_{i \leq k} \text{variance}_i \geq .70\}$$

WORKED EXAMPLE

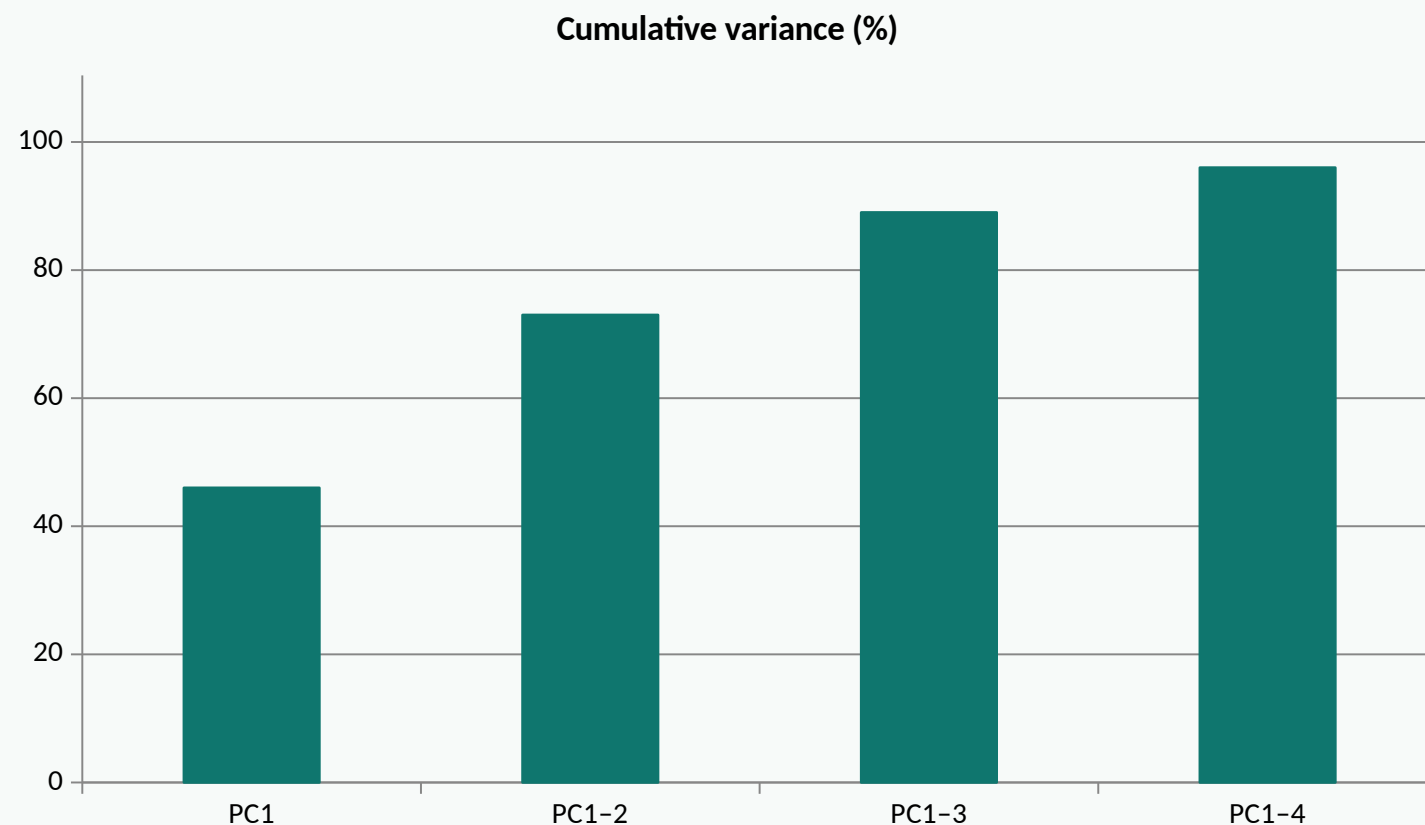
46% + 27% = 73% → retain 2 components

READOUT

**cumulative explained
variance**

Show units, denominator, and the comparison baseline.

Retained variance and loadings · evidence



WHAT THE DATA SHOWS

The selected n is the first threshold crossing and is recorded with the data version.

Evidence becomes a decision only after scope and uncertainty are stated.

Retained variance and loadings · failure and control

FAILURE MODE

Choosing 70% mechanically when the discarded component contains rare critical behaviour.

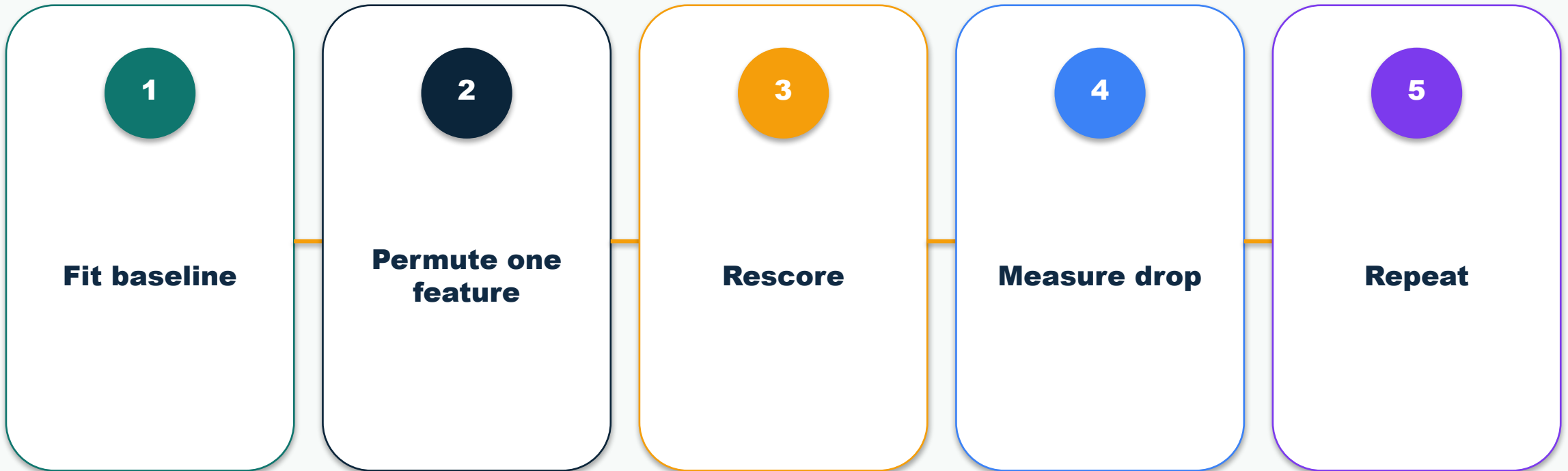
CONTROL

Review reconstruction errors and downstream performance.

VERIFICATION

The selected n is the first threshold crossing and is recorded with the data version.

Feature ranking · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Feature ranking · concrete contract

**importance**

score decrease

**repeats**

uncertainty

**correlation**

shared credit

**target**

defined

Feature ranking · worked calculation

$$\text{importance}_j = \text{score}(X) - \text{score}(\text{permute } X_j)$$

WORKED EXAMPLE

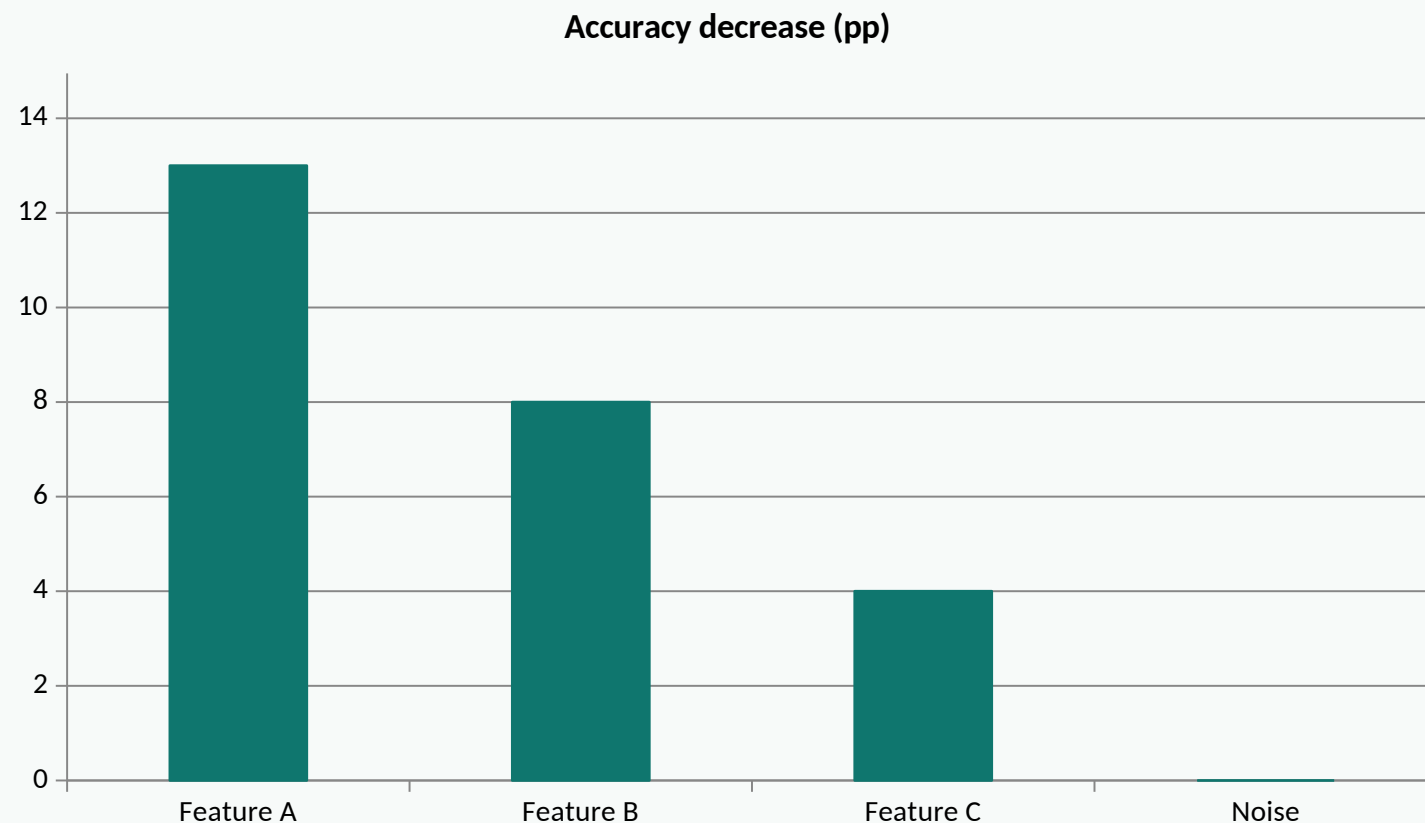
.91 baseline – .78 permuted = .13 importance

READOUT

**permutation
importance**

Show units, denominator, and the comparison baseline.

Feature ranking · evidence



WHAT THE DATA SHOWS

Importance is positive, repeatable, and connected to an auditable model score.

Evidence becomes a decision only after scope and uncertainty are stated.

Feature ranking · failure and control

FAILURE MODE

Correlated variables split or mask importance and ranking is mistaken for causality.

CONTROL

Group correlated features and validate rankings on hold-out data.

VERIFICATION

Importance is positive, repeatable, and connected to an auditable model score.

Lab 12 · Retain variance with PCA

CENTRAL APP

**PCA & feature
ranking**

<http://localhost:8501>

INPUT

Lab 12 mock CSV

MECHANISM

PCA & feature ranking

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 12 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Retain variance with PCA produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Lab 13 · Rank predictive features

CENTRAL APP

**PCA & feature
ranking**

<http://localhost:8501>

INPUT

Lab 13 mock CSV

MECHANISM

PCA & feature ranking

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 13 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Rank predictive features produces a reproducible artifact and a defensible interpretation—not only a screenshot.

TOPIC 07

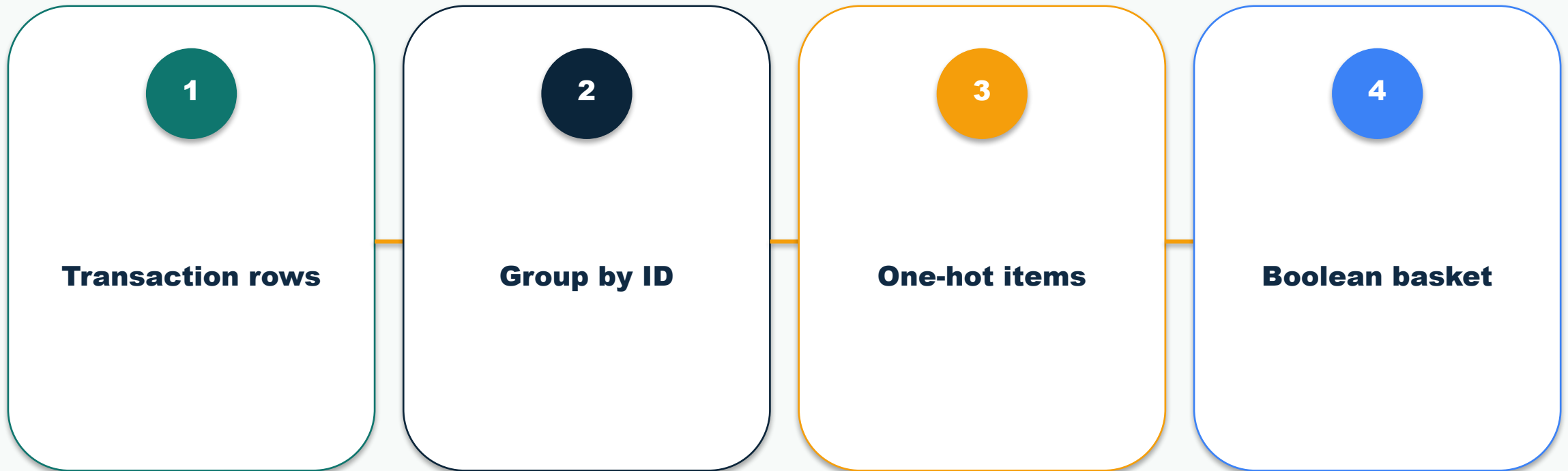
Association Analysis

LO7 · A3 · K9 · From baskets to defensible co-occurrence rules

Mechanism → contract → calculation → evidence → control



Transaction-to-basket encoding · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Transaction-to-basket encoding · concrete contract

**transaction_id**

grouping key

**item**

canonical label

**cell**

present/absent

**duplicates**

collapse

Transaction-to-basket encoding · worked calculation

$\text{basket}[t,i]=1$ if item i occurs in transaction t

WORKED EXAMPLE

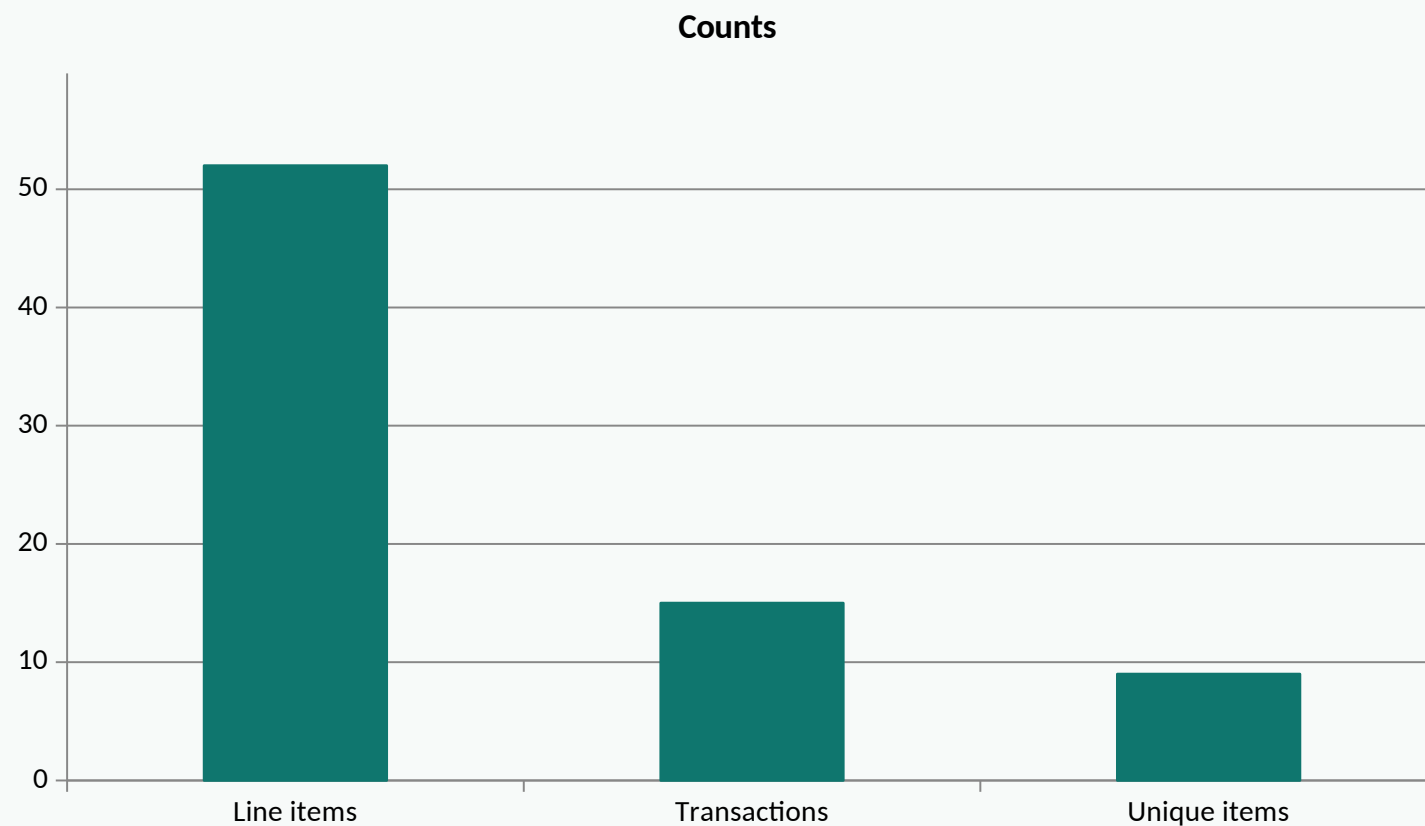
T001 milk+diapers+beer → three True cells

READOUT

basket density

Show units, denominator, and the comparison baseline.

Transaction-to-basket encoding · evidence



WHAT THE DATA SHOWS

One encoded row exists per transaction and item counts reconcile.

Evidence becomes a decision only after scope and uncertainty are stated.

Transaction-to-basket encoding · failure and control

FAILURE MODE

Treating purchase quantity as repeated independent transactions.

CONTROL

Define the basket unit and collapse duplicate item lines per transaction.

VERIFICATION

One encoded row exists per transaction and item counts reconcile.

Support · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Support · concrete contract

#

support(A)

#

support($A \cap B$)

#

minimum support

{ }

denominator

all transactions

Support · worked calculation

$$\text{support}(A) = \text{count}(A) / N$$

WORKED EXAMPLE

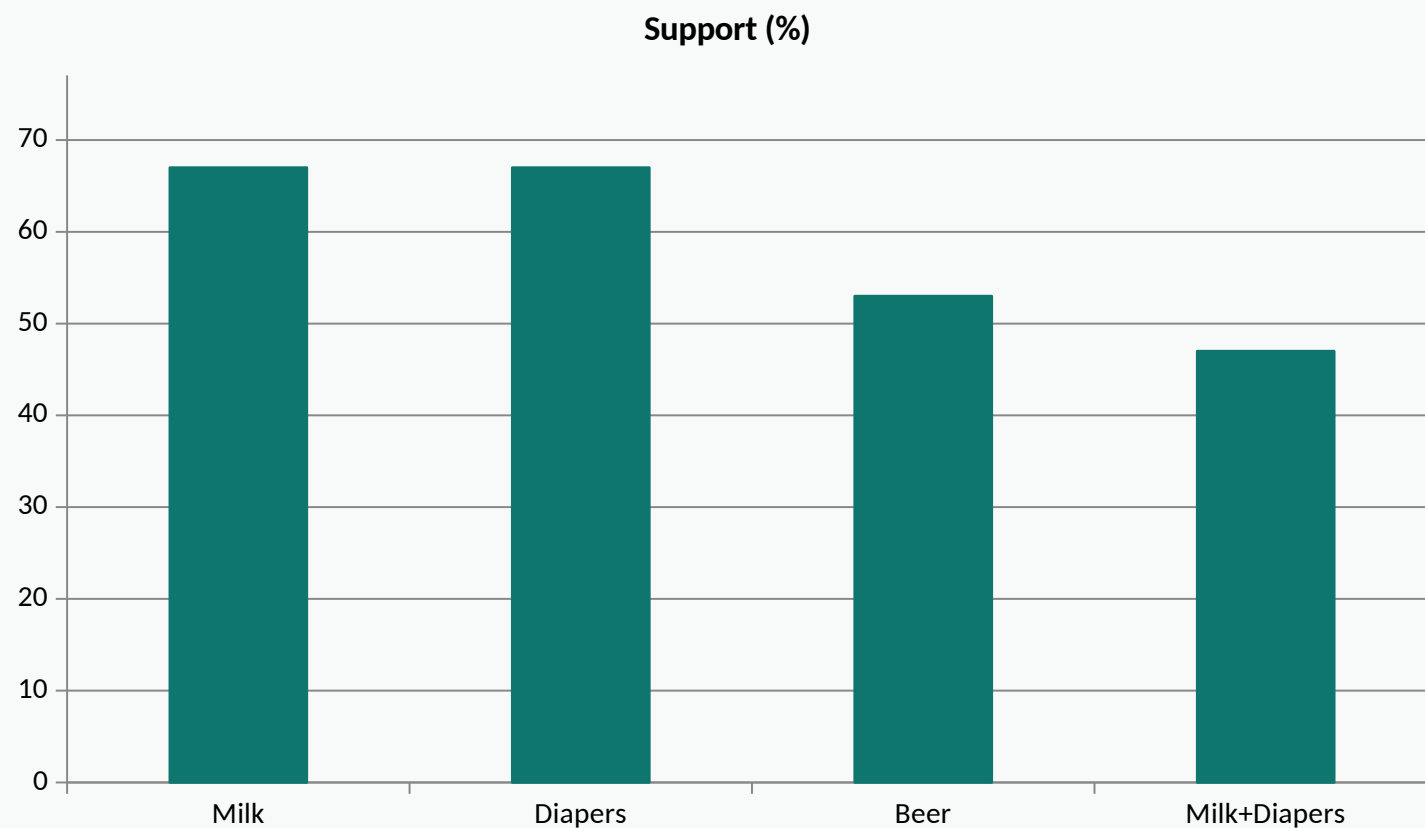
beer appears in 8 of 15 baskets → support=.533

READOUT

support

Show units, denominator, and the comparison baseline.

Support · evidence



WHAT THE DATA SHOWS

Manual counts for selected itemsets match the mined result.

Evidence becomes a decision only after scope and uncertainty are stated.

Support · failure and control

FAILURE MODE

A high minimum support removes valuable niche patterns.

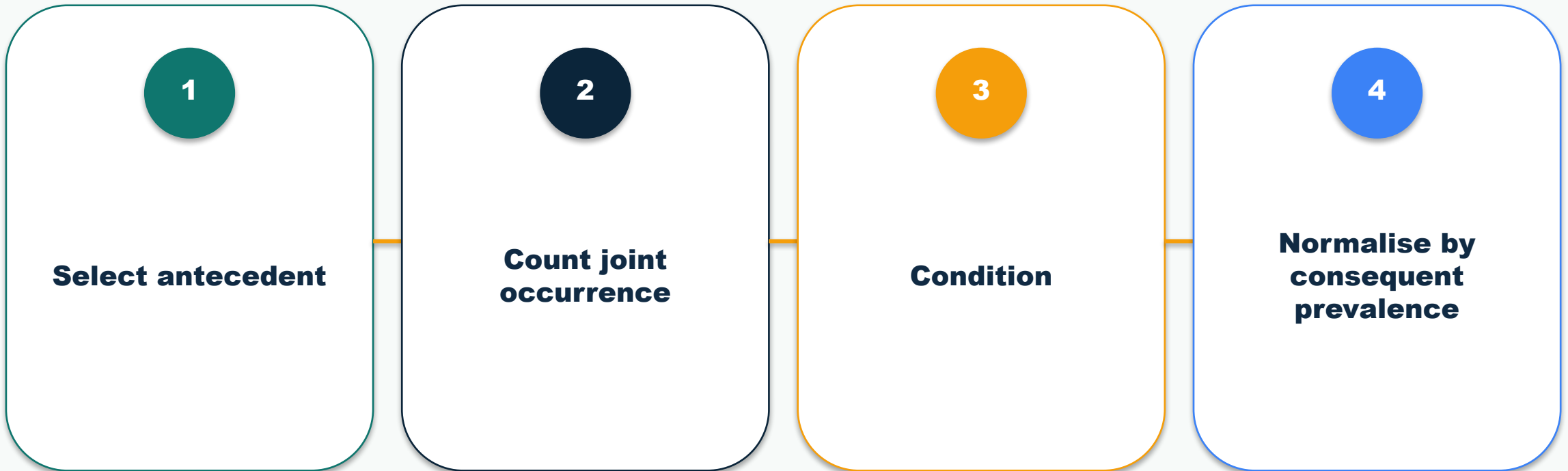
CONTROL

Tune support to transaction volume and decision value.

VERIFICATION

Manual counts for selected itemsets match the mined result.

Confidence and lift · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Confidence and lift · concrete contract

#

$$\text{confidence}(A \rightarrow B) = \text{support}(A \cap B) / \text{support}(A)$$

#

$$\text{lift} = \text{confidence} / \text{support}(B)$$

{ }

$$\text{lift} > 1$$

positive association

#

direction matters

Confidence and lift · worked calculation

$$\text{lift}(A \rightarrow B) = \text{support}(A \cap B) / (\text{support}(A) \text{support}(B))$$

WORKED EXAMPLE

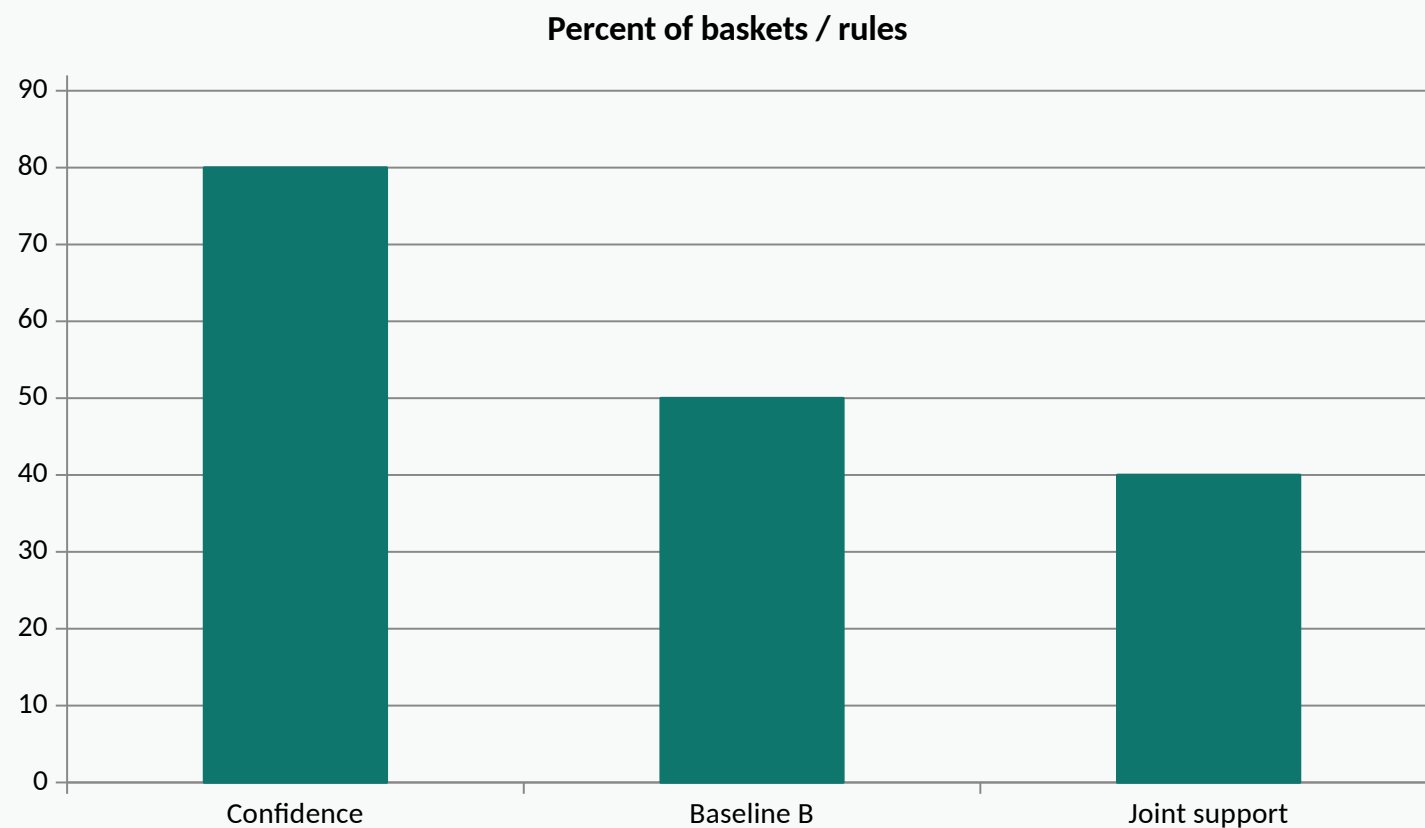
$$.40 / (.50 \times .50) = 1.60$$

READOUT

confidence and lift

Show units, denominator, and the comparison baseline.

Confidence and lift · evidence



WHAT THE DATA SHOWS

The rule includes support, confidence, lift, and basket counts.

Evidence becomes a decision only after scope and uncertainty are stated.

Confidence and lift · failure and control

FAILURE MODE

High confidence for a universally popular item is mistaken for a useful rule.

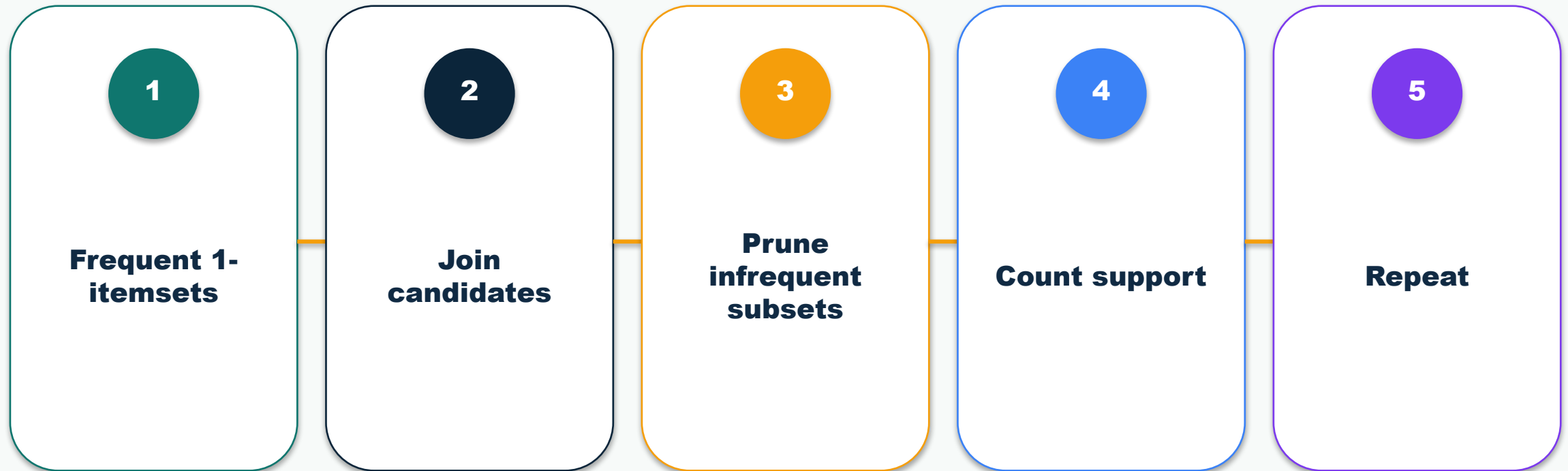
CONTROL

Require lift above 1 and assess absolute support.

VERIFICATION

The rule includes support, confidence, lift, and basket counts.

Apriori candidate pruning · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Apriori candidate pruning · concrete contract

#

downward closure

#

candidate level k

#

minimum support

#

frequent itemset

Apriori candidate pruning · worked calculation

if itemset S is infrequent, every superset of S is infrequent

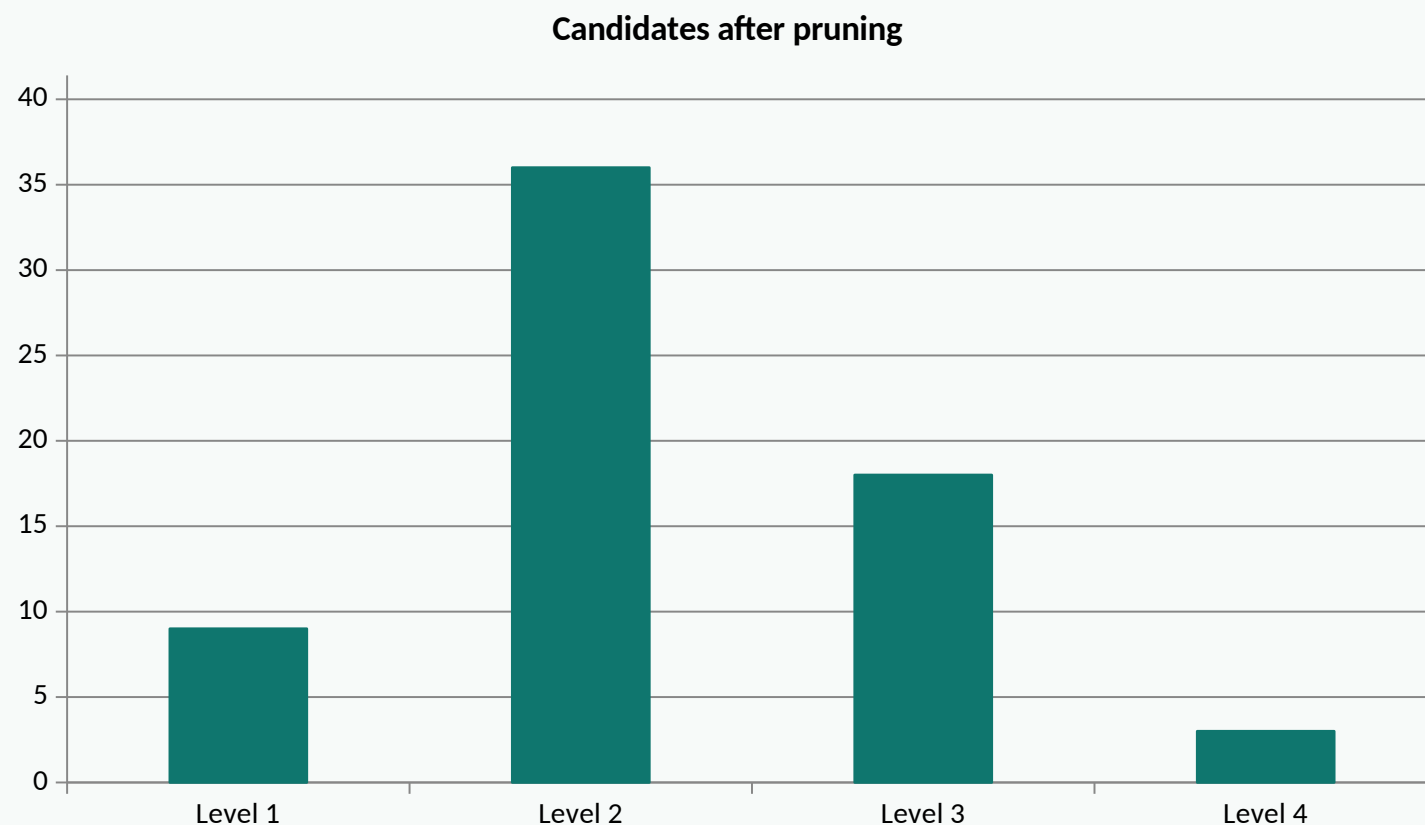
WORKED EXAMPLE

{milk,diapers} below threshold → prune all triples containing both

READOUT
candidate count by level

Show units, denominator, and the comparison baseline.

Apriori candidate pruning · evidence



WHAT THE DATA SHOWS

Candidate counts shrink after pruning and results reproduce at the same threshold.

Evidence becomes a decision only after scope and uncertainty are stated.

Apriori candidate pruning · failure and control

FAILURE MODE

A low threshold creates a combinatorial explosion of candidates.

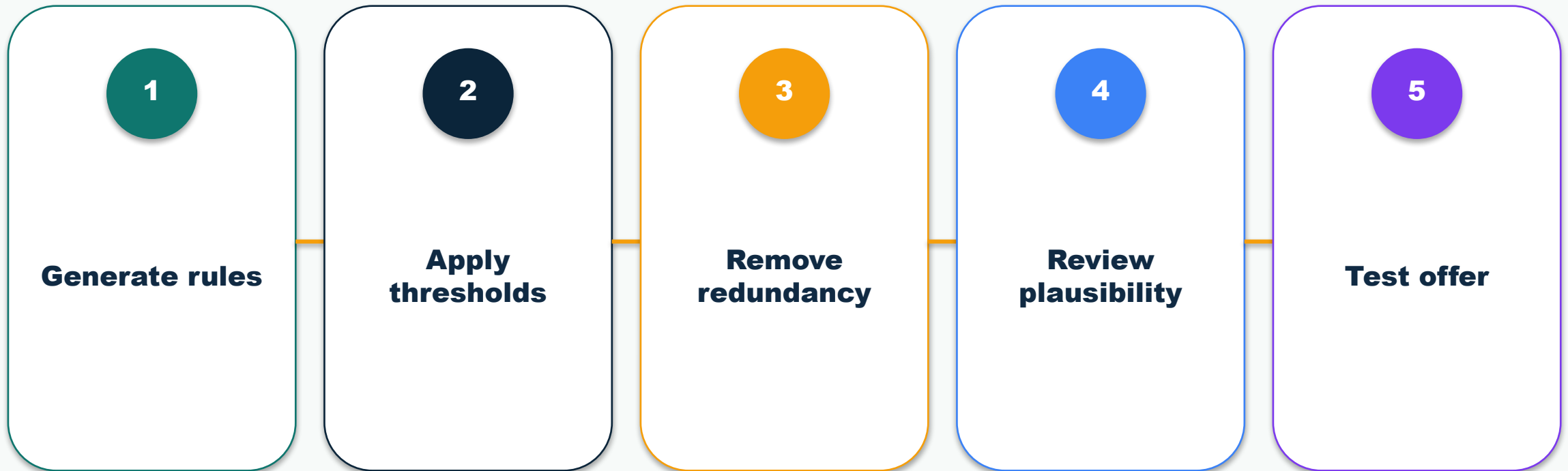
CONTROL

Start with a defensible support floor and cap itemset length.

VERIFICATION

Candidate counts shrink after pruning and results reproduce at the same threshold.

Rule filtering and business action · runtime mechanism



Trace the named state at each boundary; a missing state breaks reproducibility.

Rule filtering and business action · concrete contract



antecedent

trigger basket



consequent

suggested item



leverage

excess joint rate



action

controlled test

Rule filtering and business action · worked calculation

$$\text{leverage} = \text{support}(A \cap B) - \text{support}(A) \text{support}(B)$$

WORKED EXAMPLE

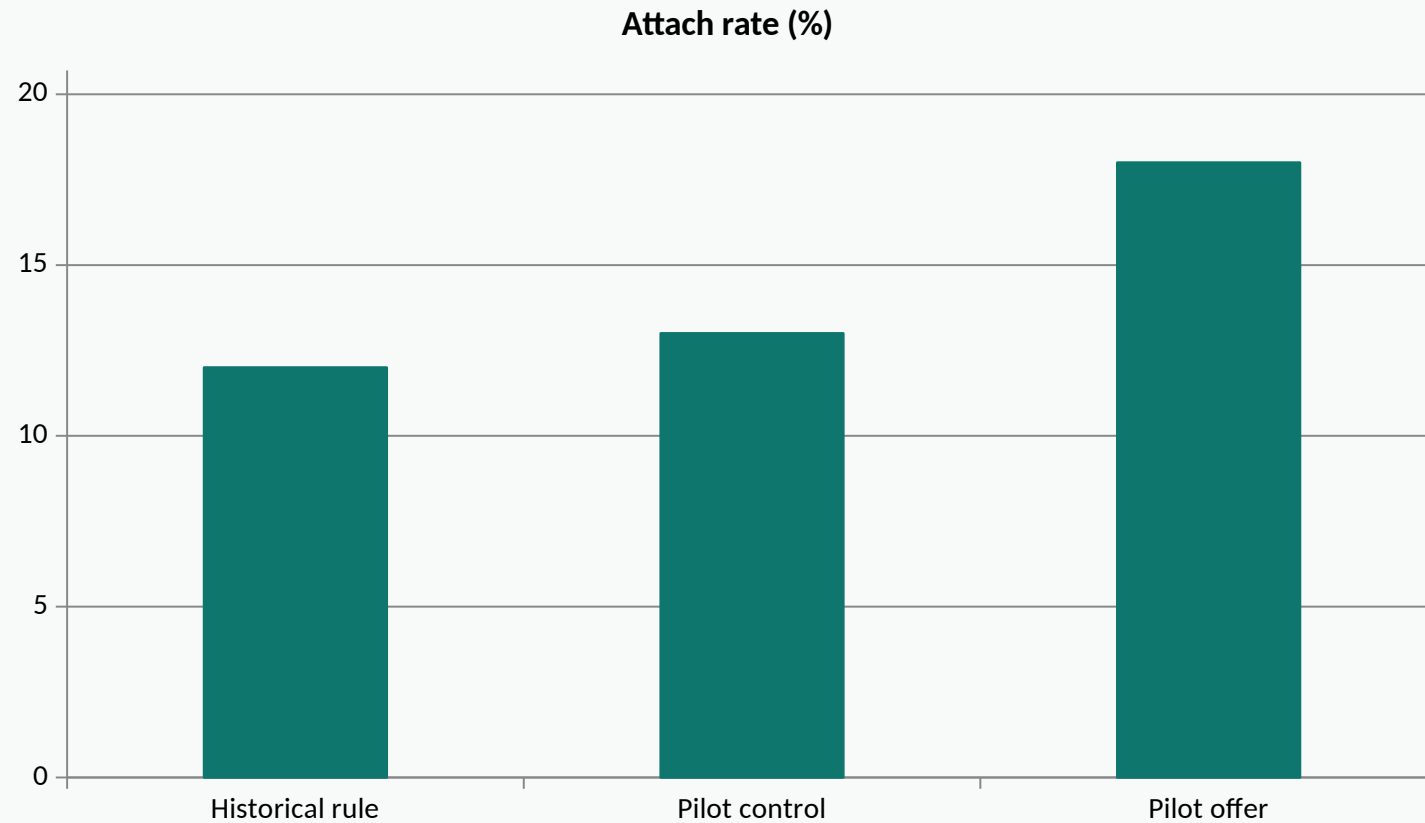
$$.40 - (.50 \times .50) = .15$$

READOUT

**incremental basket
outcome**

Show units, denominator, and the comparison baseline.

Rule filtering and business action · evidence



WHAT THE DATA SHOWS

The action has an owner, audience, guardrail, and measured outcome.

Evidence becomes a decision only after scope and uncertainty are stated.

Rule filtering and business action · failure and control

FAILURE MODE

Deploying a bundle solely because lift is high on a tiny sample.

CONTROL

Require minimum counts and validate with a controlled promotion.

VERIFICATION

The action has an owner, audience, guardrail, and measured outcome.

Lab 14 · Mine market-basket rules

CENTRAL APP

Association rules

<http://localhost:8501>

INPUT

Lab 14 mock CSV

MECHANISM

Association rules

EVIDENCE

metric + chart + interpretation

The full click path, commands, checks, troubleshooting, and cleanup are in the Learner Guide and lab folder.

Lab 14 · acceptance evidence

1

Record inputs

2

**Capture
parameters**

3

Export results

4

Explain limits

PASS: Mine market-basket rules produces a reproducible artifact and a defensible interpretation—not only a screenshot.

Python mechanism reference

The v16 technical topics are retained below as runnable Python concepts and checks.

CSV import and data table

MECHANISM

Parser assigns types; validation quarantines malformed rows.

VERIFY

Schema + row-count reconciliation

Scatter plots and interactive subsets

INPUT

typed data

PYTHON

**Selections must retain stable
row identifiers.**

PROOF

**Selected + excluded
= source**

Select columns and rows

MECHANISM

Feature role and row predicate are explicit contracts.

VERIFY

No target leakage

Merge and concatenate

INPUT

typed data

PYTHON

**Join keys control horizontal
integration; concat stacks
compatible schemas.**

PROOF

**Cardinality and
totals reconcile**

Sampling, transpose, and randomisation

MECHANISM

Sampling estimates; transpose changes unit-of-analysis; shuffling breaks order dependence.

VERIFY

Seed and row identity recorded

Feature statistics and save data

INPUT

typed data

PYTHON

**Profile values before export;
persist a cleaned dataset with
provenance.**

PROOF

**Hash, schema, and
row count**

Decision tree

MECHANISM

Recursive impurity reduction forms axis-aligned leaves.

VERIFY

Depth and leaf support

Logistic regression

INPUT

typed data

PYTHON

**Linear log-odds become
calibrated class probabilities.**

PROOF

**Calibration plus
threshold**

K-nearest neighbours

MECHANISM

Scaled distance and neighbourhood size define the vote.

VERIFY

Sensitivity to k and scaling

Support vector machine and kernel

INPUT

typed data

PYTHON

**Margin maximisation
separates classes; kernels
change similarity space.**

PROOF

**C, gamma, and held-
out AUC**

Gaussian Naive Bayes

MECHANISM

Class priors multiply conditional Gaussian likelihoods.

VERIFY

Likelihood assumptions checked

Random forest

INPUT

typed data

PYTHON

**Bootstrap trees plus random
feature subsets reduce
variance.**

PROOF

**Out-of-bag or CV
score**

Stacking classifier

MECHANISM

Out-of-fold base predictions train a meta-learner without leakage.

VERIFY

Meta-features are out-of-fold

AdaBoost

INPUT

typed data

PYTHON

**Successive weak learners
reweight previously
misclassified cases.**

PROOF

**Learning rate and
error trajectory**

Save and load model

MECHANISM

Persist preprocessing and estimator together as one versioned pipeline.

VERIFY

Round-trip prediction equality

Classification threshold

INPUT

typed data

PYTHON

**A score becomes a class only
after applying an operating
threshold.**

PROOF

**Cost matrix and
capacity**

Release checklist

FAILURE MODE

A good-looking output is mistaken for a valid model.

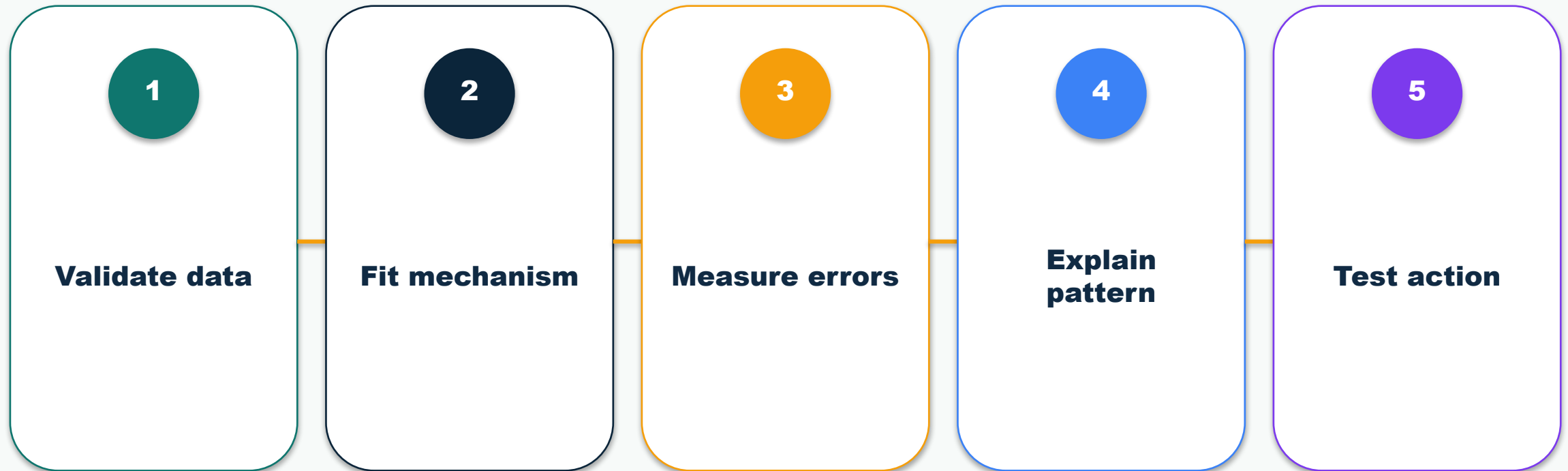
CONTROL

Recheck scope, leakage, baseline, metric, error costs, and reproducibility.

VERIFICATION

Every lab artifact has a file, parameter record, chart, and interpretation.

From pattern to responsible action



Trace the named state at each boundary; a missing state breaks reproducibility.

THANK YOU

Keep the evidence reproducible.

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